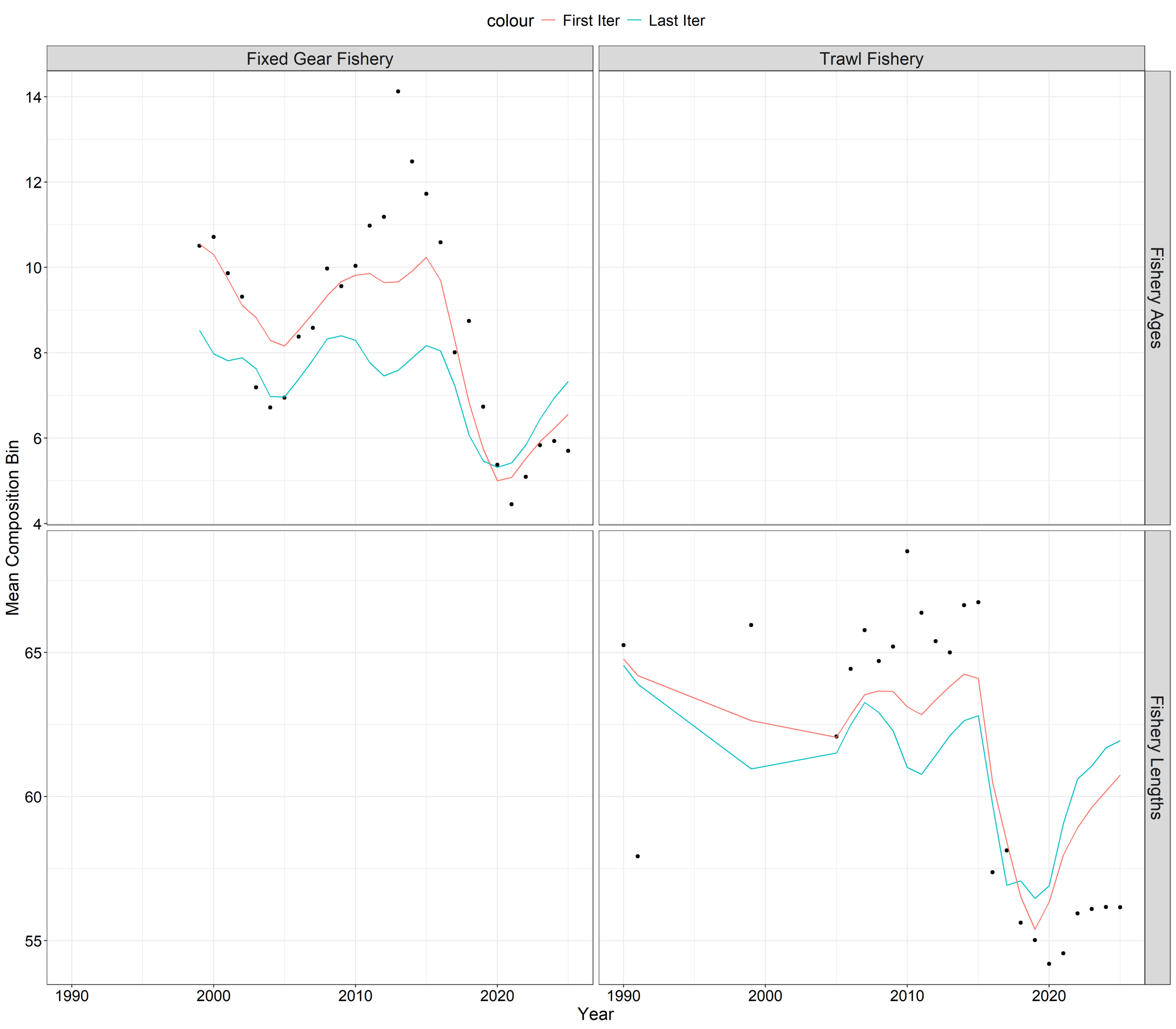
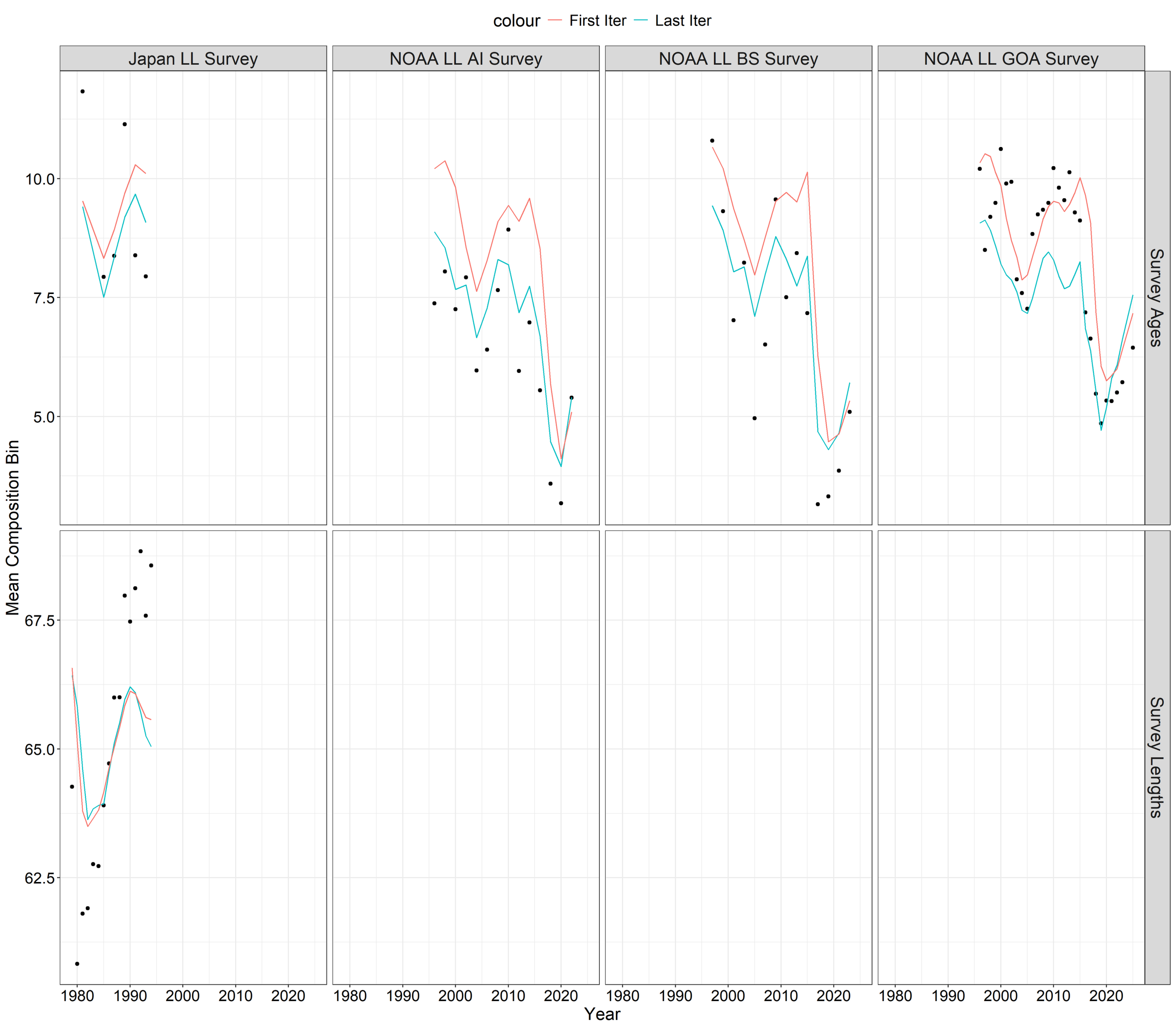
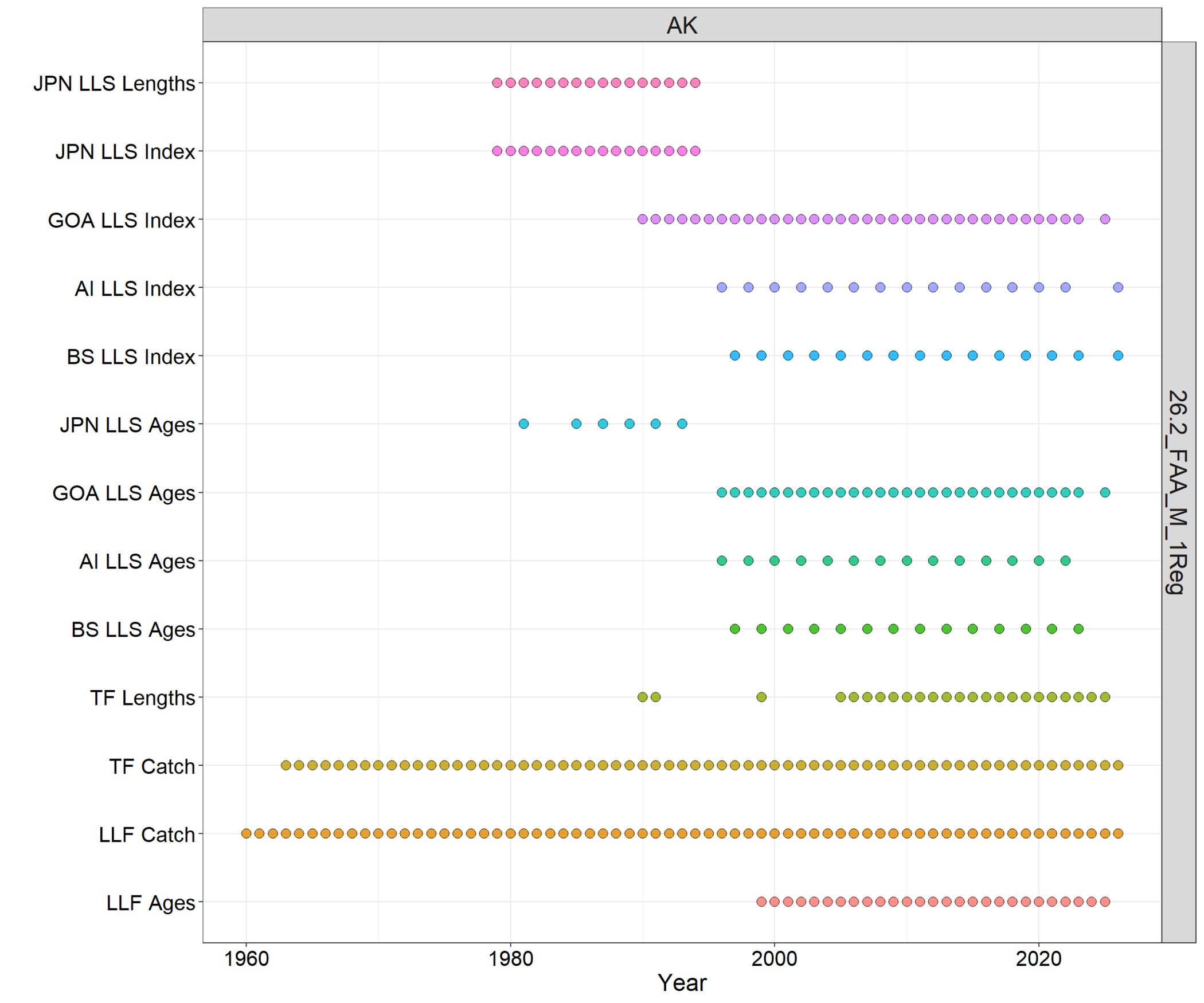
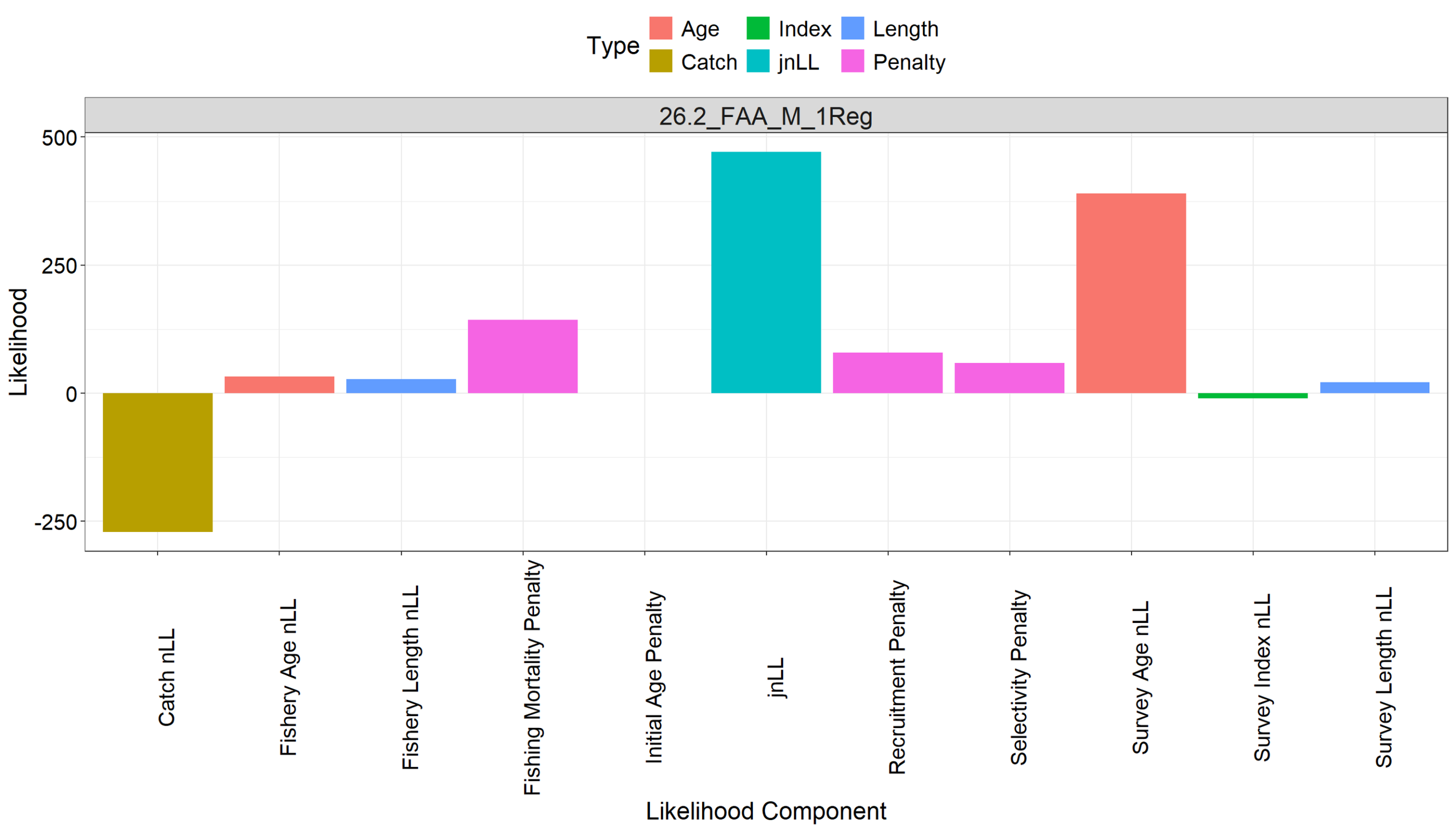

colour
First Iter
Last Iter
Fixed Gear Fishery
Trawl Fishery
Fishery Ages
Fishery Lengths
Mean Composition Bin
14
12
10
8
6
4
65
60
55
1990
2000
2010
2020
Year


colour
First Iter
Last Iter
Japan LL Survey
NOAA LL AI Survey
NOAA LL BS Survey
NOAA LL GOA Survey
Survey Ages
Survey Lengths
Mean Composition Bin
10.0
7.5
5.0
67.5
65.0
62.5
1980
1990
2000
2010
2020
Year


AK
JPN LLS Lengths
JPN LLS Index
GOA LLS Index
AI LLS Index
BS LLS Index
JPN LLS Ages
GOA LLS Ages
AI LLS Ages
BS LLS Ages
TF Lengths
TF Catch
LLF Catch
LLF Ages
26.2_FAA_M_1Reg
1960
1980
2000
2020
Year


Type
Age
Index
Length
Catch
jnLL
Penalty
26.2_FAA_M_1Reg
500
250
0
-250
Likelihood
Catch nLL
Fishery Age nLL
Fishery Length nLL
Fishing Mortality Penalty
Initial Age Penalty
jnLL
Recruitment Penalty
Selectivity Penalty
Survey Age nLL
Survey Index nLL
Survey Length nLL
Likelihood Component

Numbers-at-Age

Sex — Female ····· Male

Age- 1 | Age- 2 | Age- 3 | Age- 4 | Age- 5 | Age- 6

Age- 7 | Age- 8 | Age- 9 | Age- 10 | Age- 11 | Age- 12

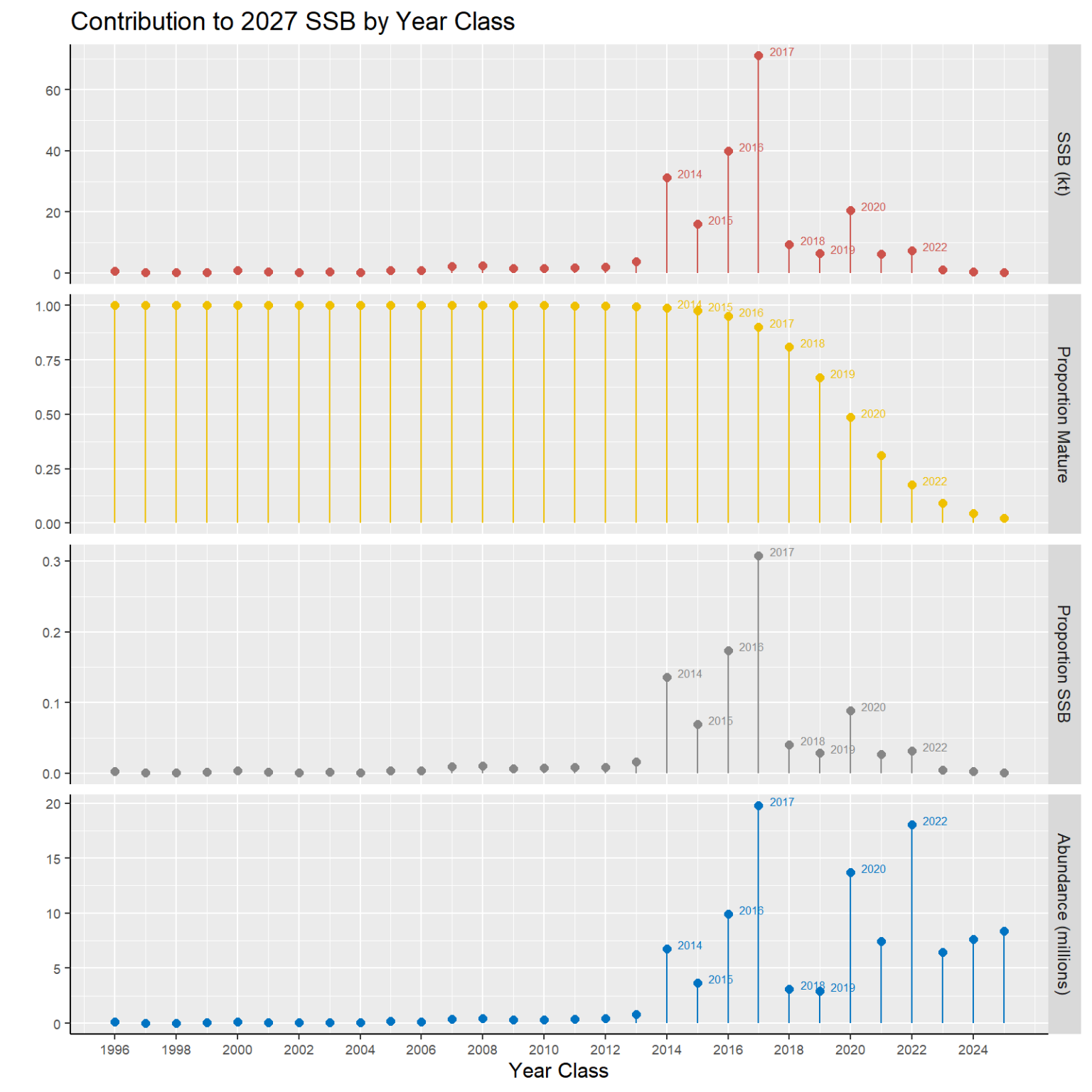
Age- 13 | Age- 14 | Age- 15 | Age- 16 | Age- 17 | Age- 18

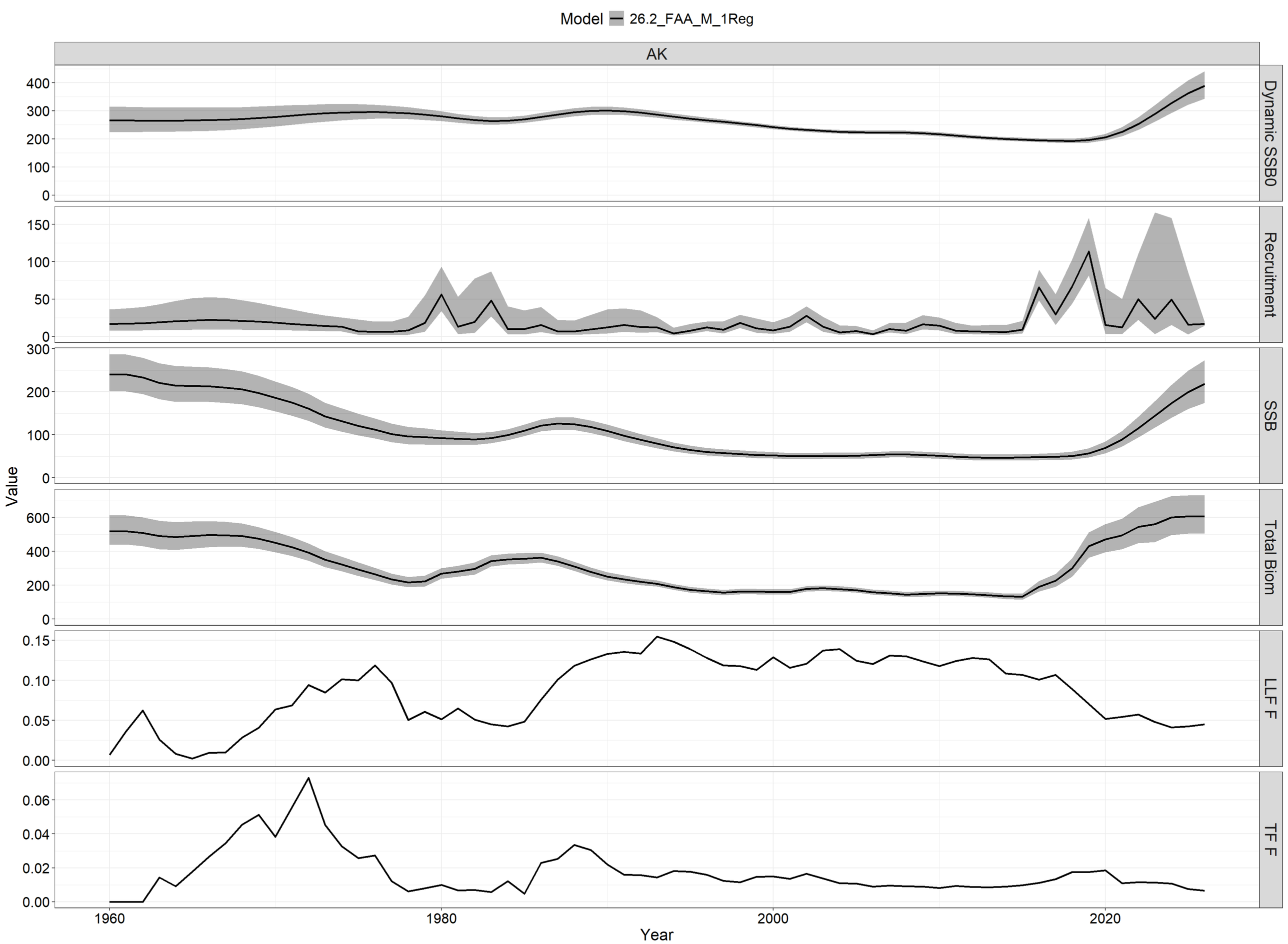
Age- 19 | Age- 20 | Age- 21 | Age- 22 | Age- 23 | Age- 24

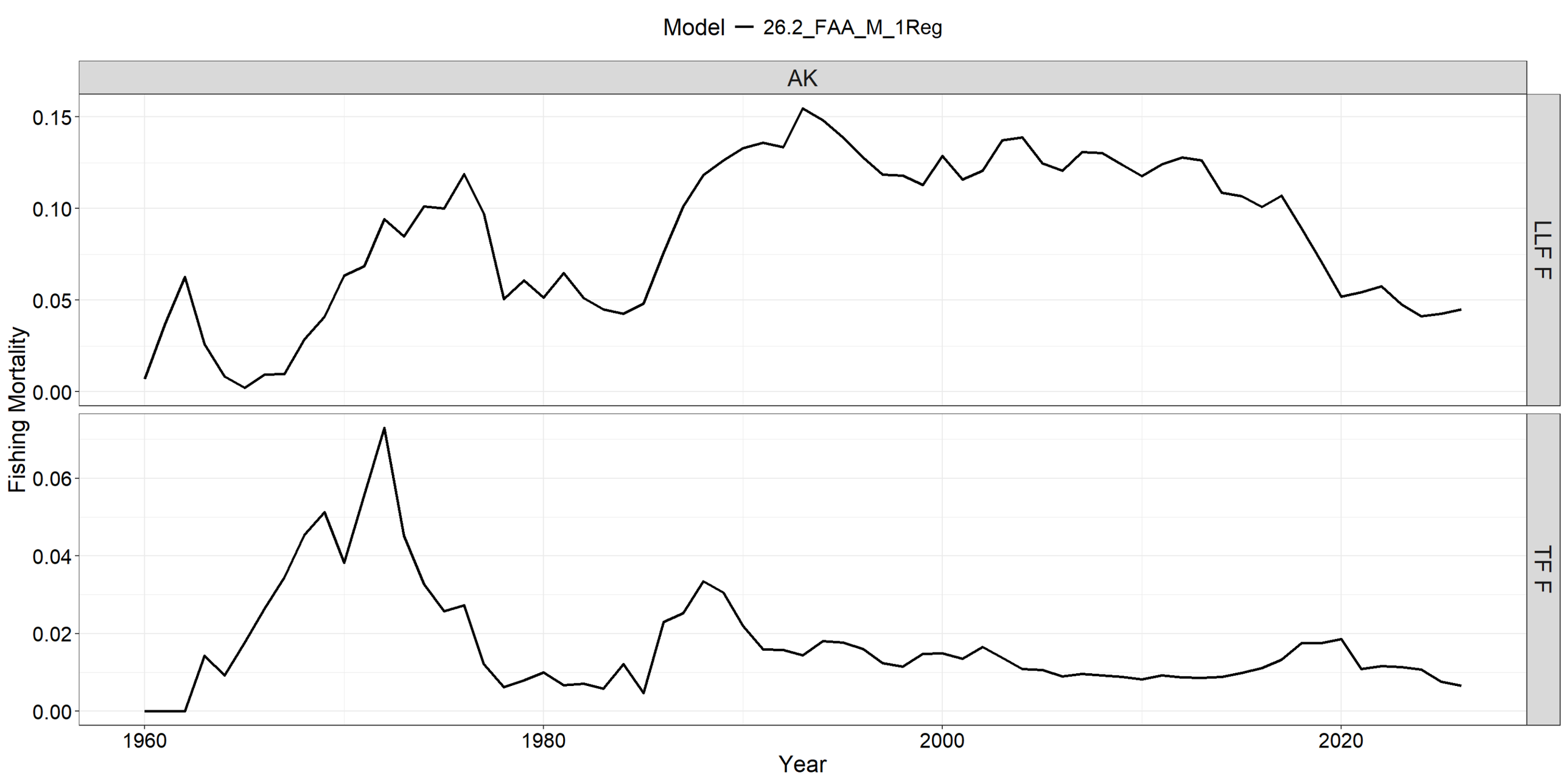
Age- 25 | Age- 26 | Age- 27 | Age- 28 | Age- 29 | Age- 30

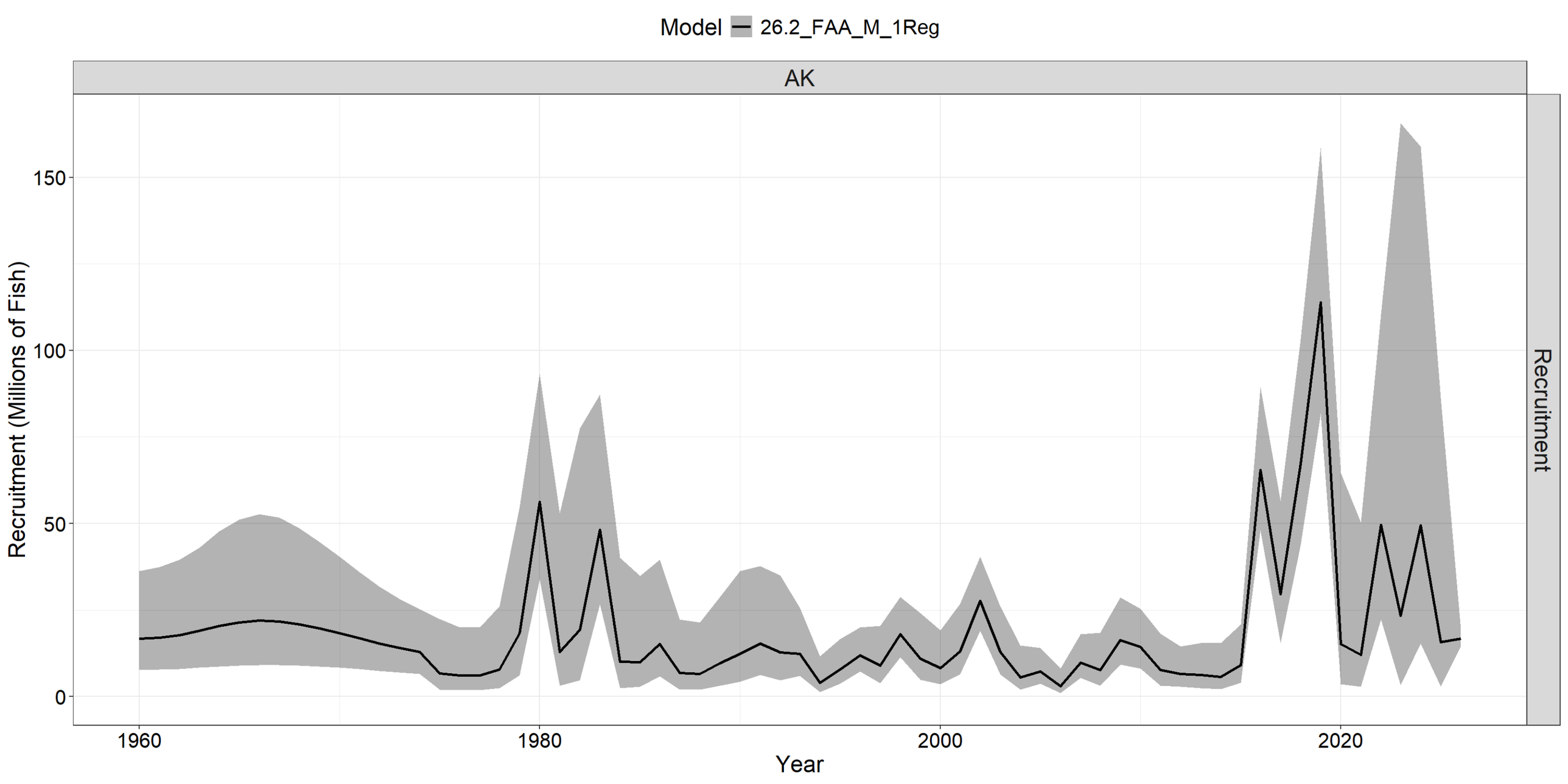
Numbers-at-Age (millions)

Year

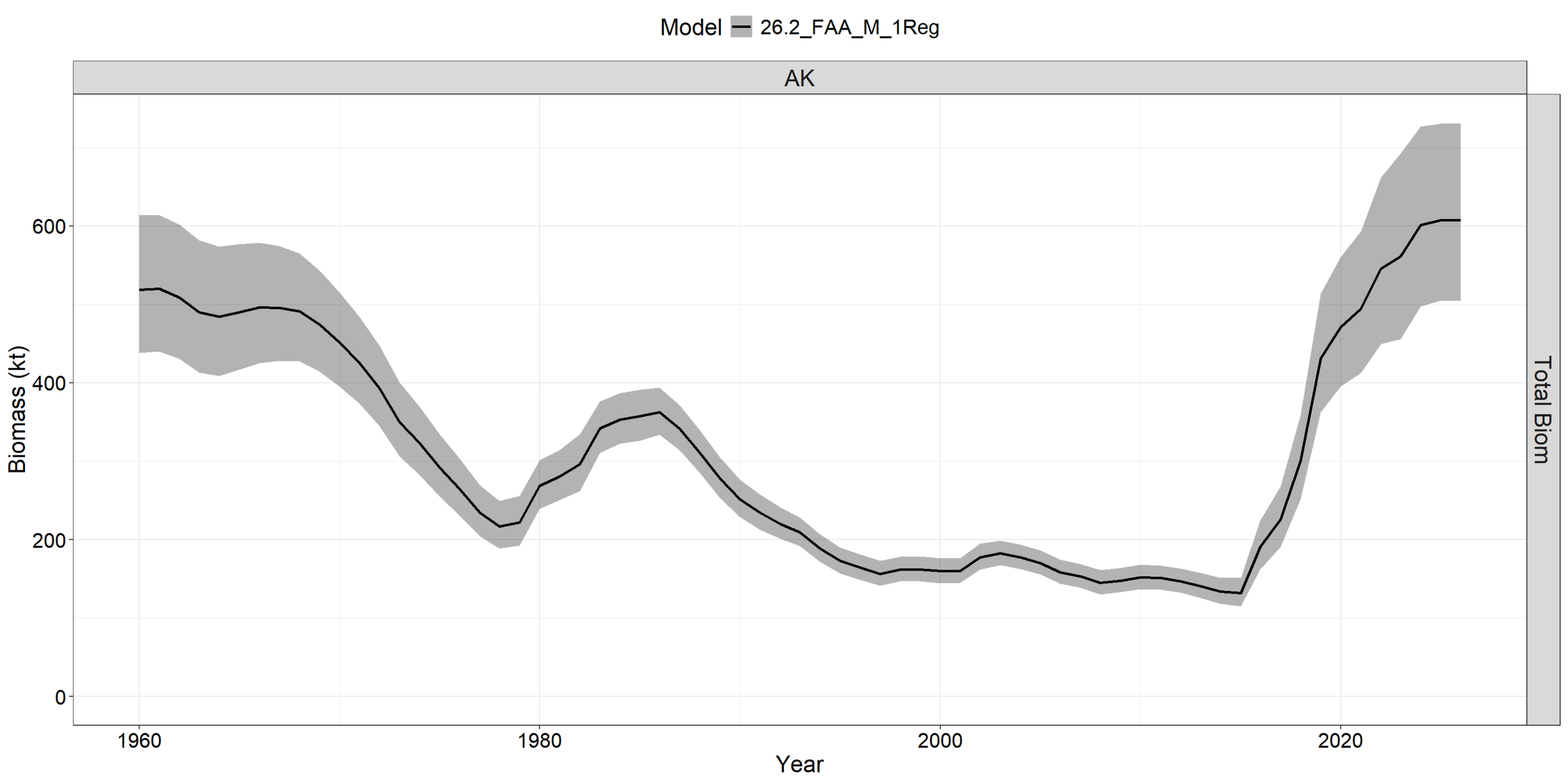

Contribution to 2027 SSB by Year Class
SSB (kt)
Proportion Mature
Proportion SSB
Abundance (millions)
Year Class
1996
1998
2000
2002
2004
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
2014
2015
2016
2017
2018
2019
2020
2022

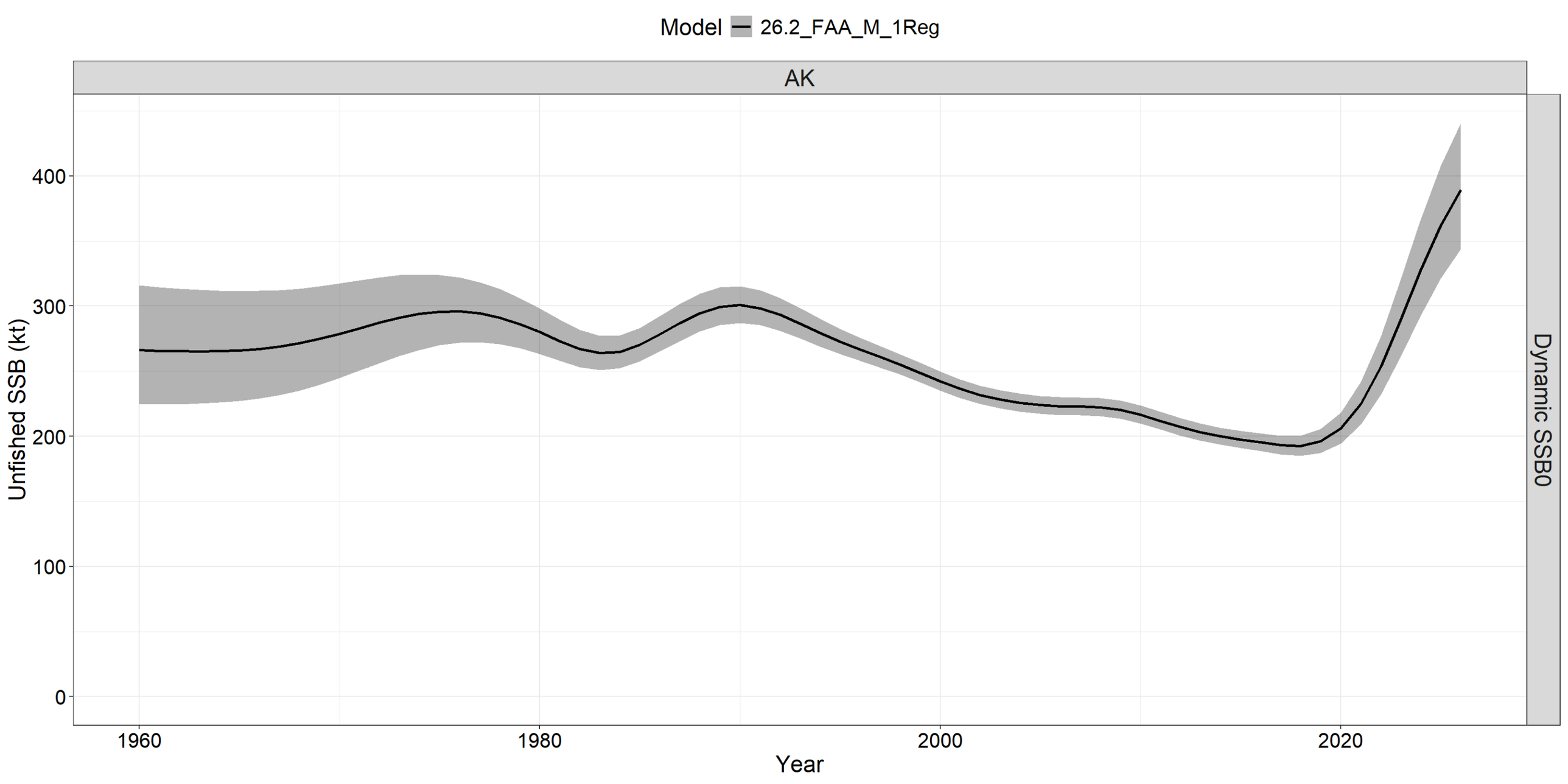

Model
26.2_FAA_M_1Reg
AK
Dynamic SSB0
Recruitment
SSB
Total Biom
LLF F
TF F
Value
Year
0
100
200
300
400
50
150
600
0.00
0.02
0.04
0.06
0.10
0.15
1960
1980
2000
2020


Model — 26.2_FAA_M_1Reg
AK
LLF F
TF F
Fishing Mortality
0.15
0.10
0.05
0.00
0.06
0.04
0.02
0.00
1960
1980
2000
2020
Year

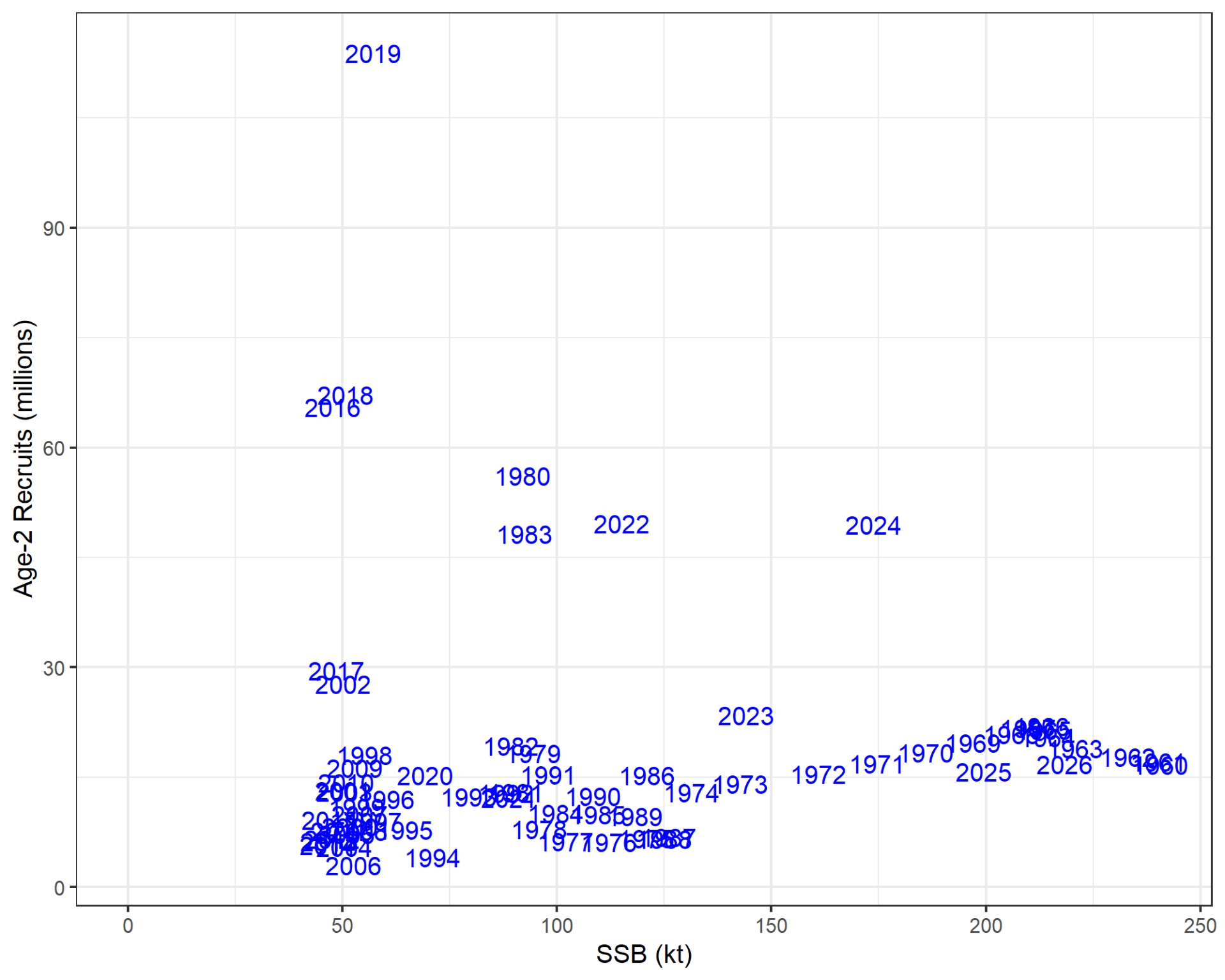

Model
26.2_FAA_M_1Reg
AK
Recruitment
Recruitment (Millions of Fish)
150
100
50
0
1960
1980
2000
2020
Year

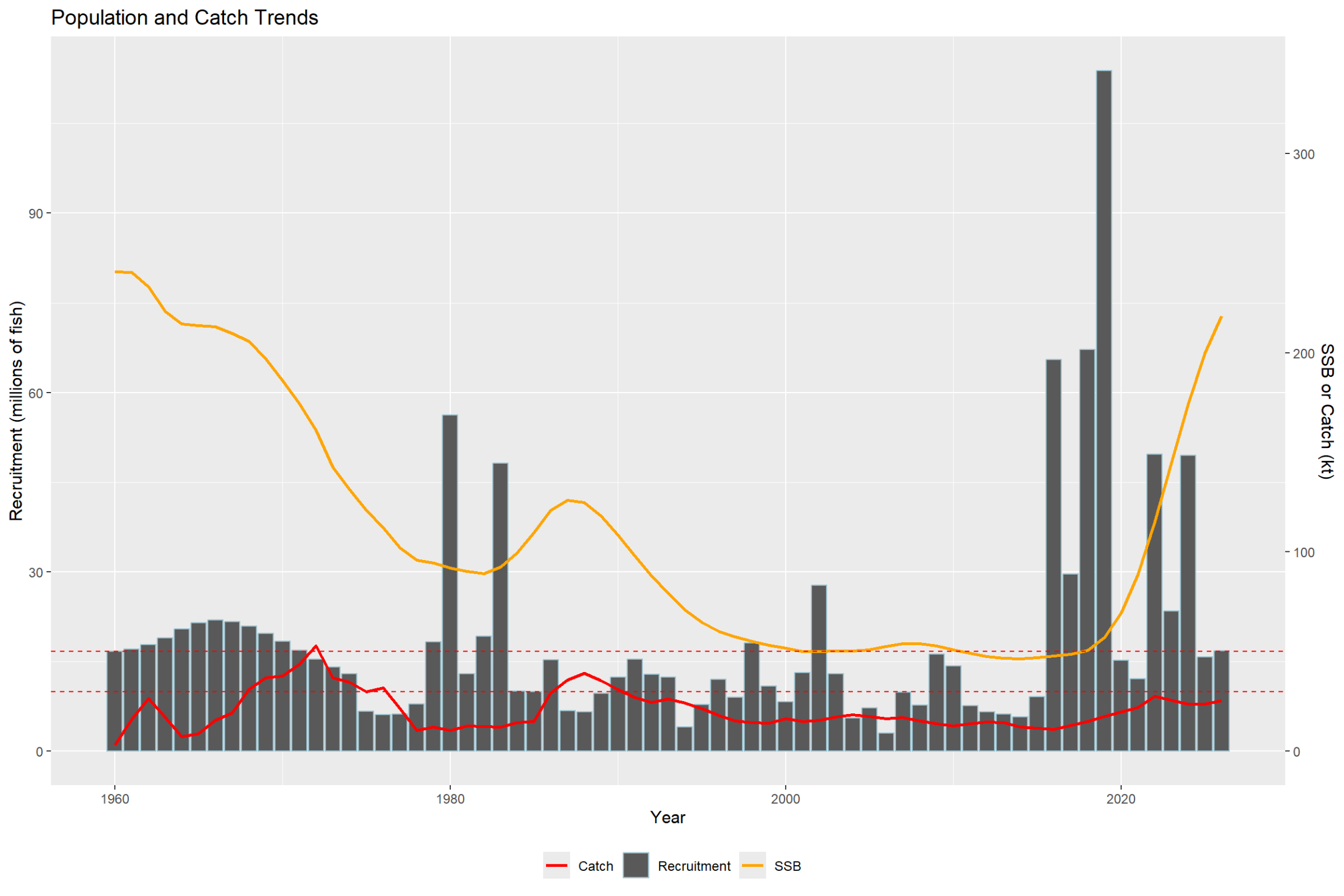

Model
26.2_FAA_M_1Reg
AK
SSB
SSB (kt)
300
200
100
0
1960
1980
2000
2020
Year

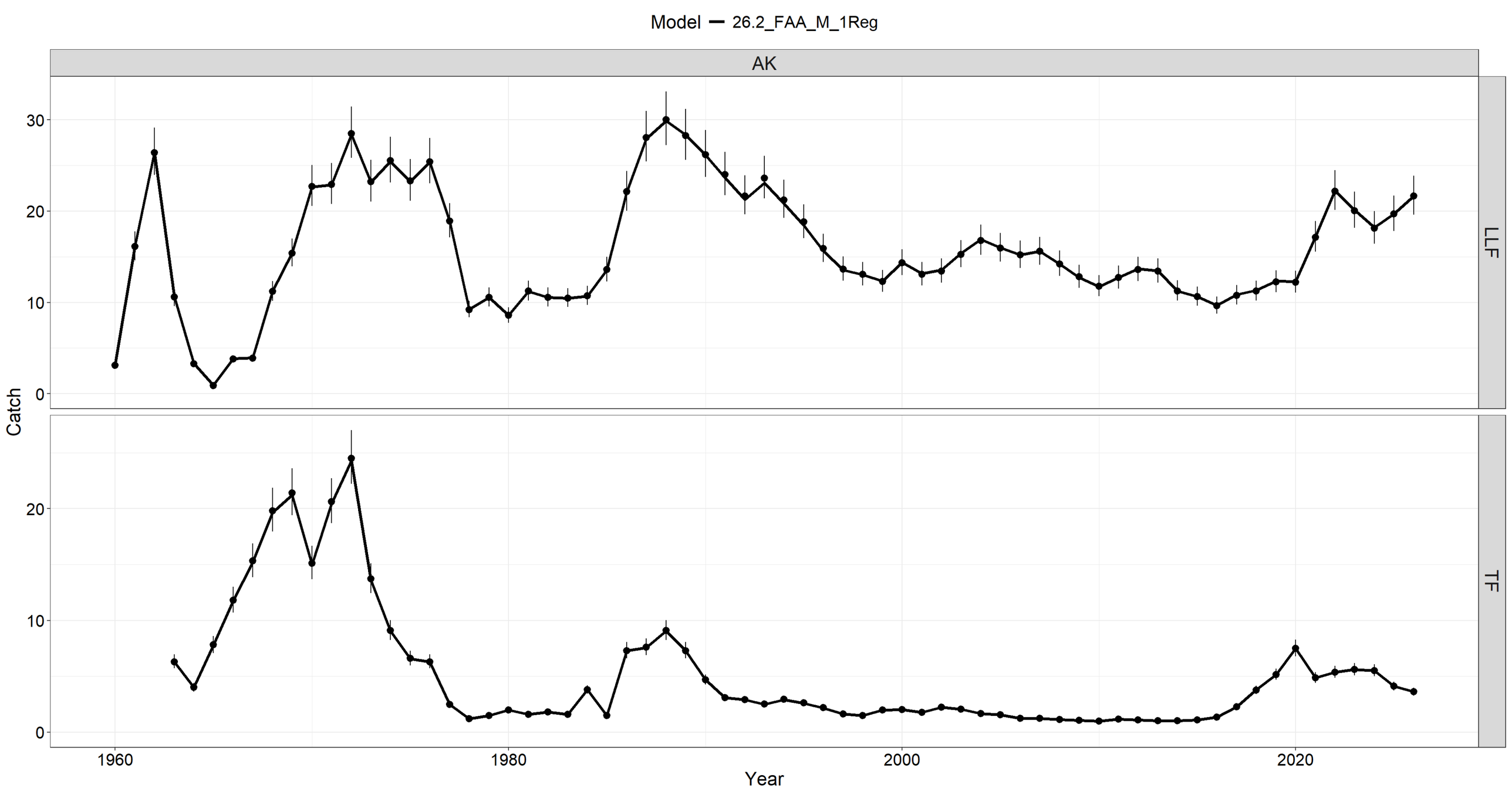

Model
26.2_FAA_M_1Reg
AK
Total Biom
Biomass (kt)
0
200
400
600
1960
1980
2000
2020
Year


Model
26.2_FAA_M_1Reg
AK
Dynamic SSB0
Unfished SSB (kt)
400
300
200
100
0
1960
1980
2000
2020
Year


Type — Population 1 Dynamic SSB0 -- Population 1 SSB
Model — 26.2_FAA_M_1Reg
AK
400
300
200
100
0
SSB (kt)
1960
1980
2000
2020
Year


2019
2018
2016
1980
1983
2022
2024
2017
2002
2023
1998
2009
2020
1991
1986
1974
1973
1972
1971
1970
1969
1963
2025
2026
1990
1994
2006
0
30
60
90
0
50
100
150
200
250
Age-2 Recruits (millions)
SSB (kt)


Population and Catch Trends
Recruitment (millions of fish)
SSB or Catch (kt)
0
30
60
90
0
100
200
300
1960
1980
2000
2020
Year
Catch
Recruitment
SSB


Model — 26.2_FAA_M_1Reg
AK
LLF
TF
Catch
Year
0
10
20
30
1960
1980
2000
2020

Model — 26.2_FAA_M_1Reg

BS LLS

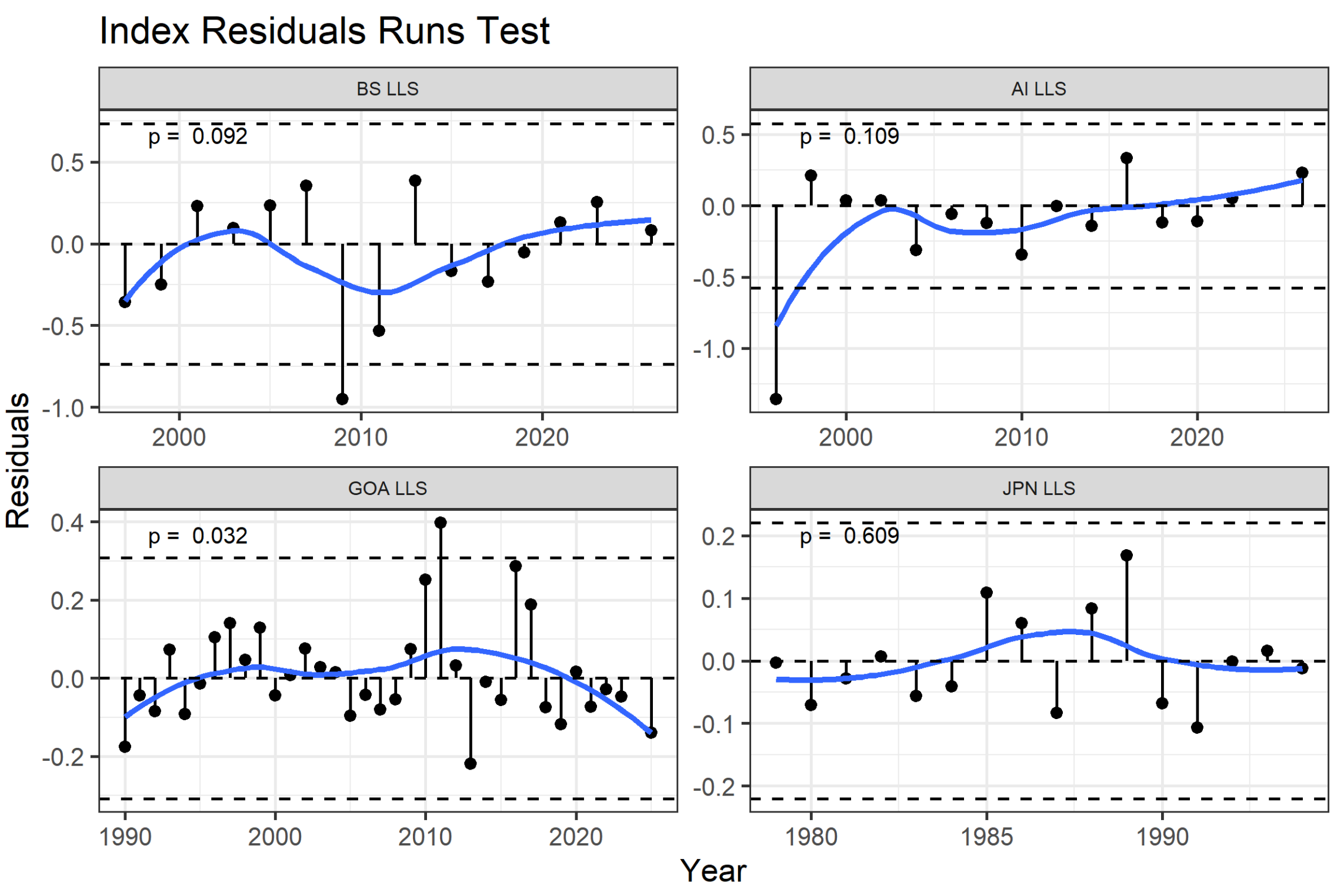
AI LLS

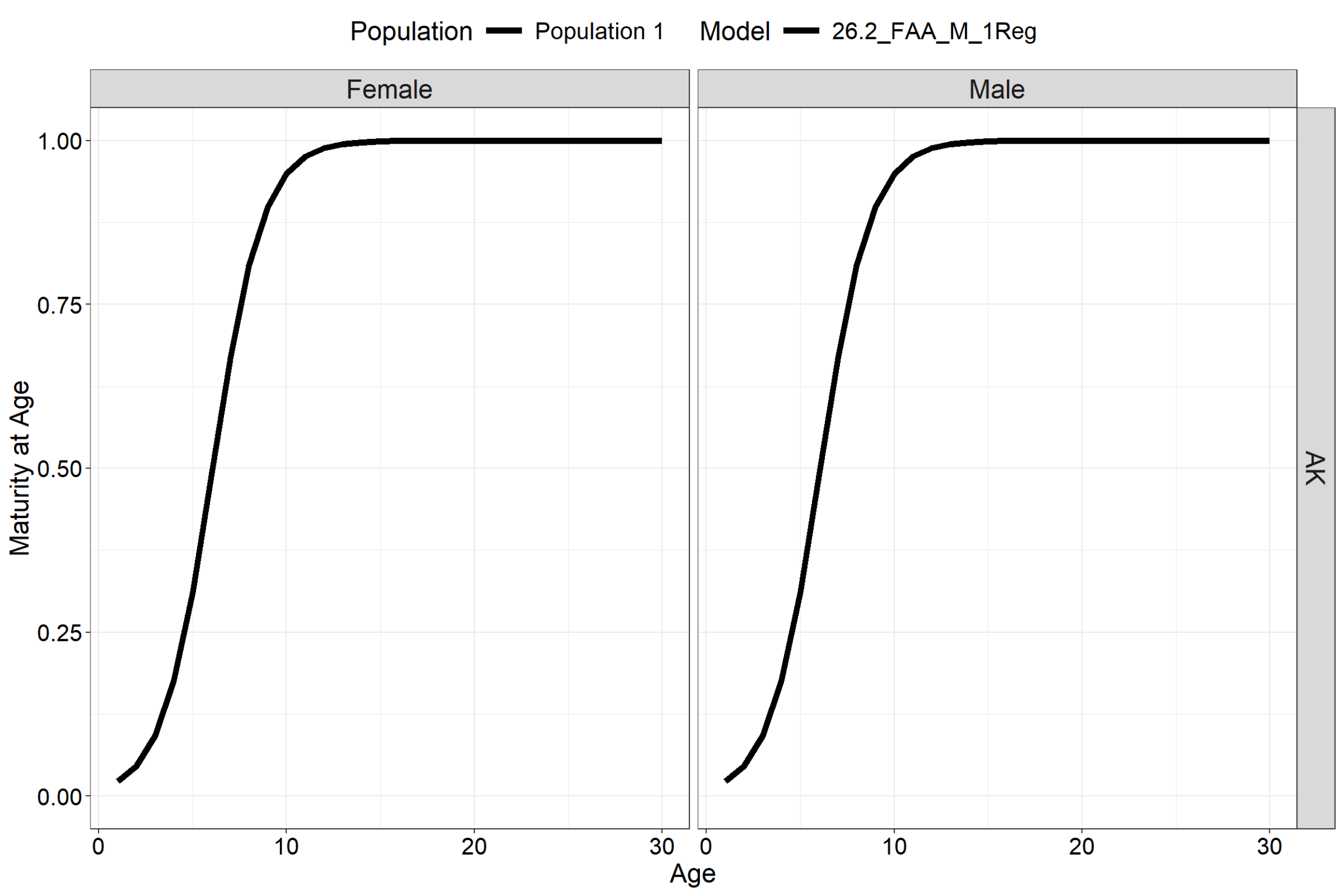
GOA LLS

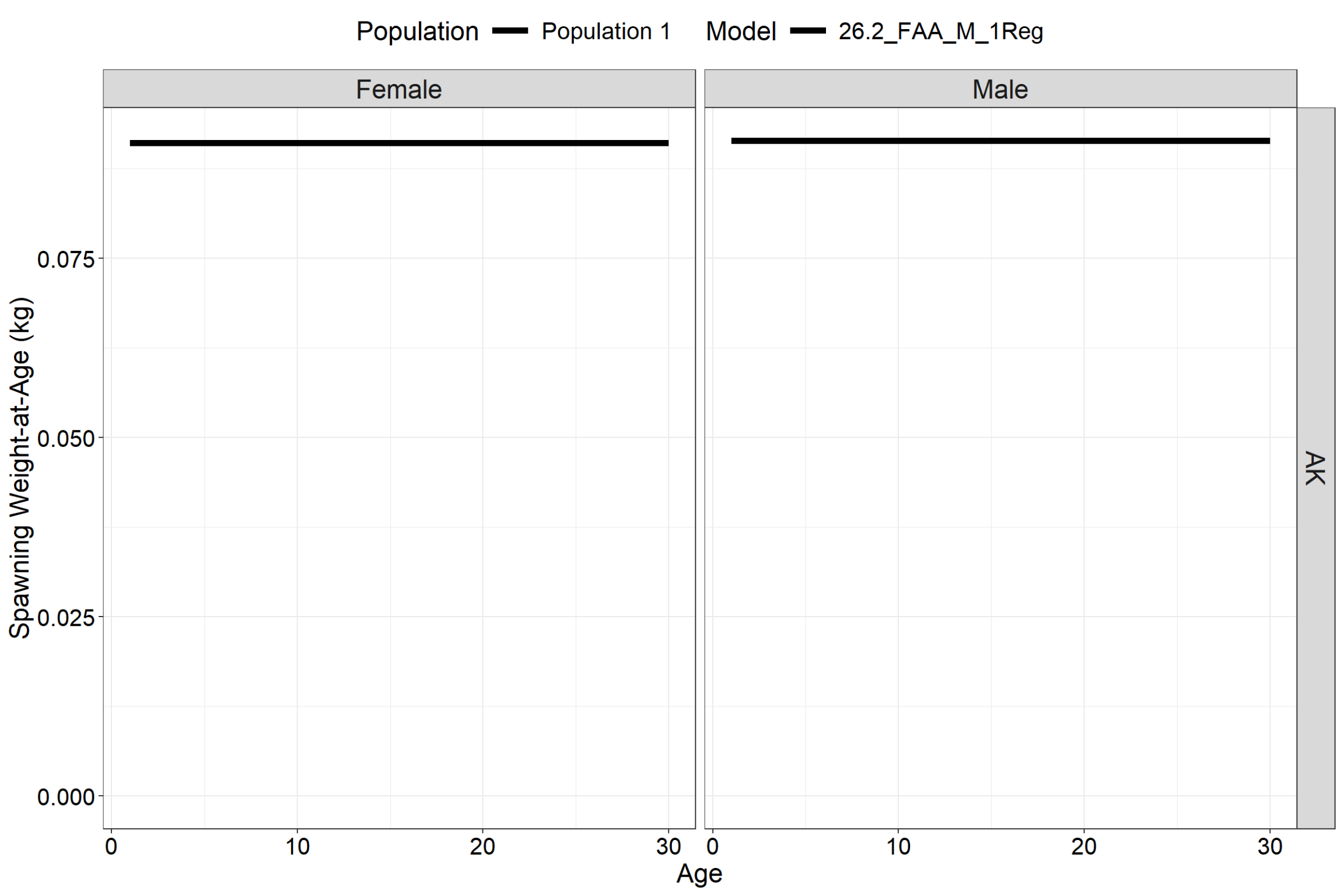
JPN LLS

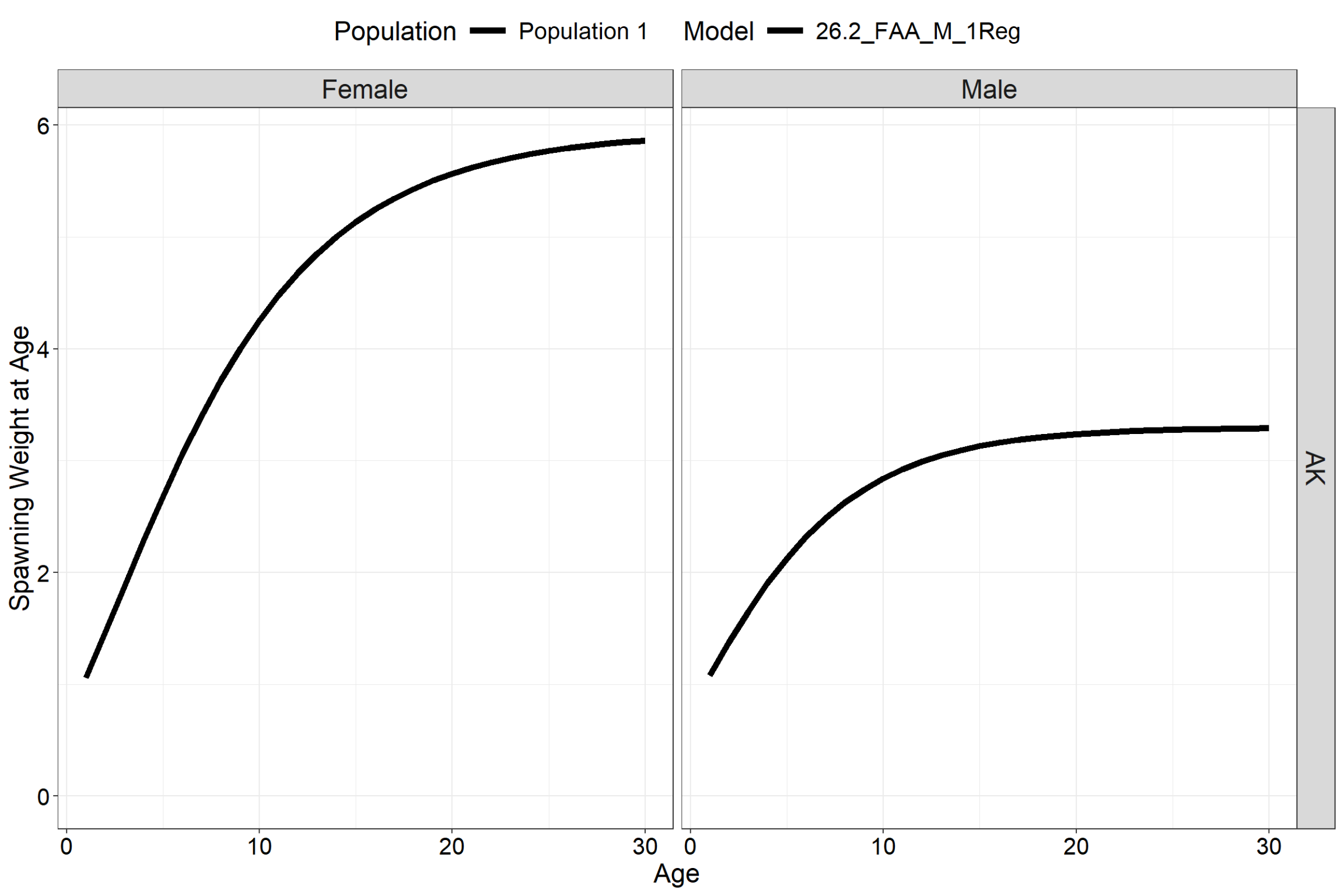
RPNs (1000s)

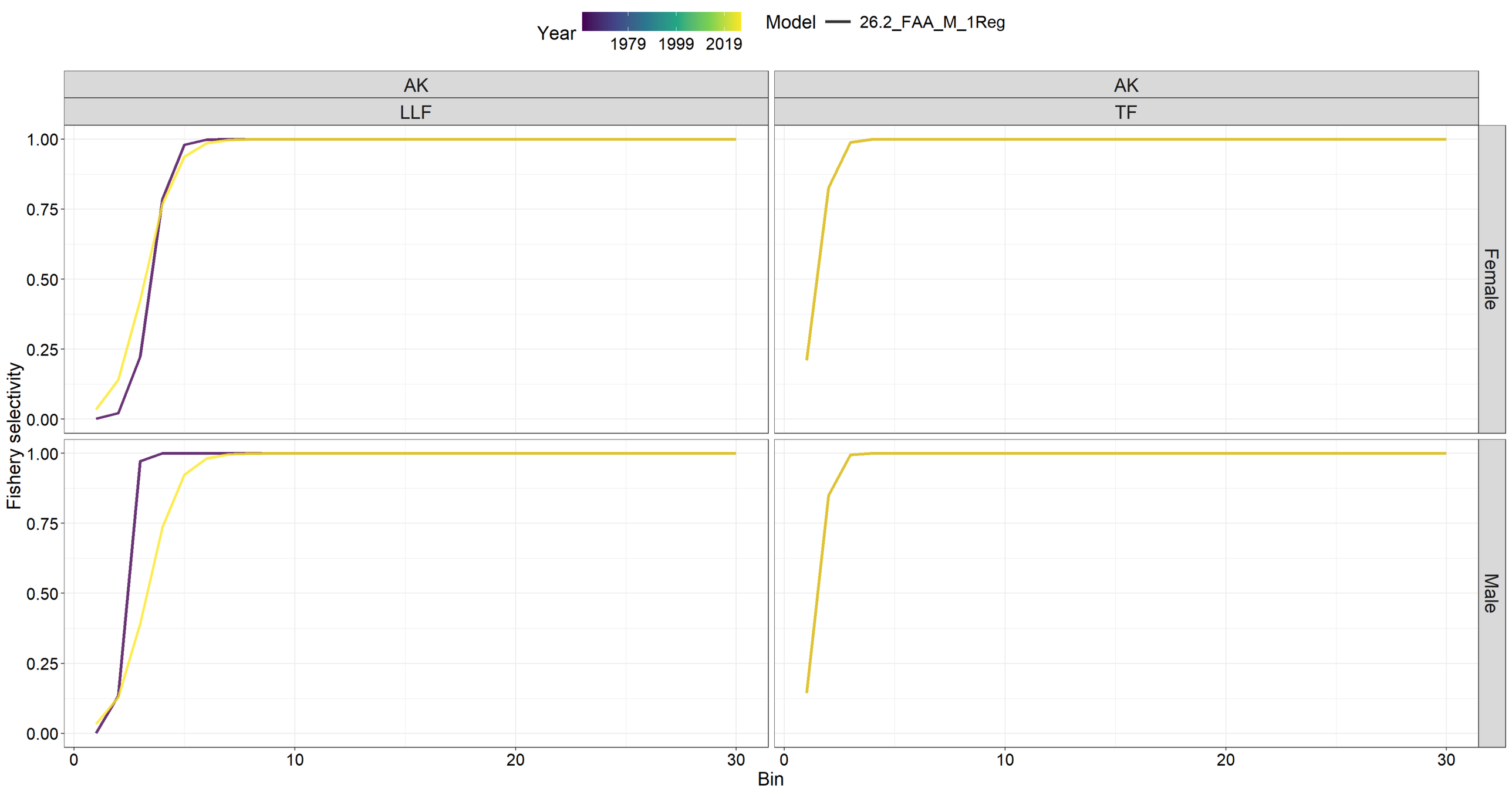
Year

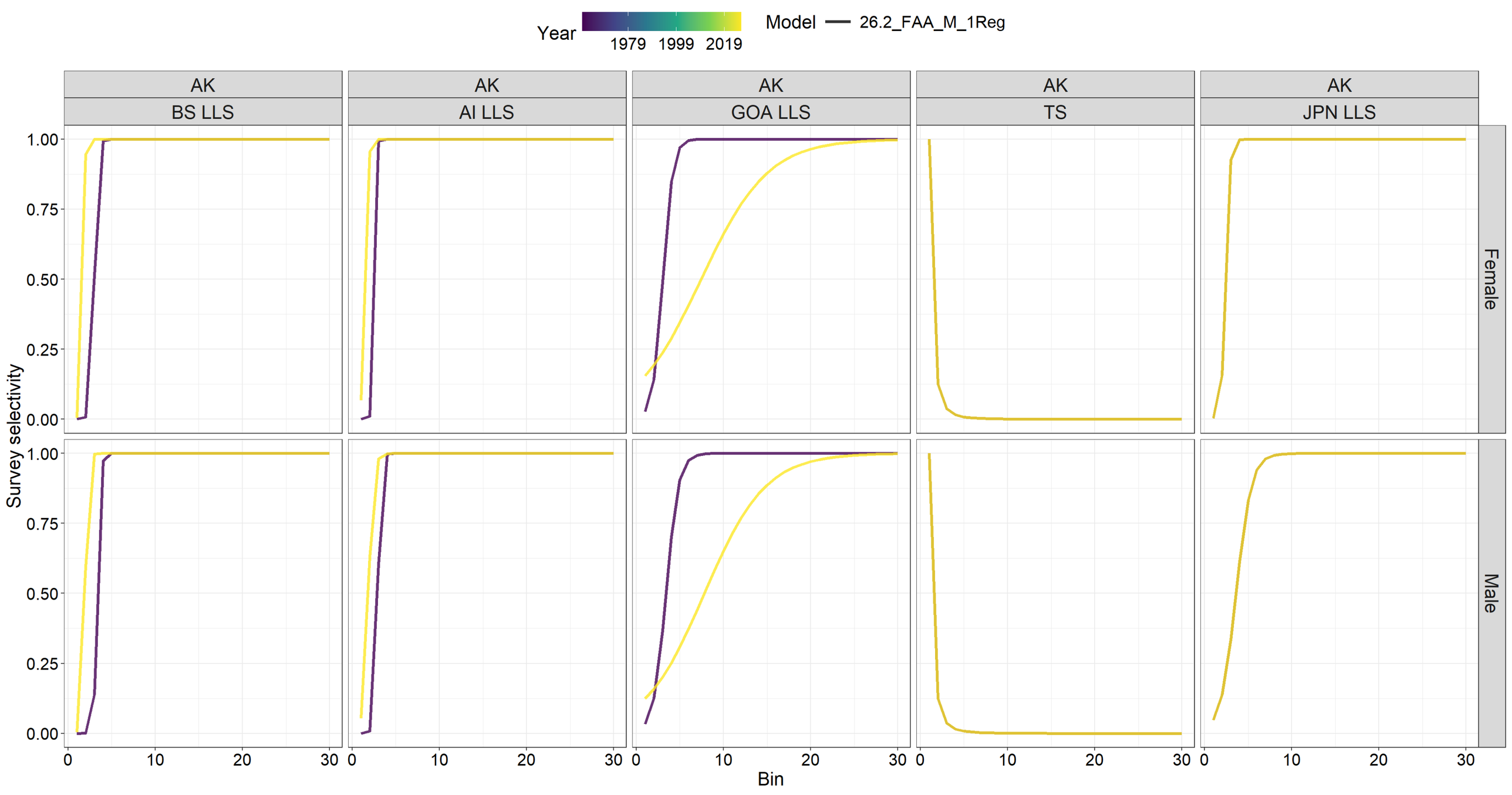

Index Residuals Runs Test
BS LLS
p = 0.092
AI LLS
p = 0.109
GOA LLS
p = 0.032
JPN LLS
p = 0.609
Residuals
Year


Population
Population 1
Model
26.2_FAA_M_1Reg
Female
Male
AK
Maturity at Age
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Age

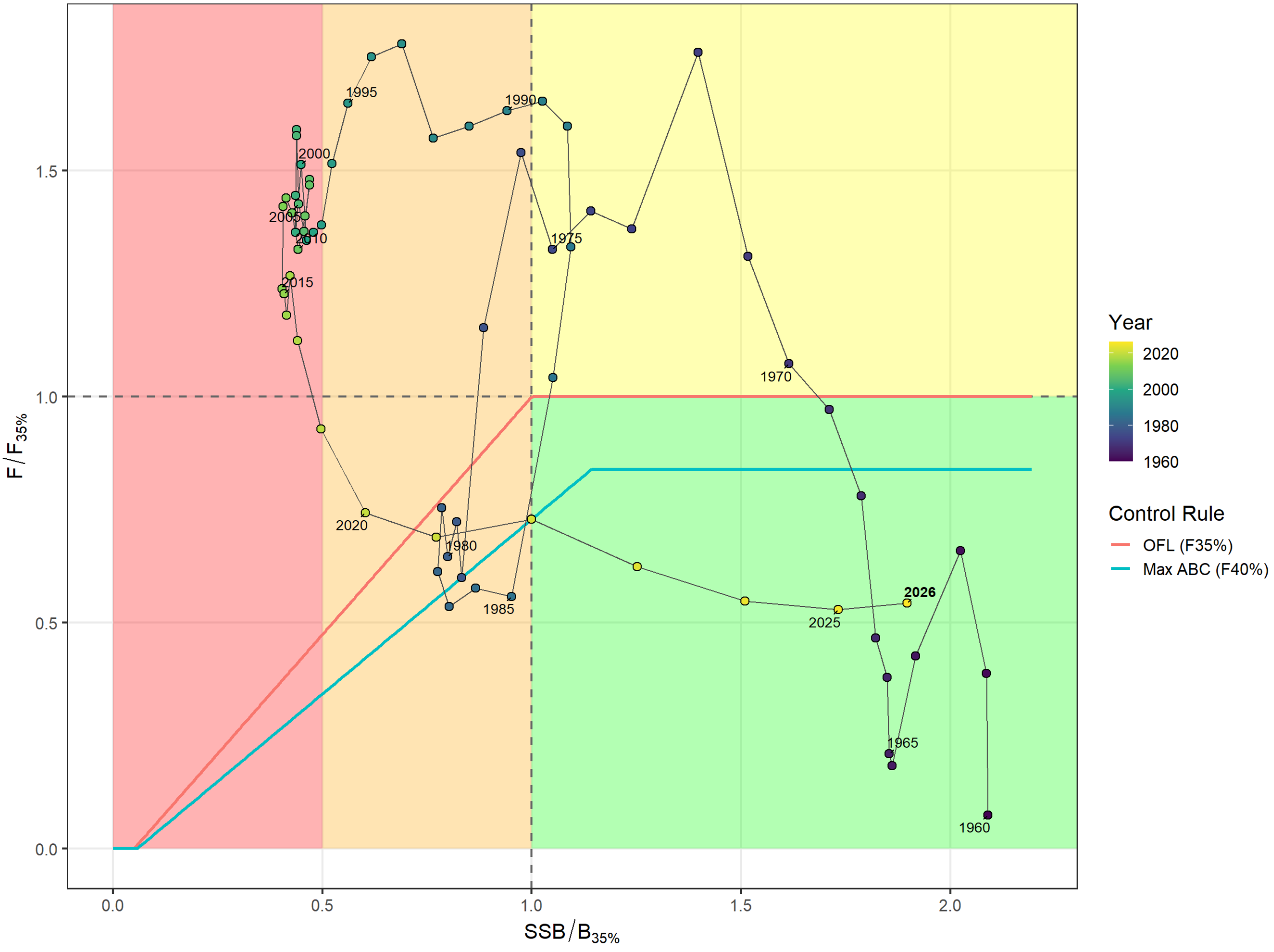

Population
Population 1
Model
26.2_FAA_M_1Reg
Female
Male
AK
Spawning Weight-at-Age (kg)
0.075
0.050
0.025
0.000
0
10
20
30
Age


Population
Population 1
Model
26.2_FAA_M_1Reg
Female
Male
AK
Spawning Weight at Age
Age
0
2
4
6
0
10
20
30

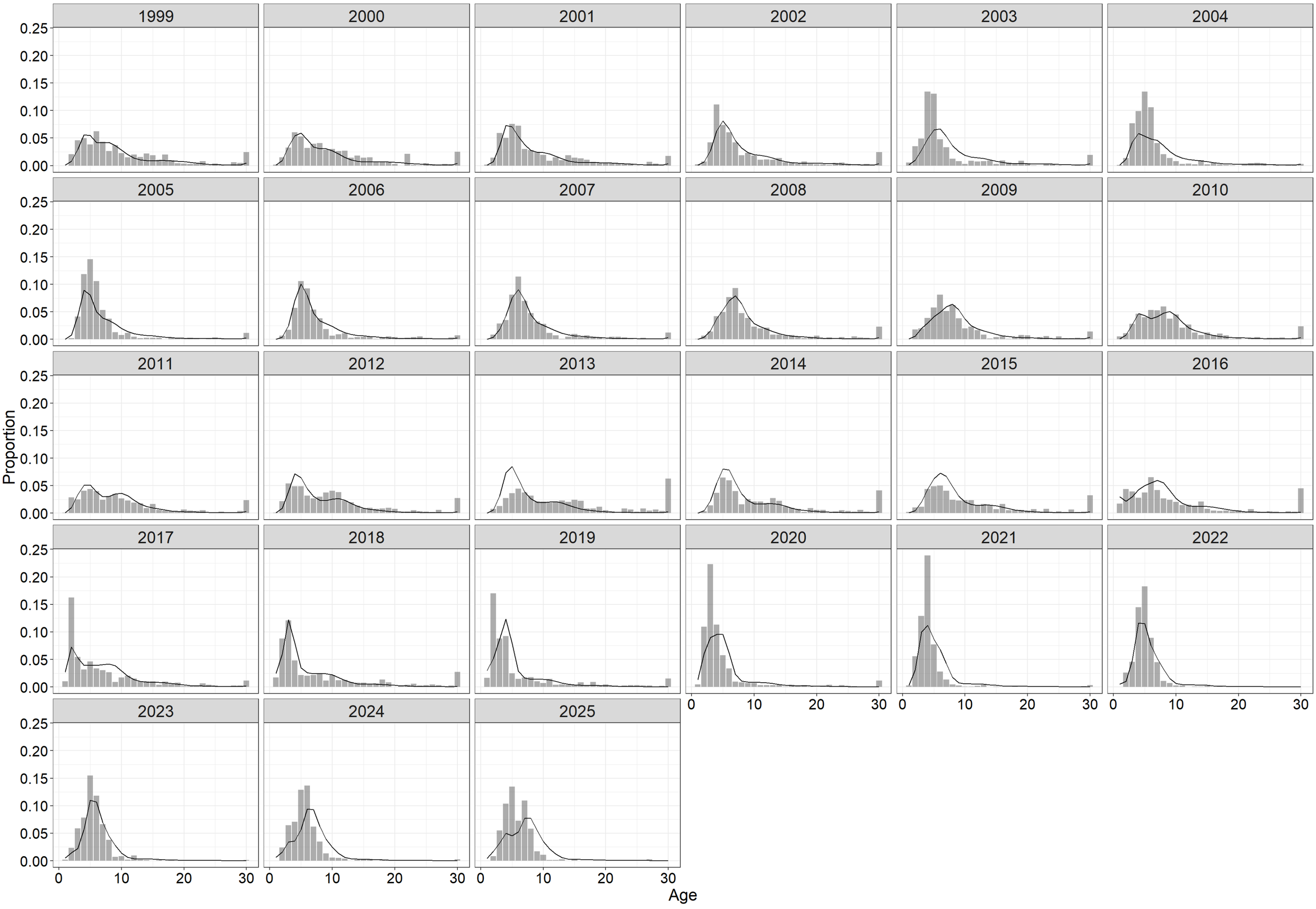

Year
1979
1999
2019
Model
26.2_FAA_M_1Reg
AK
LLF
AK
TF
Female
Male
Fishery selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin


Year
1979
1999
2019
Model
26.2_FAA_M_1Reg
AK
BS LLS
AK
AI LLS
AK
GOA LLS
AK
TS
AK
JPN LLS
Female
Male
Survey selectivity
1.00
0.75
0.50
0.25
0.00
0
10
20
30
Bin

Phase Plane


1995
1990
2000
2005
2010
2015
1975
1970
2020
1980
1985
2025
2026
1965
1960
1.5
1.0
0.5
0.0
F / F35%
0.0
0.5
1.0
1.5
2.0
SSB / B35%
Year
2020
2000
1980
1960
Control Rule
OFL (F35%)
Max ABC (F40%)

Fixed Gear Fishery Age Comps Female AK



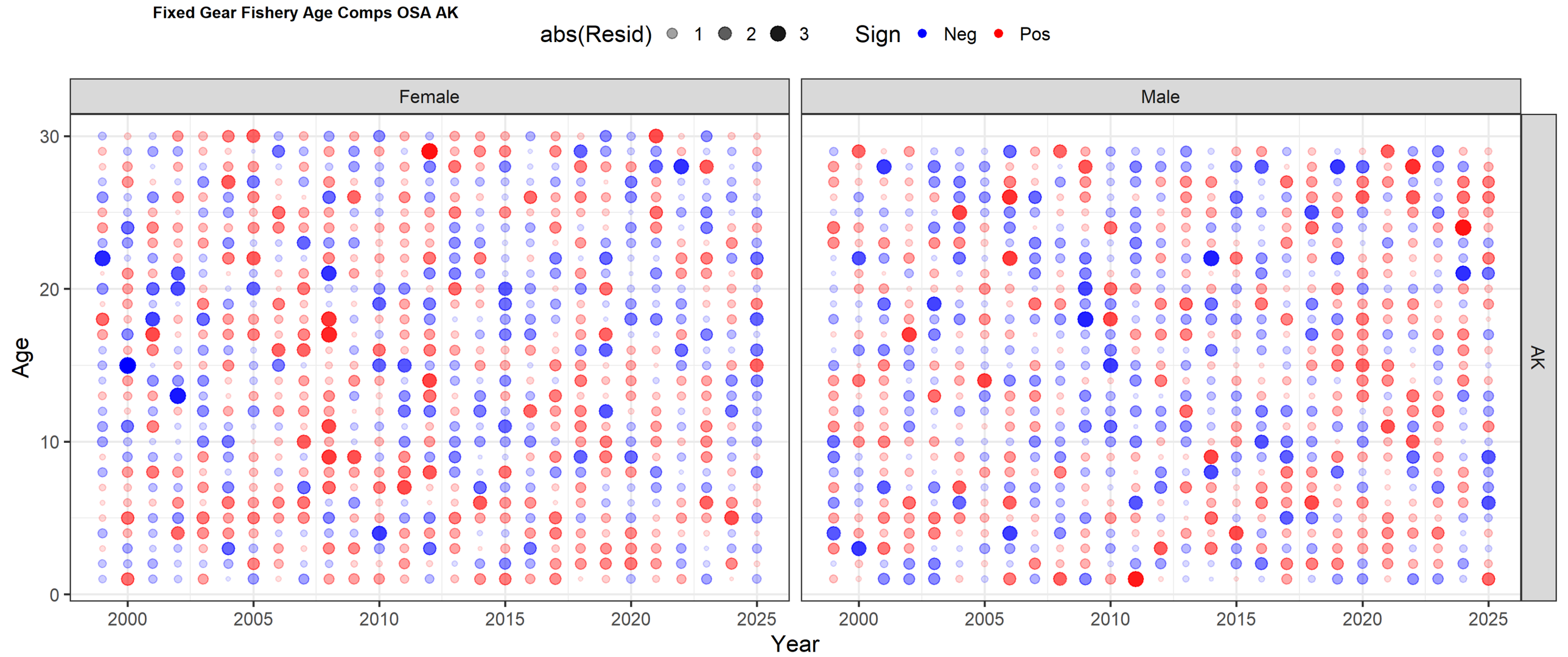
Fixed Gear Fishery Age Comps Male AK

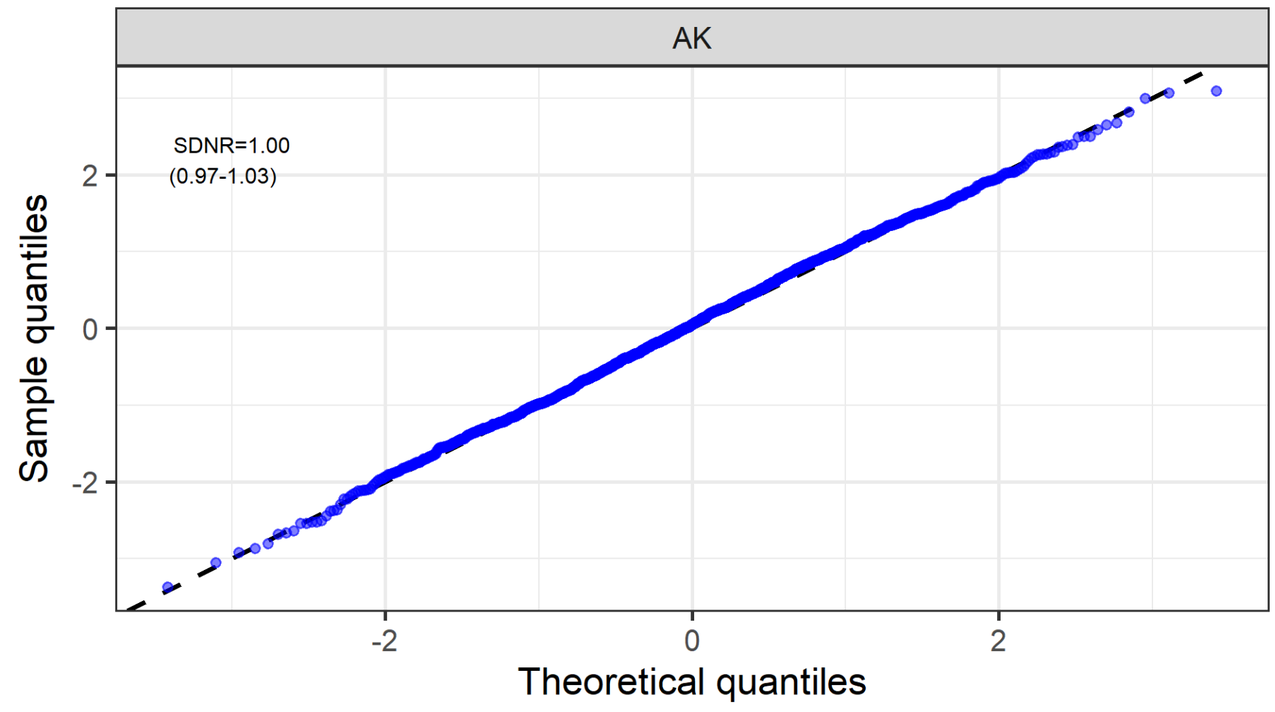
1999 2000 2001 2002 2003 2004
2005 2006 2007 2008 2009 2010
2011 2012 2013 2014 2015 2016
2017 2018 2019 2020 2021 2022
2023 2024 2025

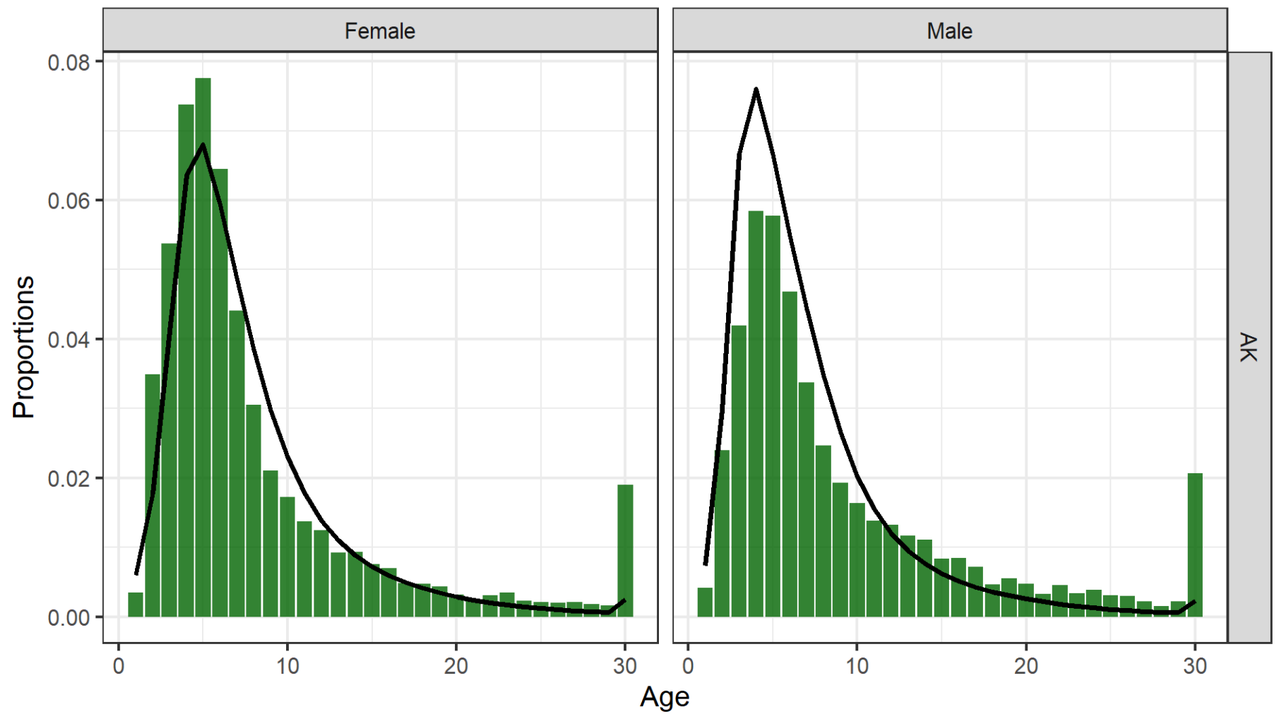
Proportion: 0.00, 0.05, 0.10, 0.15

Age: 0, 10, 20, 30

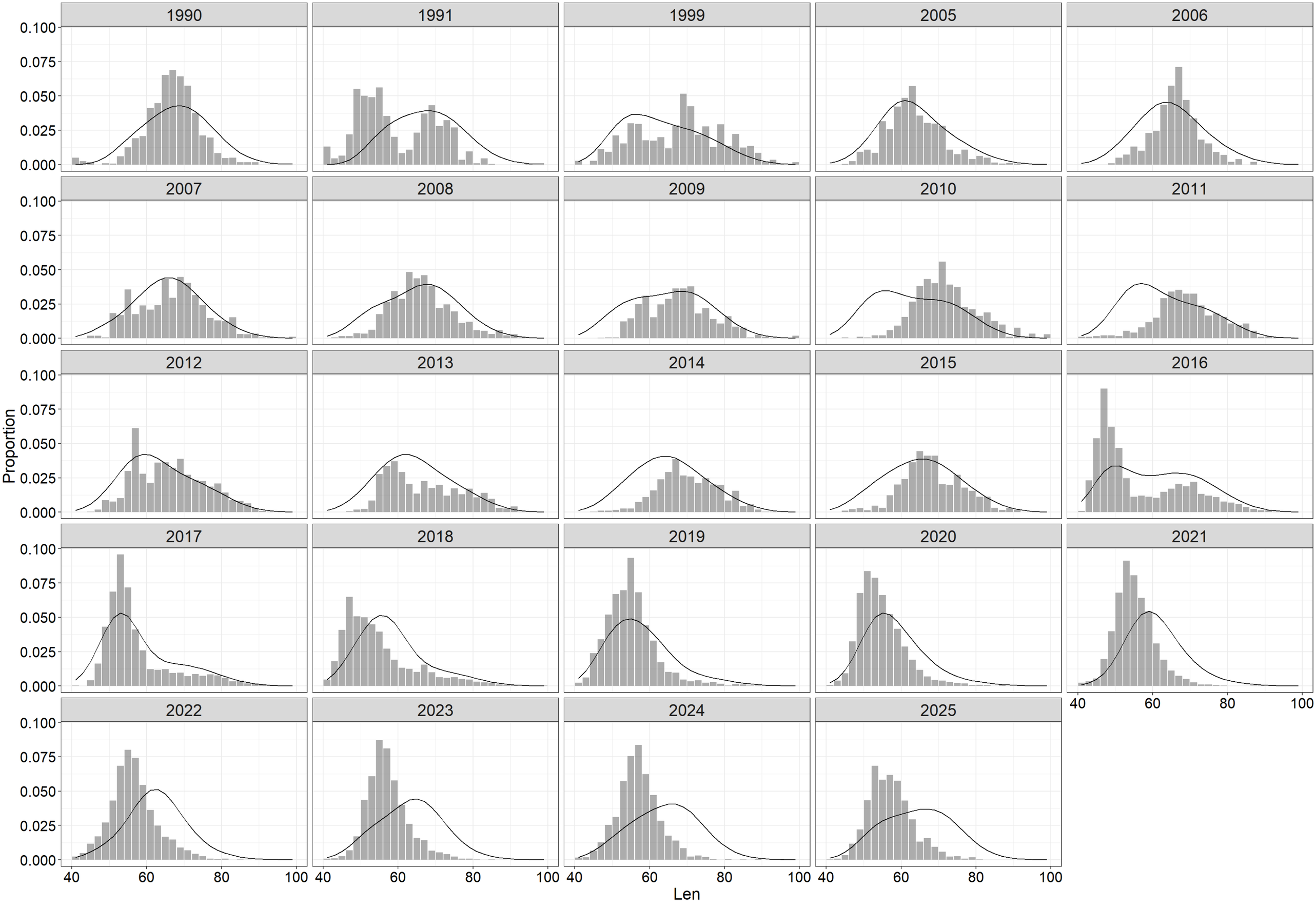
Fixed Gear Fishery Age Comps OSA AK



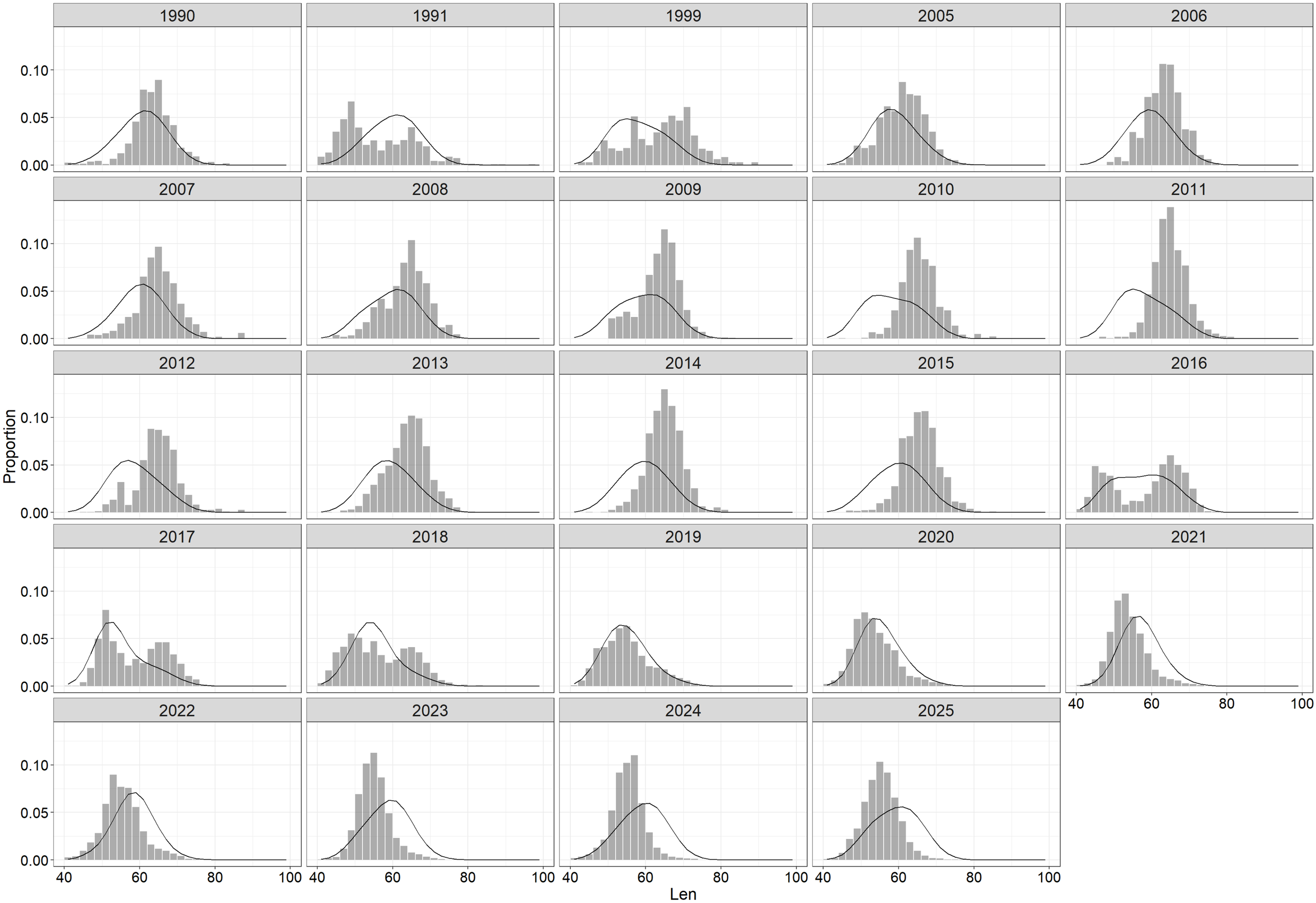




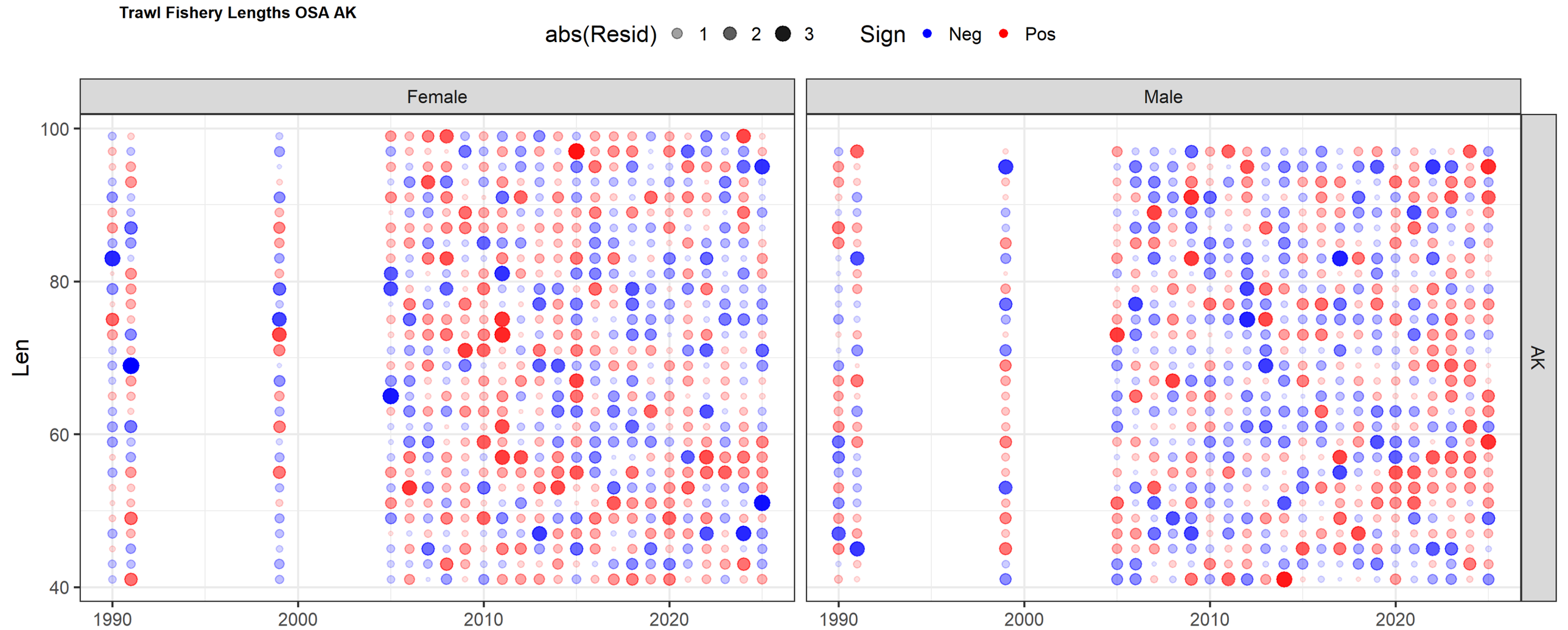
Trawl Fishery Lengths Female AK

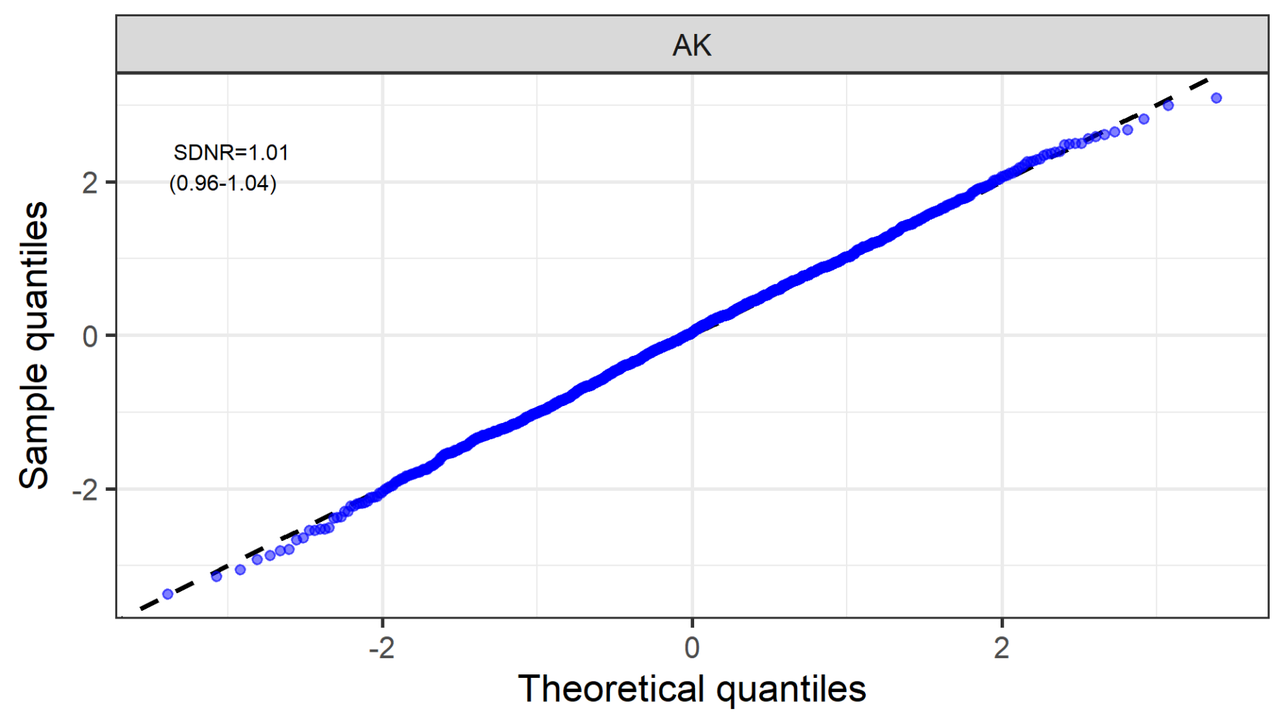


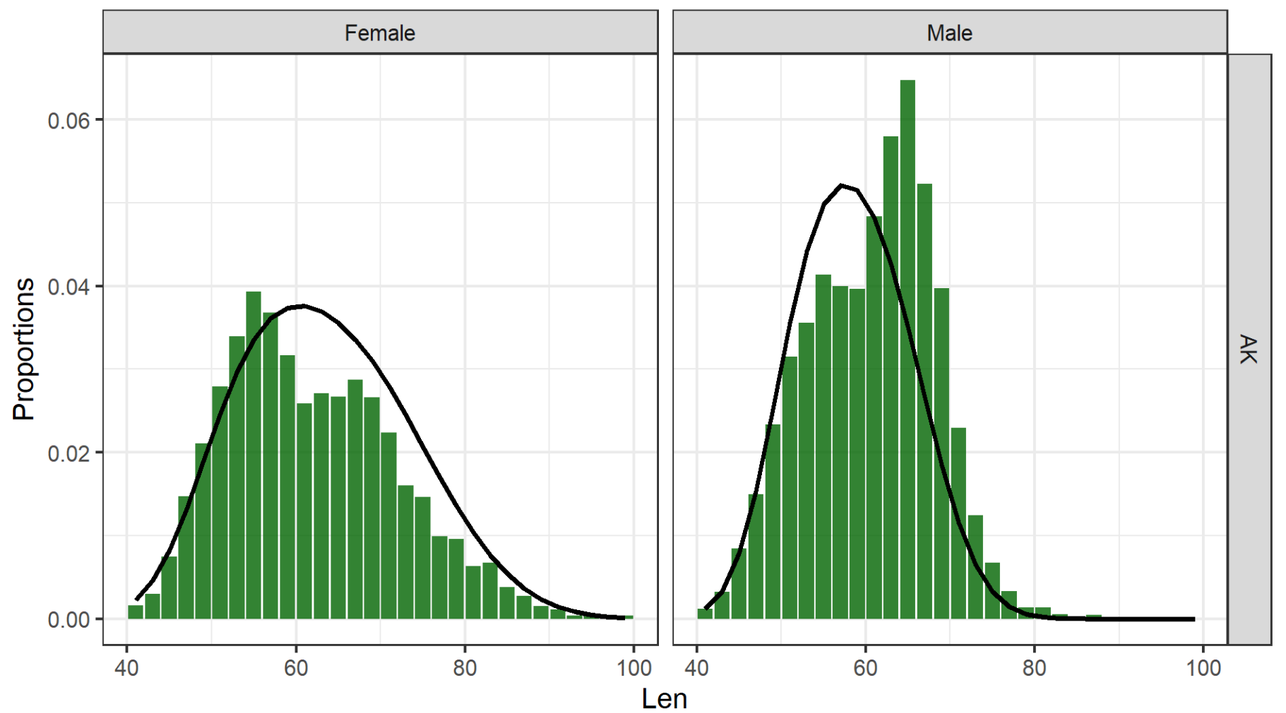
Trawl Fishery Lengths Male AK



Trawl Fishery Lengths OSA AK







BS LL Survey Age Comps Female AK

1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

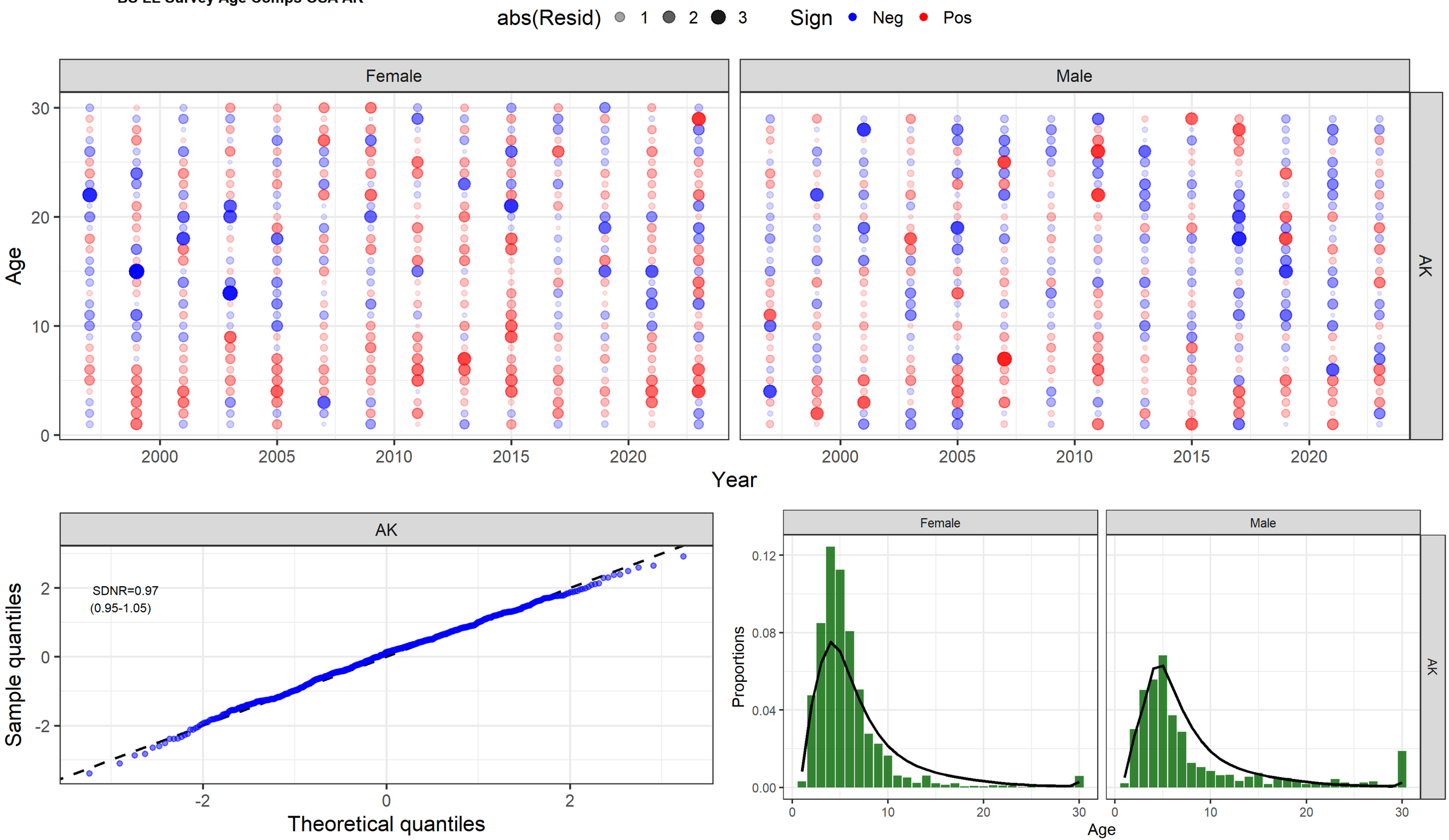
Proportion
Age

BS LL Survey Age Comps Male AK

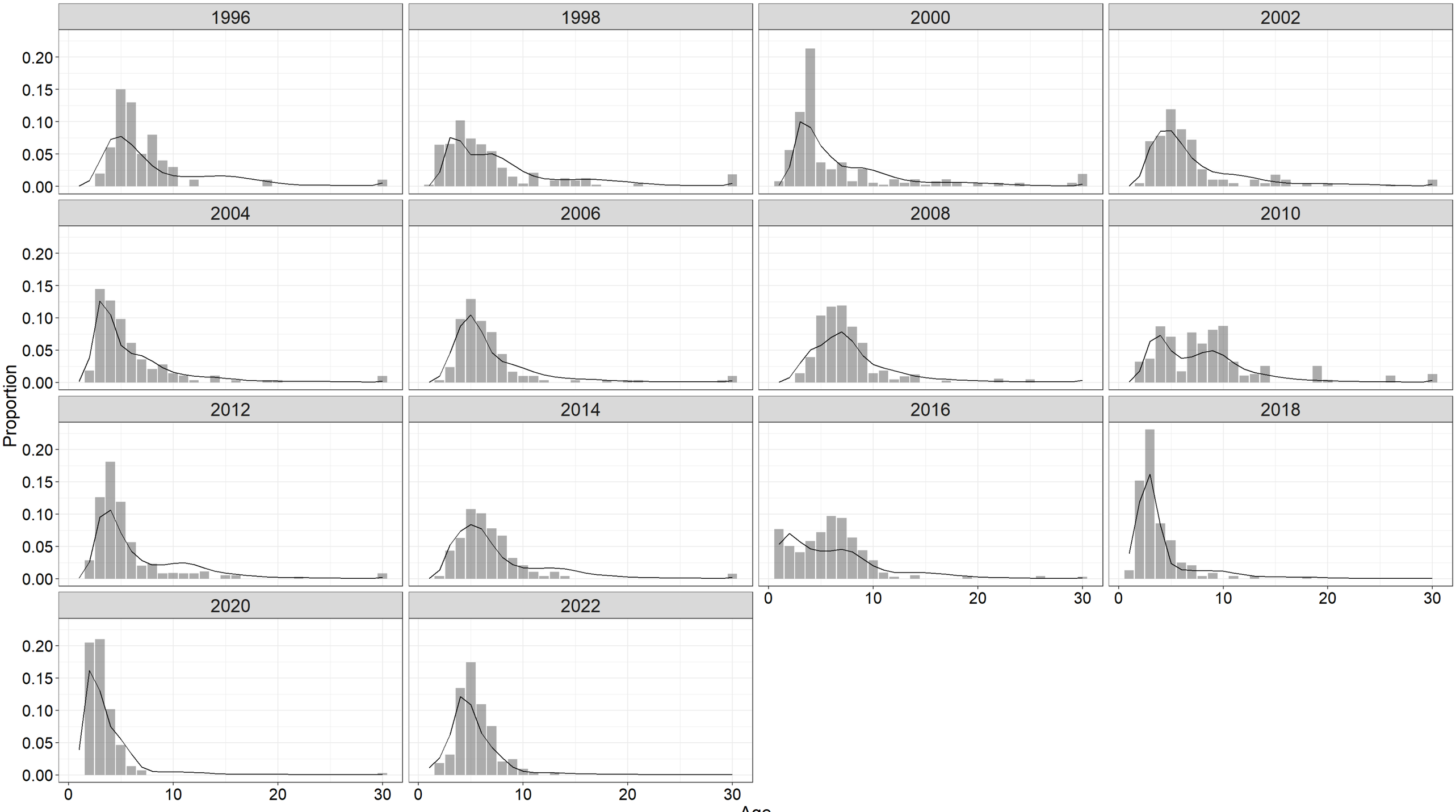
1997 1999 2001 2003
2005 2007 2009 2011
2013 2015 2017 2019
2021 2023

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

BS LL Survey Age Comps OSA AK



AI LL Survey Age Comps Female AK



AI LL Survey Age Comps Male AK

1996 1998 2000 2002
2004 2006 2008 2010
2012 2014 2016 2018
2020 2022

Proportion
0.00 0.05 0.10 0.15
0 10 20 30
Age

AI LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.96
(0.95-1.05)

Sample quantiles

Theoretical quantiles

Female

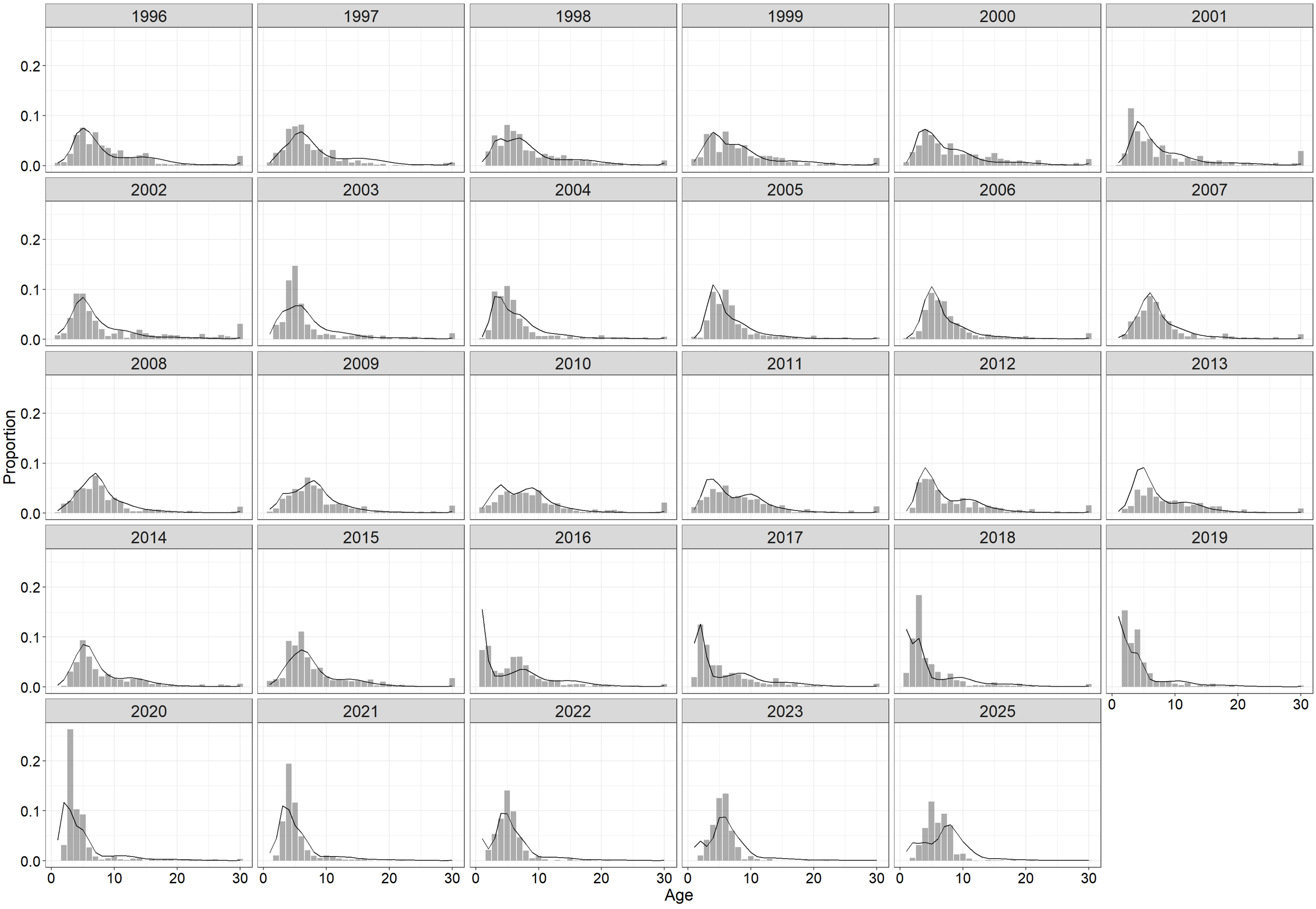
Male

AK

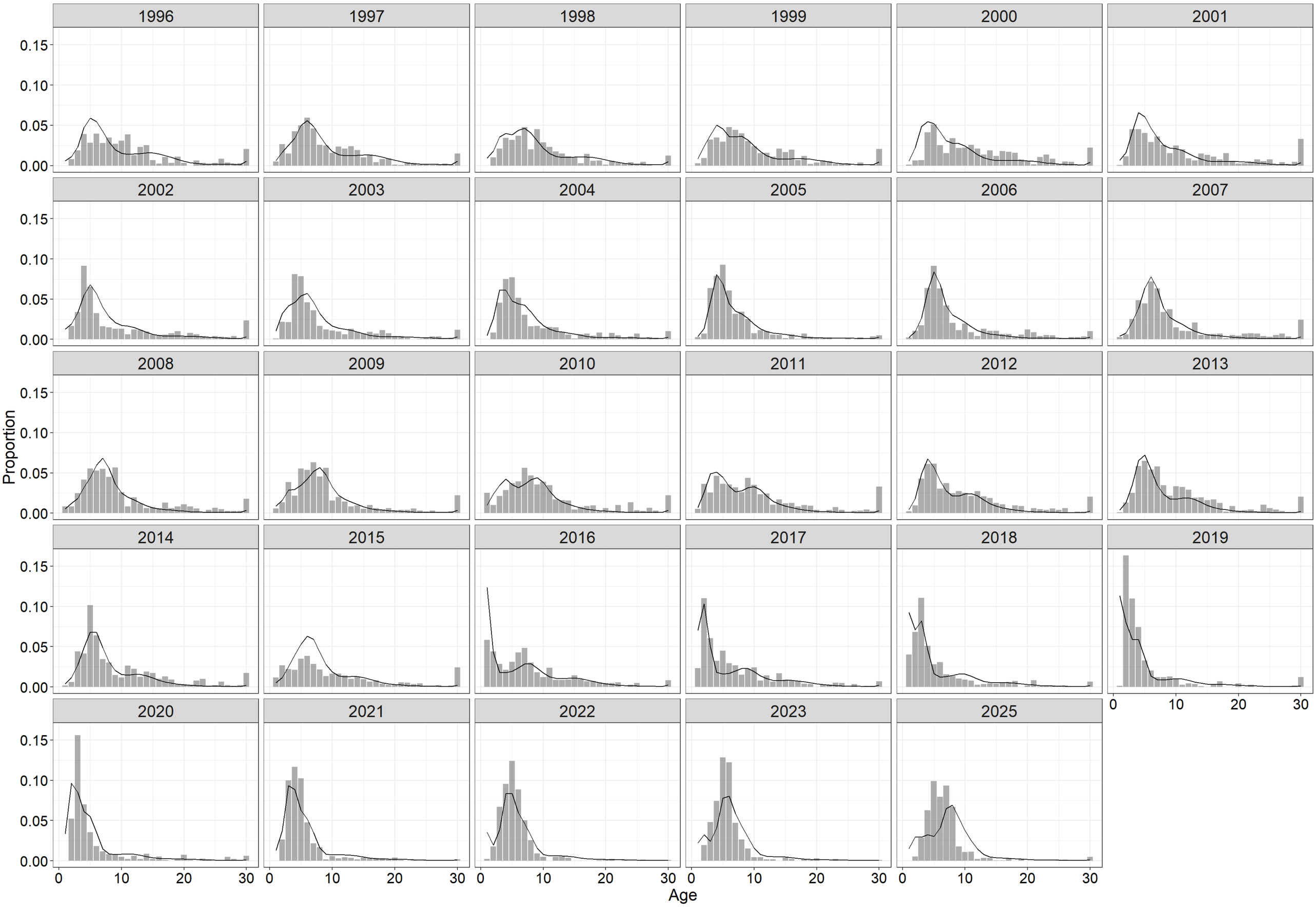
Proportions

Age

GOA LL Survey Age Comps Female AK



GOA LL Survey Age Comps Male AK



GOA LL Survey Age Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female

Male

AK

Age

Year

AK

SDNR=0.92
(0.97-1.03)

Sample quantiles

Theoretical quantiles

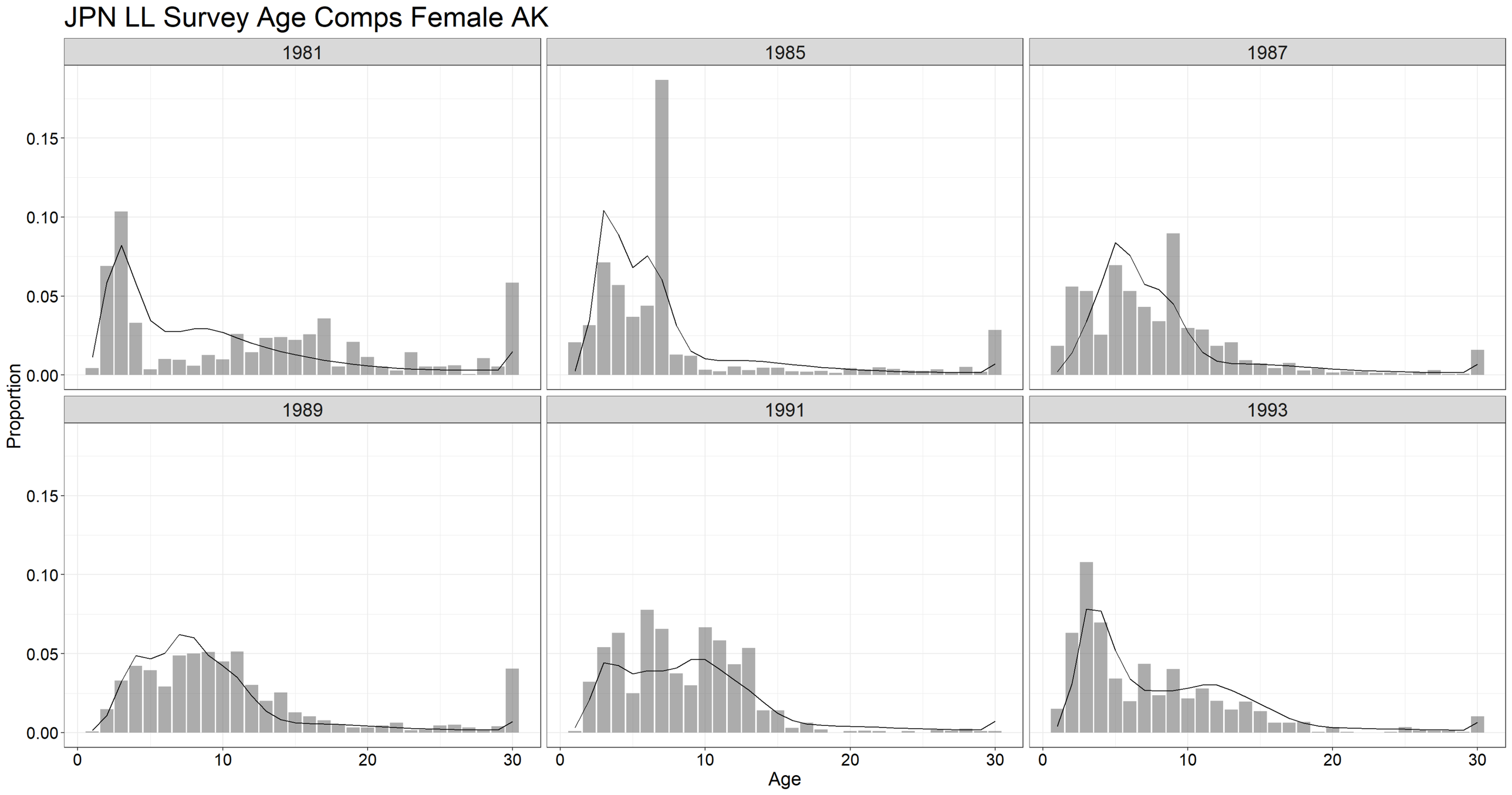
Female

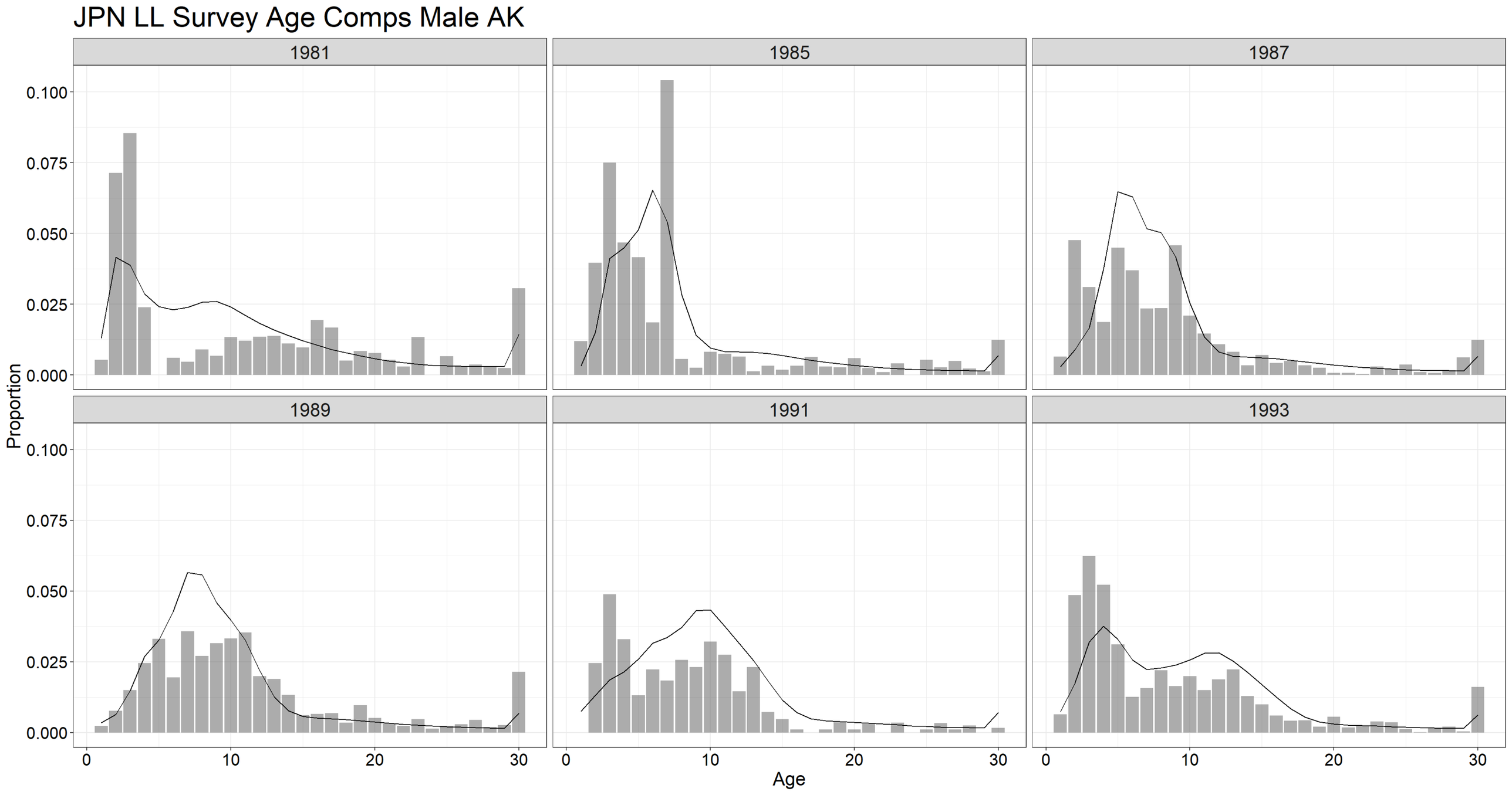
Male

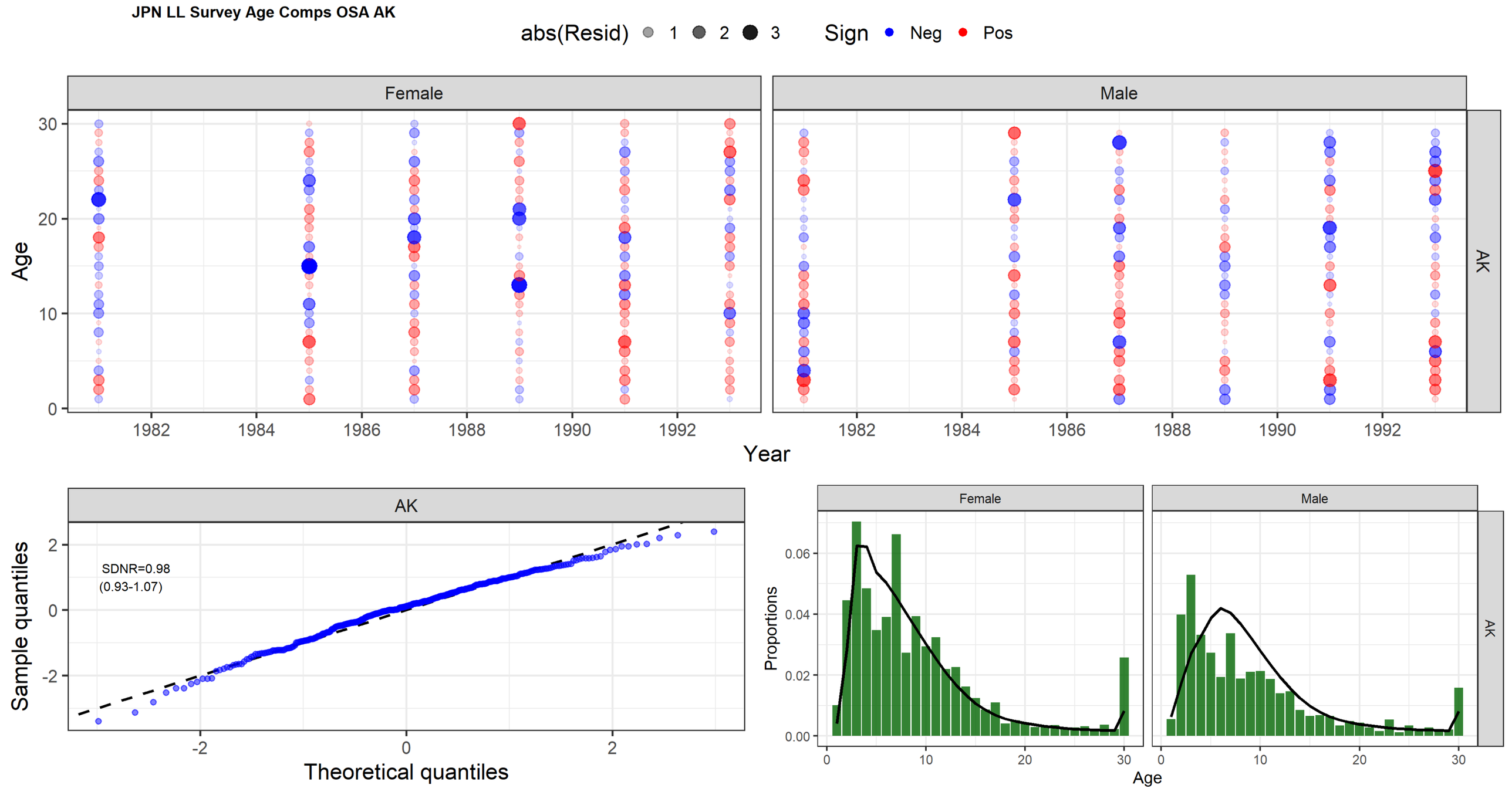
AK

Proportions

Age


JPN LL Survey Age Comps Female AK
1981
1985
1987
1989
1991
1993
Proportion
0.00
0.05
0.10
0.15
0
10
20
30
Age


JPN LL Survey Age Comps Male AK
1981
1985
1987
1989
1991
1993
0.100
0.075
0.050
0.025
0.000
Proportion
0
10
20
30
Age


JPN LL Survey Age Comps OSA AK
abs(Resid) 1 2 3
Sign Neg Pos
Female
Male
AK
Age
Year
SDNR=0.98
(0.93-1.07)
Sample quantiles
Theoretical quantiles
Proportions

JPN LL Survey Length Comps Female AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps Male AK

1979 1980 1981 1982
1983 1984 1985 1986
1987 1988 1989 1990
1991 1992 1993 1994

Proportion
0.00 0.02 0.04 0.06 0.08
40 60 80 100
Len

JPN LL Survey Length Comps OSA AK

abs(Resid) 1 2 3 Sign Neg Pos

Female | Male | AK

Len: 100, 80, 60, 40

1980 1985 1990 1980 1985 1990

Year

AK

SDNR=0.90
(0.95-1.05)

Sample quantiles: 2, 0, -2

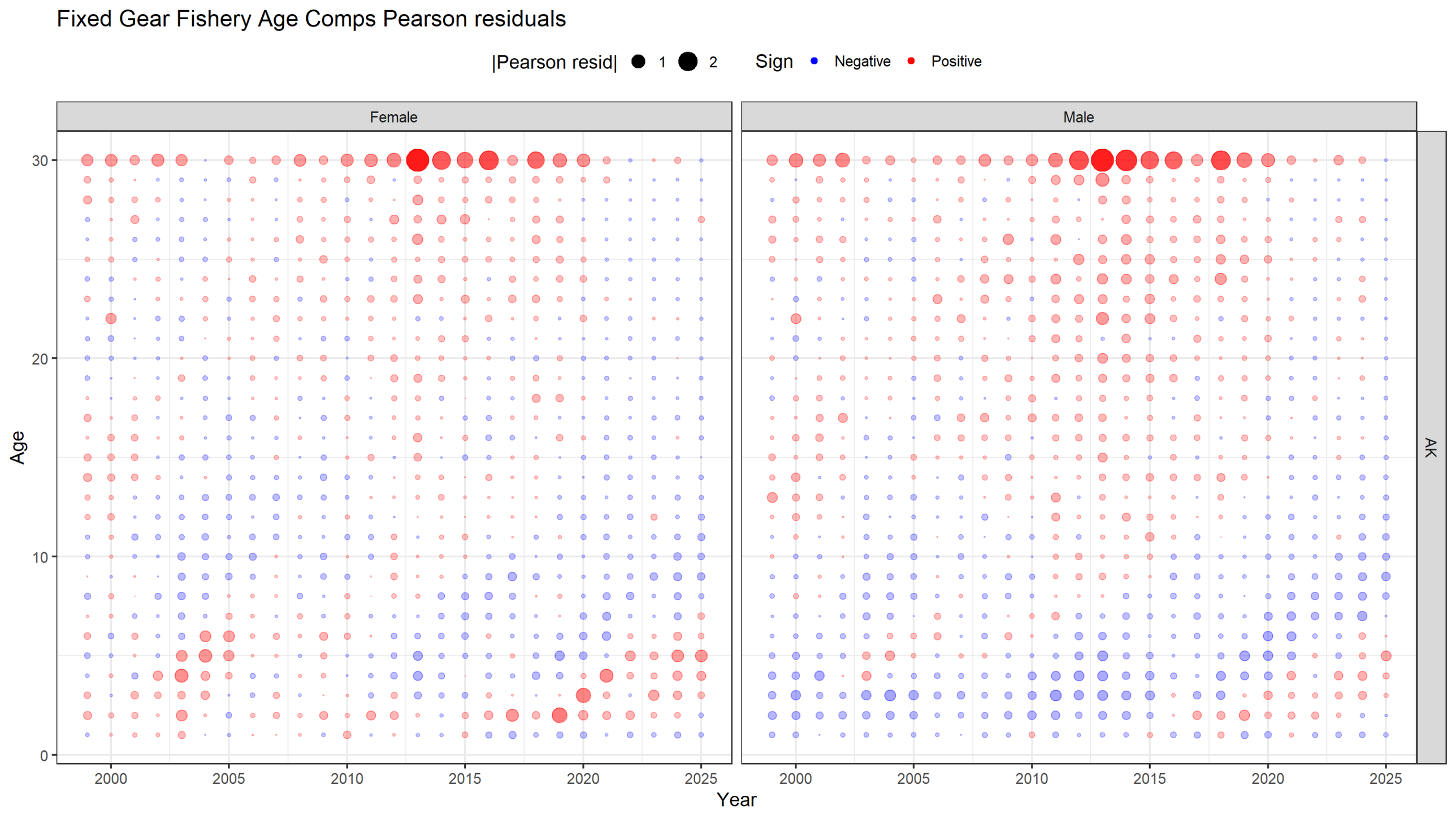
Theoretical quantiles: -2, 0, 2

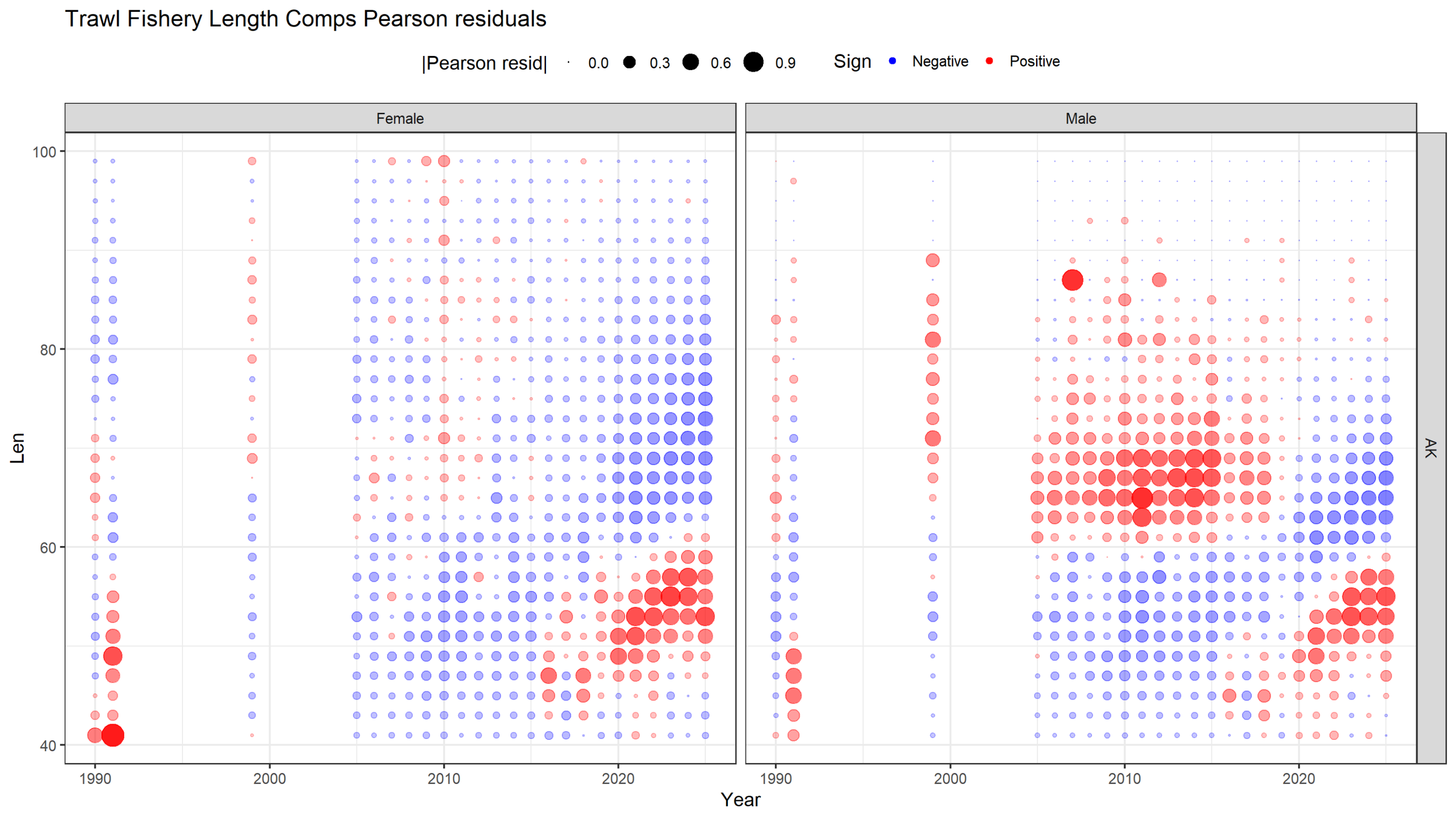
Female | Male | AK

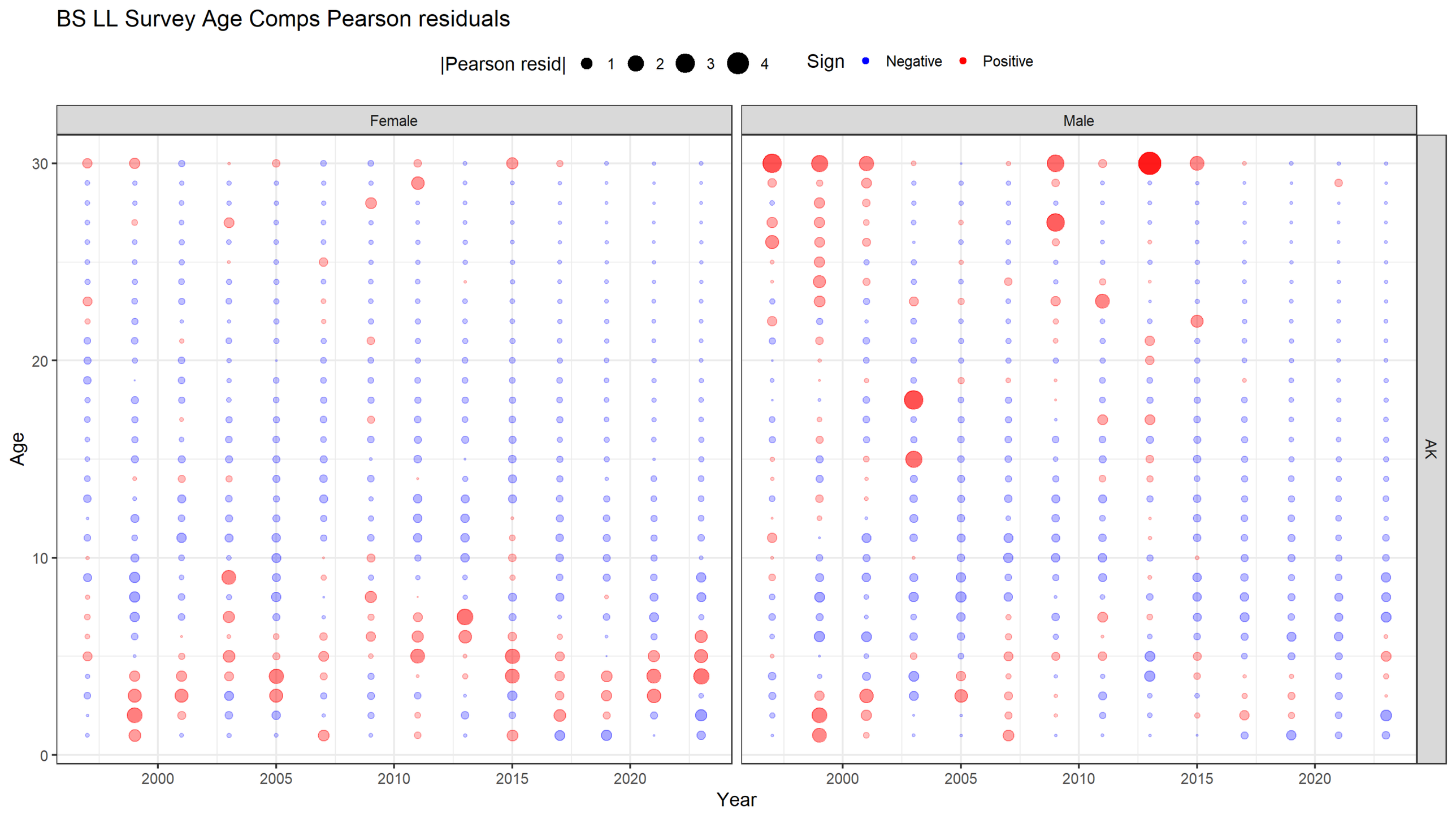
Proportions: 0.06, 0.04, 0.02, 0.00

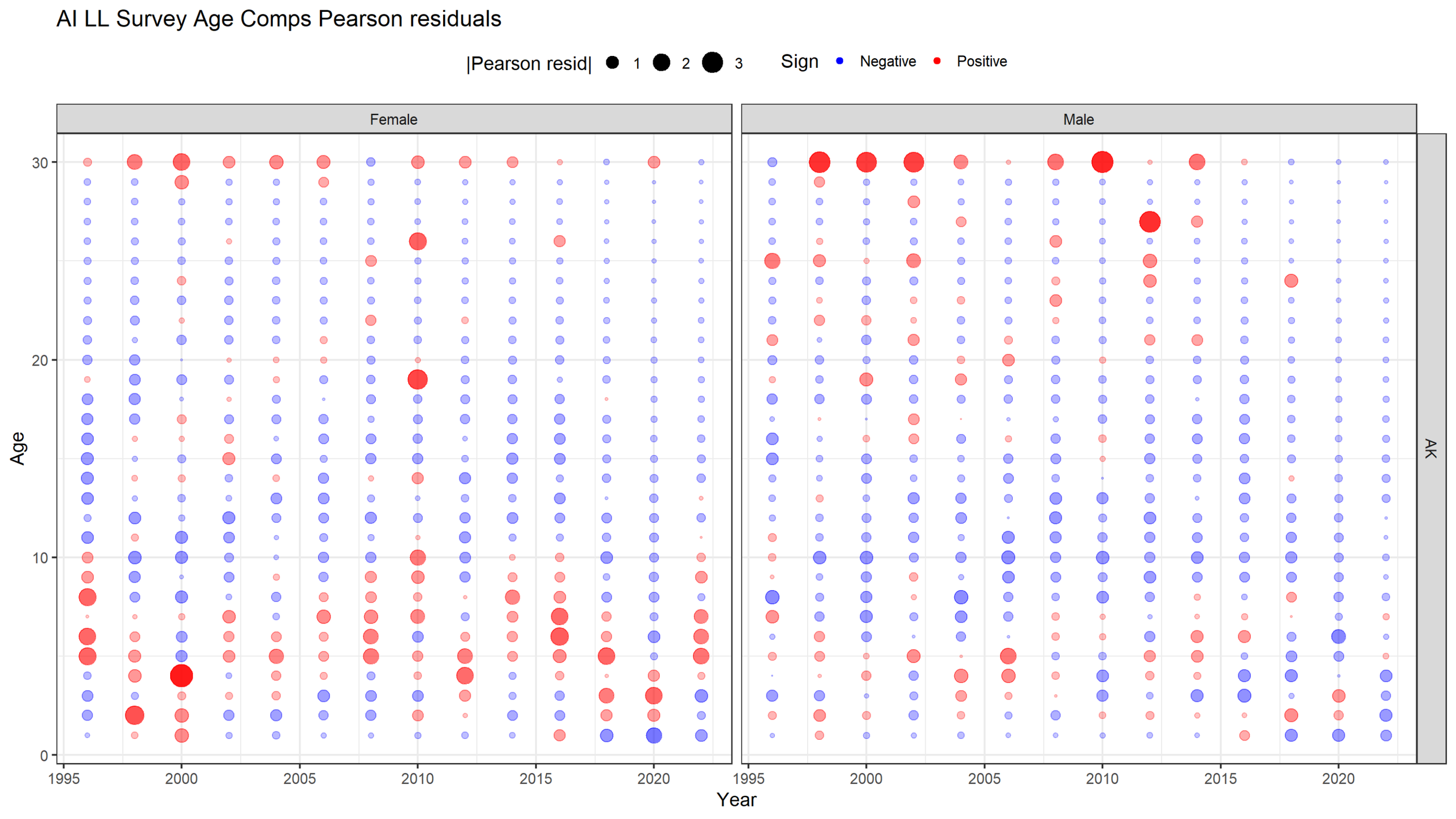
40 60 80 100 40 60 80 100

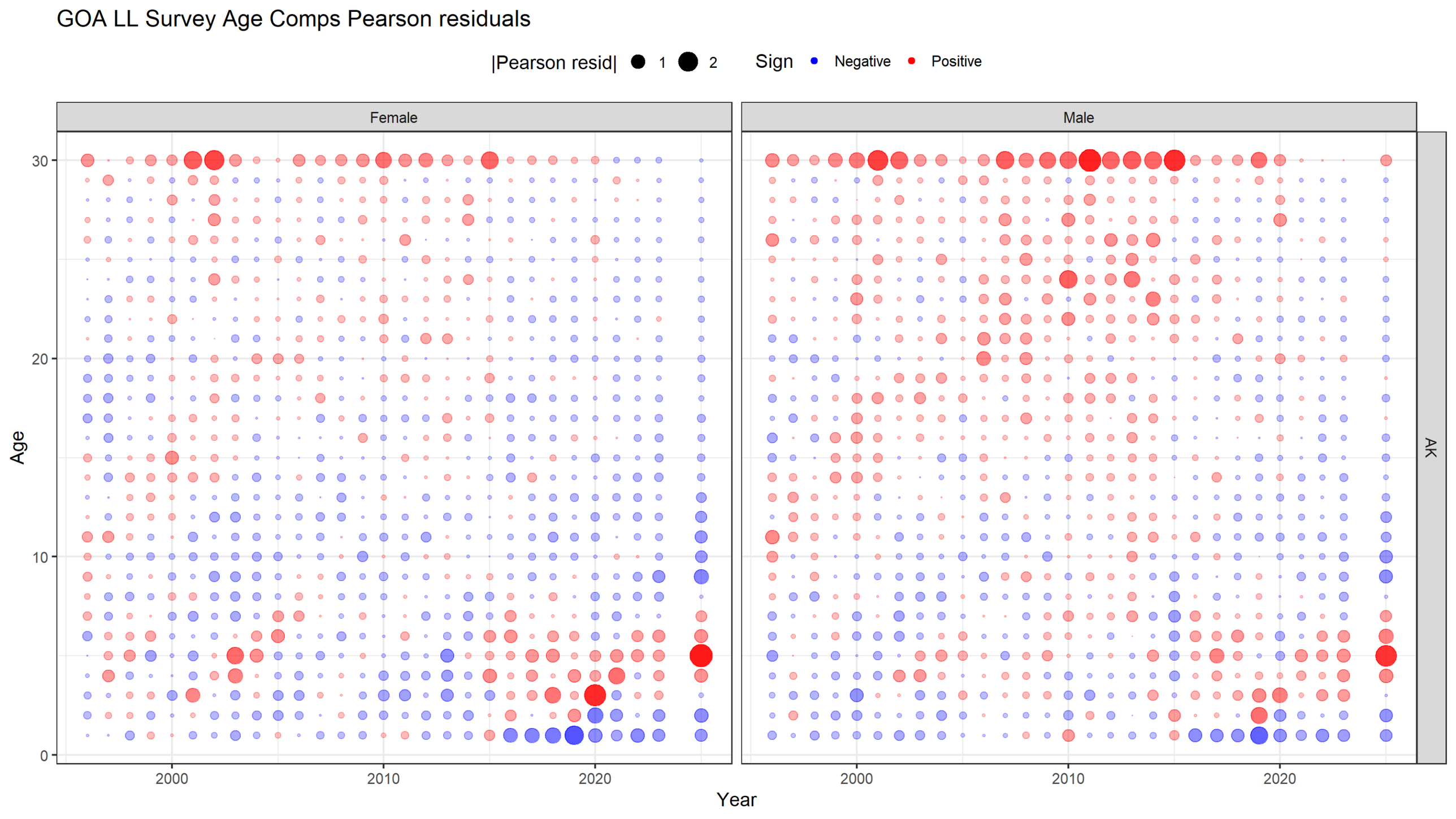
Len

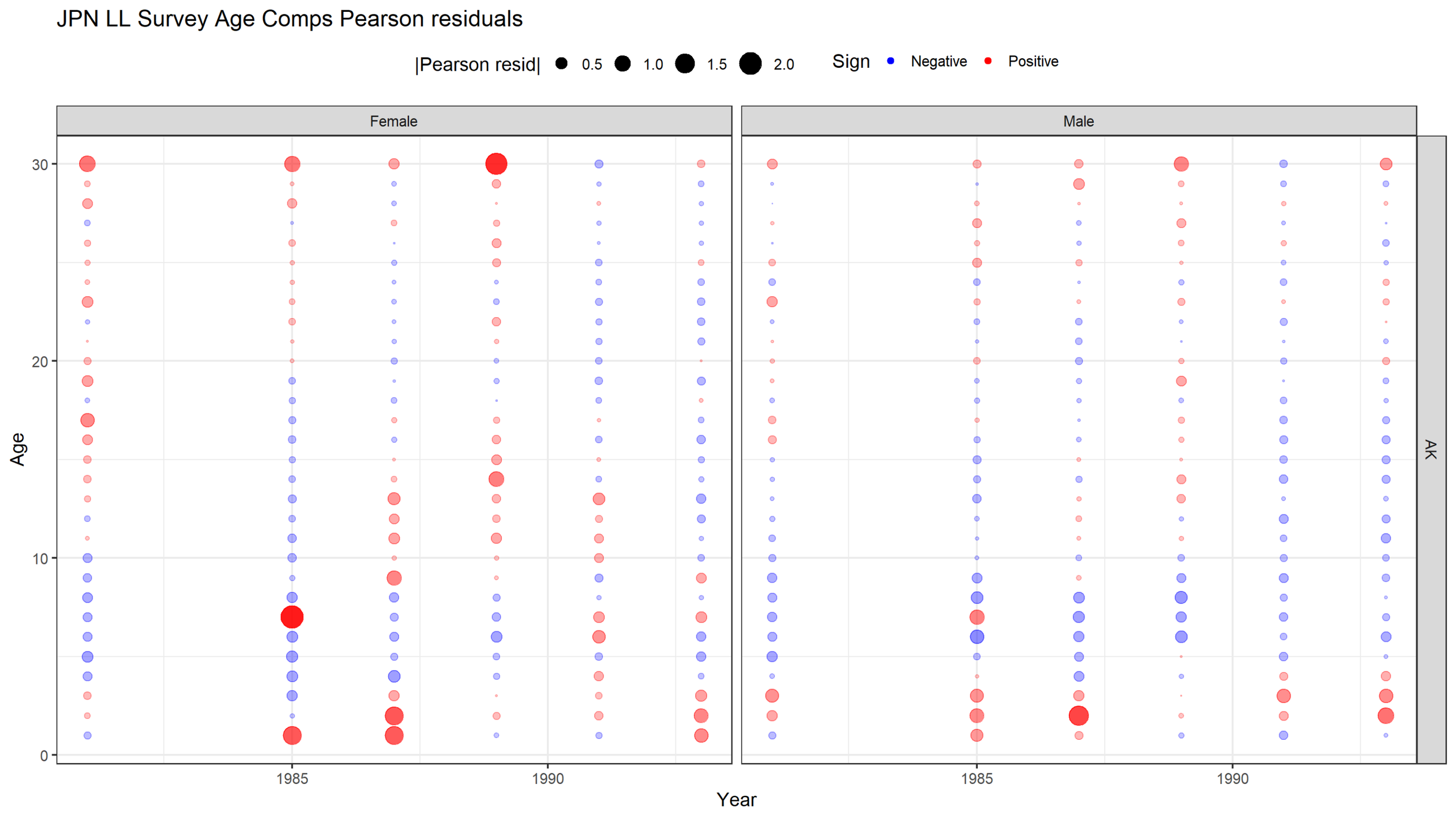

Fixed Gear Fishery Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020
2025

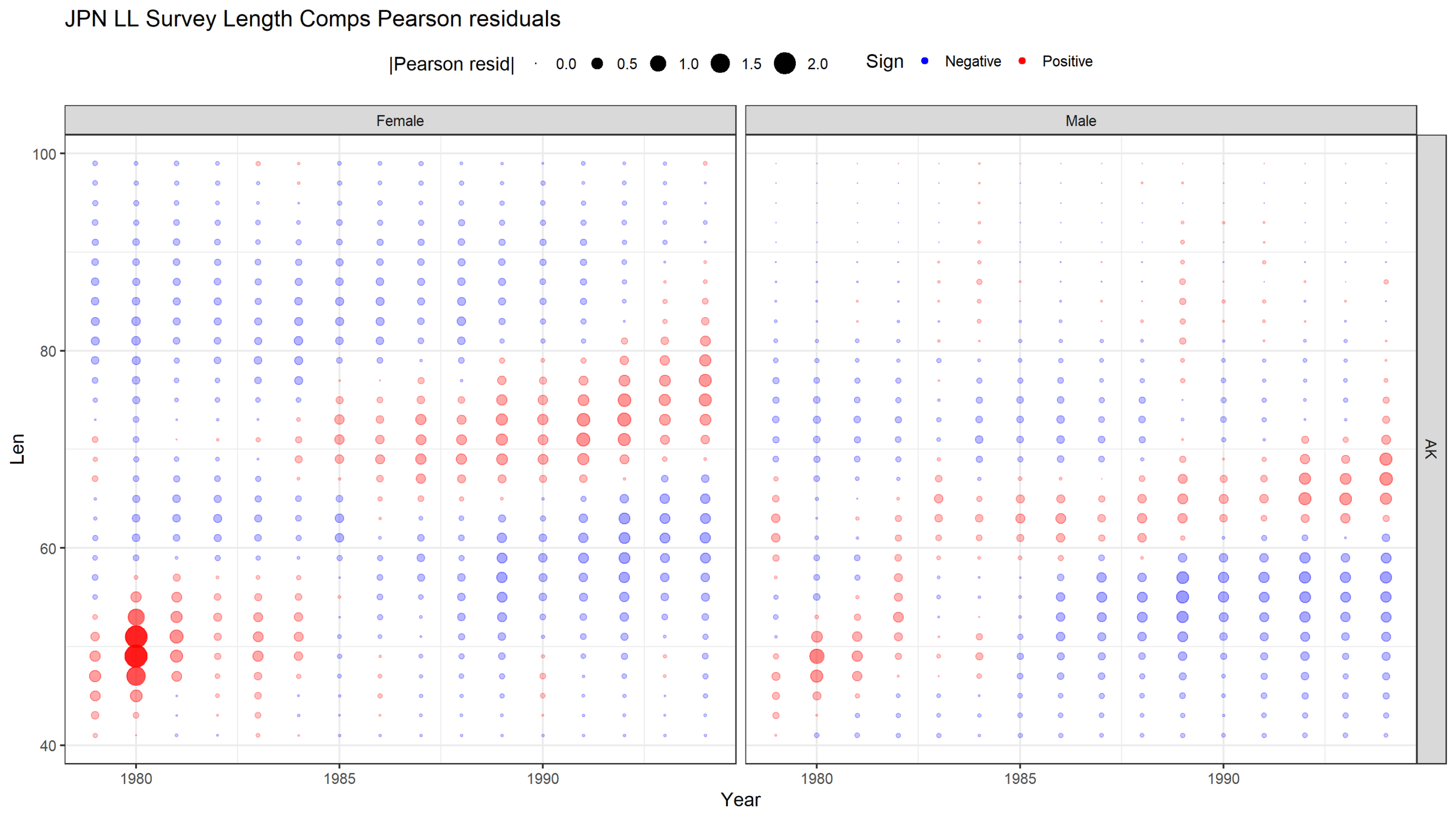

Trawl Fishery Length Comps Pearson residuals
|Pearson resid|
0.0
0.3
0.6
0.9
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1990
2000
2010
2020
Year

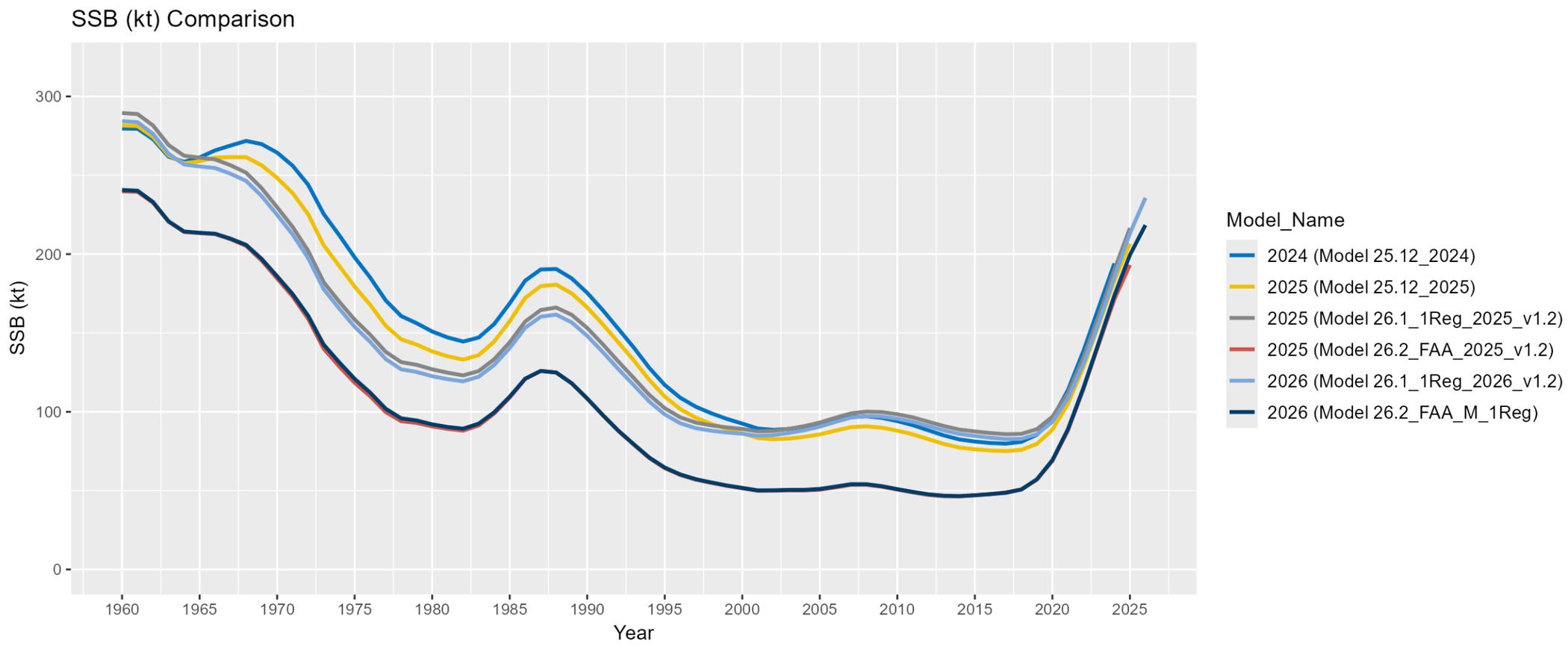

BS LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3 4
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2005
2010
2015
2020

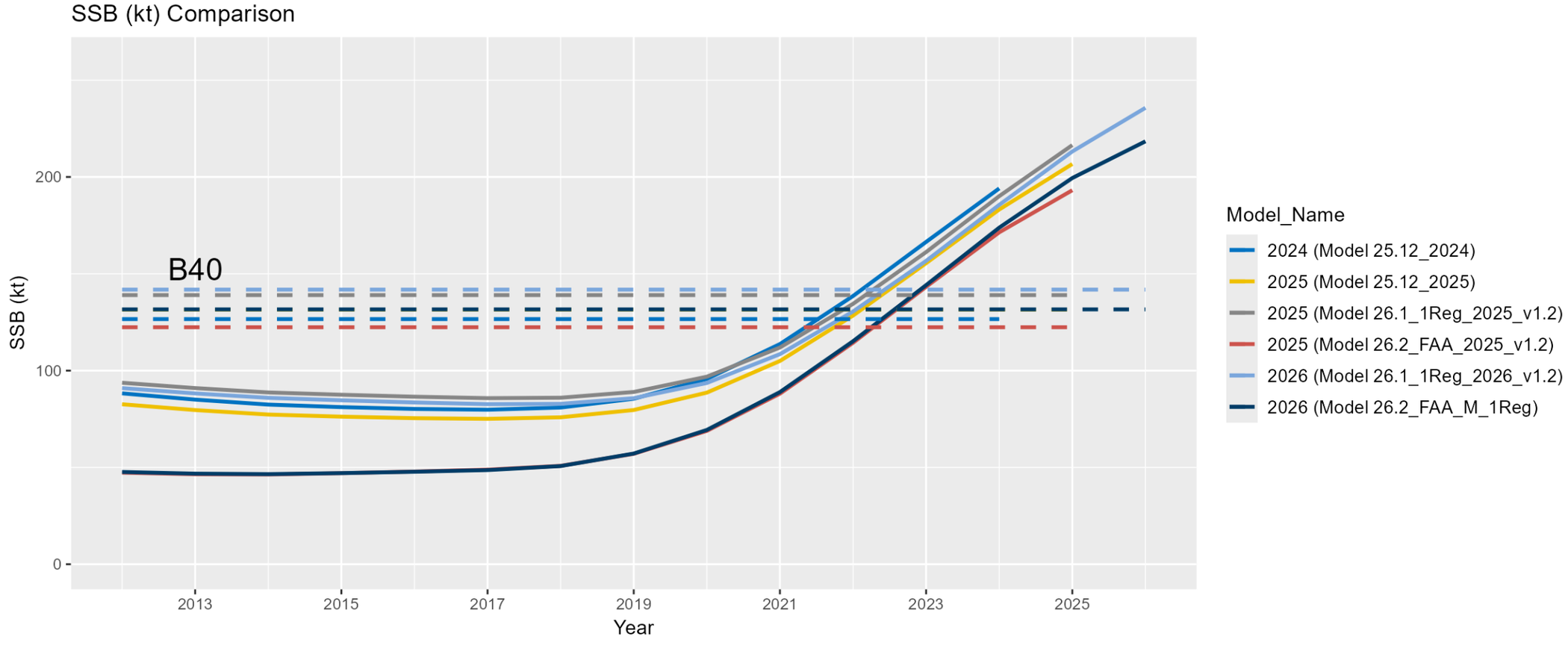

AI LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2 3
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
1995
2000
2005
2010
2015
2020

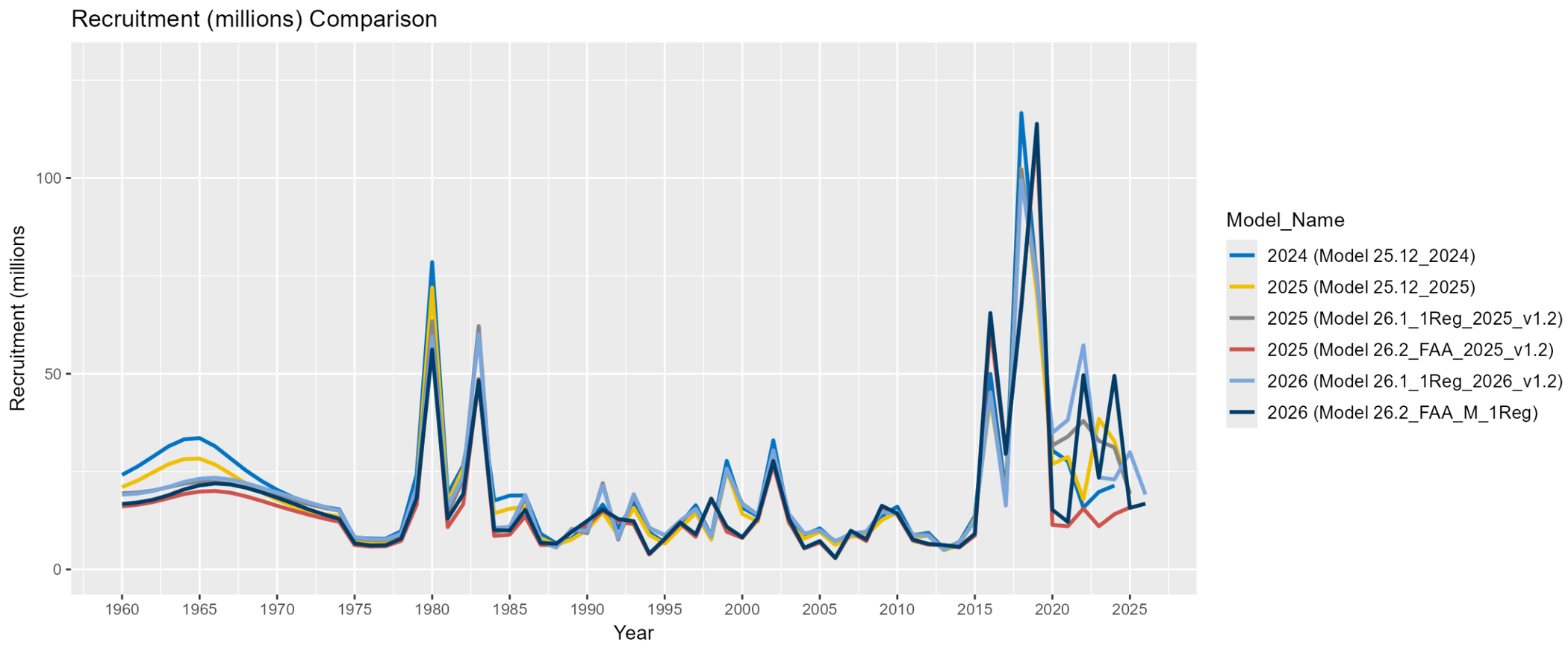

GOA LL Survey Age Comps Pearson residuals
|Pearson resid| 1 2
Sign Negative Positive
Female
Male
AK
Age
Year
0
10
20
30
2000
2010
2020

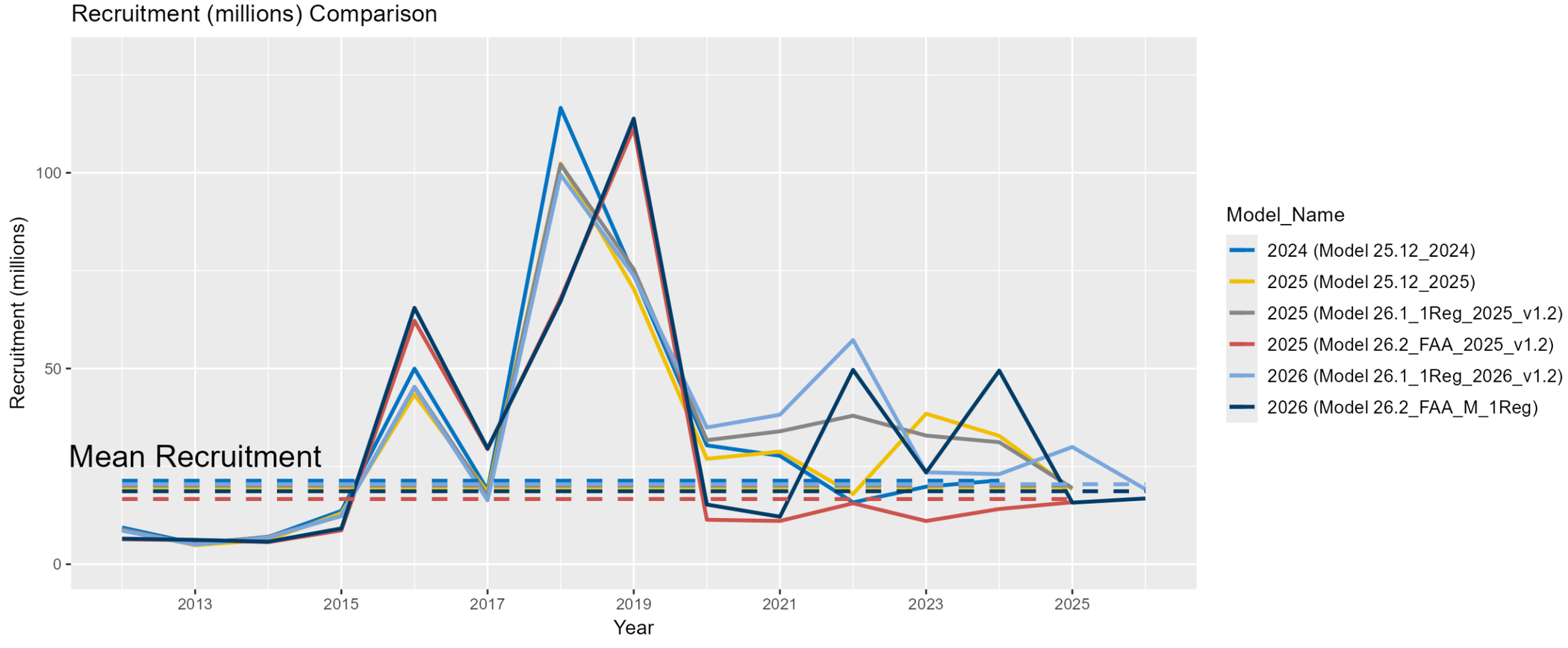

JPN LL Survey Age Comps Pearson residuals
|Pearson resid|
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Age
30
20
10
0
1985
1990
1985
1990
Year

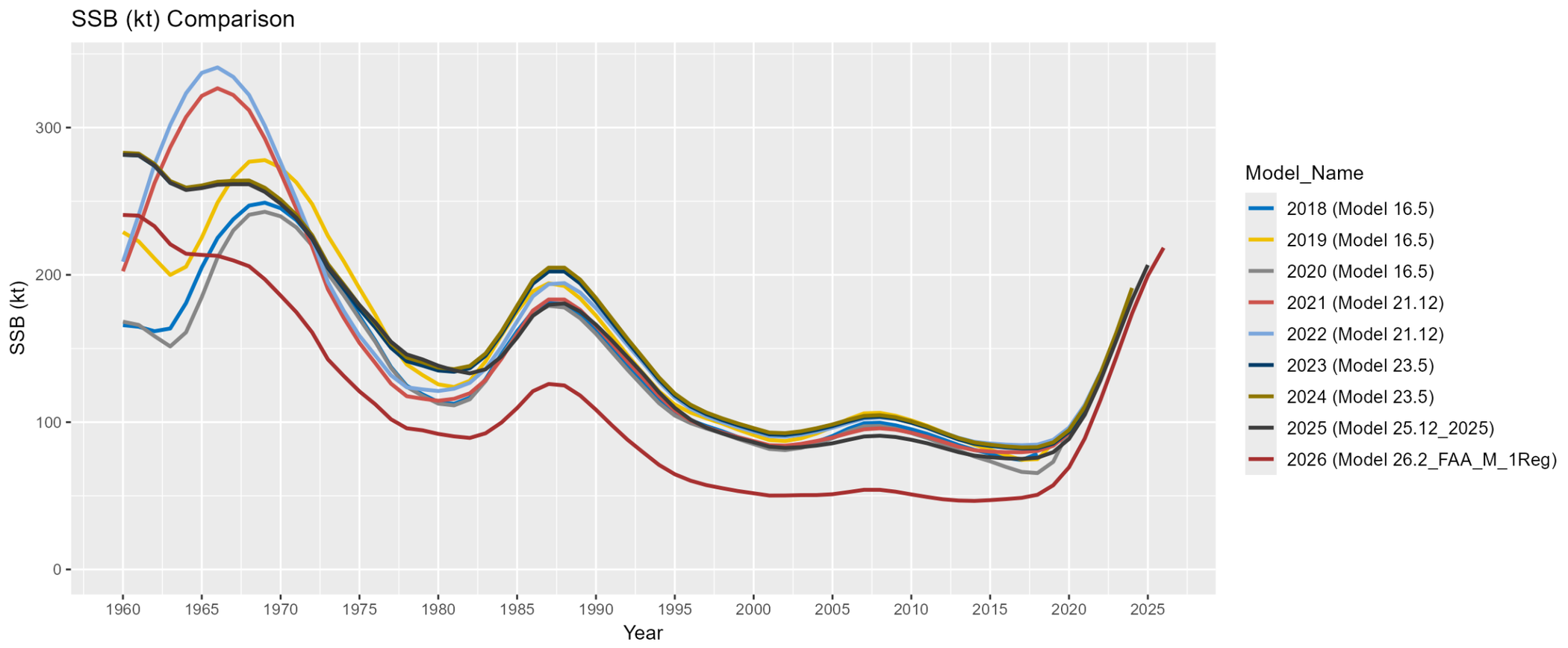

JPN LL Survey Length Comps Pearson residuals
|Pearson resid|
0.0
0.5
1.0
1.5
2.0
Sign
Negative
Positive
Female
Male
AK
Len
100
80
60
40
1980
1985
1990
Year

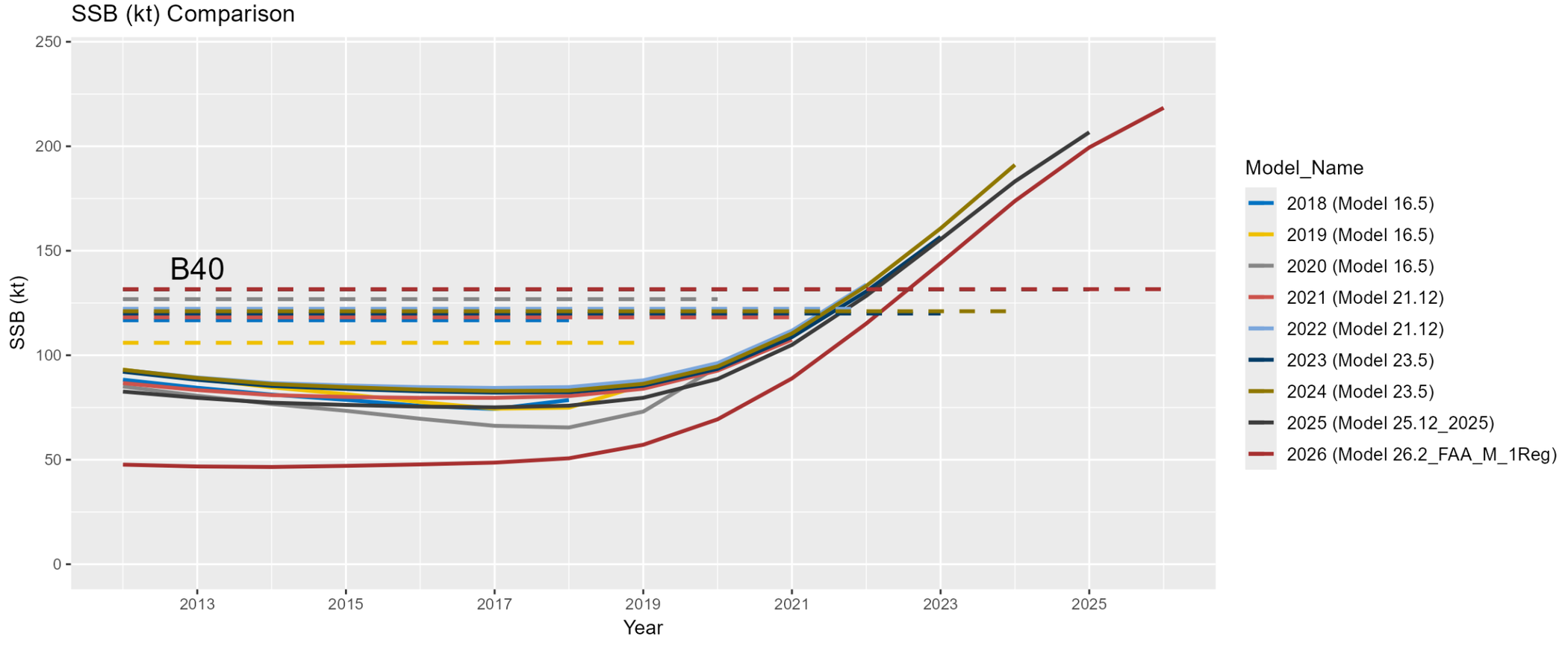

SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_M_1Reg)

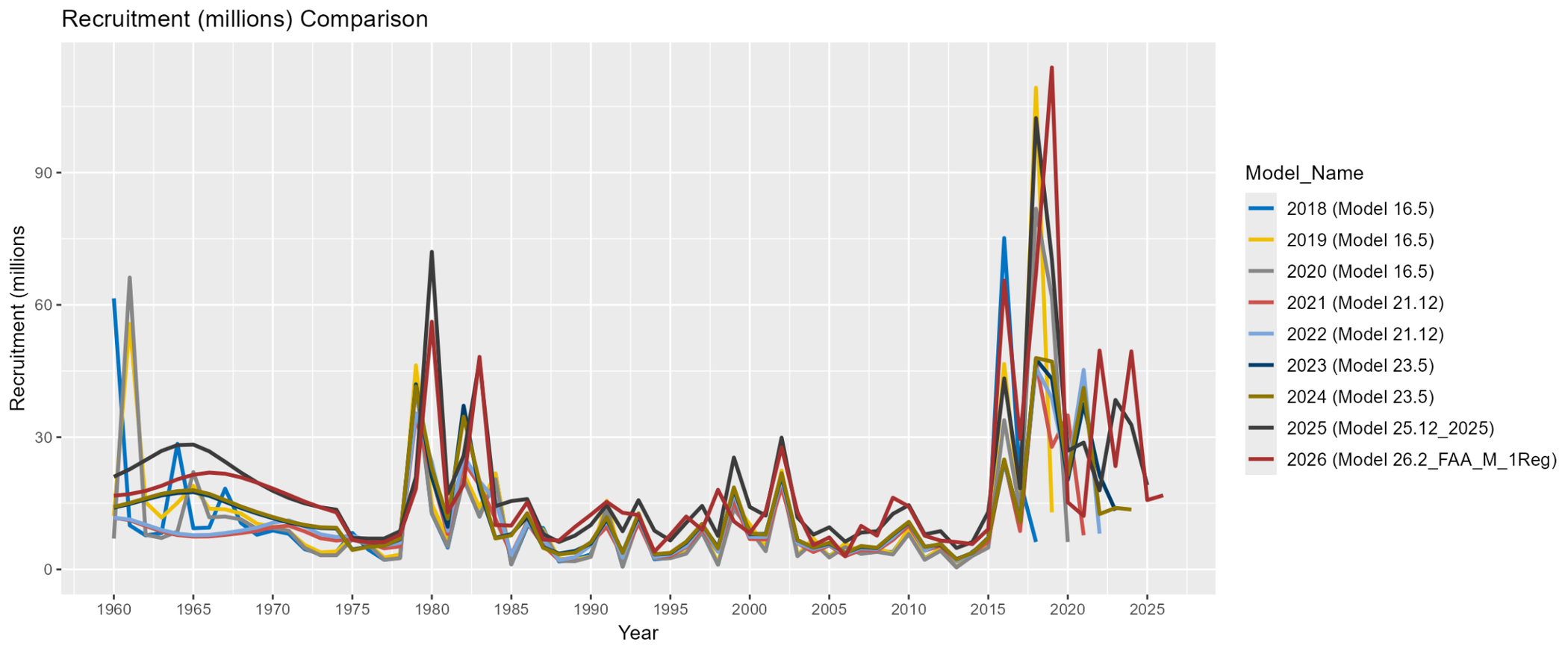

SSB (kt) Comparison
B40
SSB (kt)
200
100
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_M_1Reg)

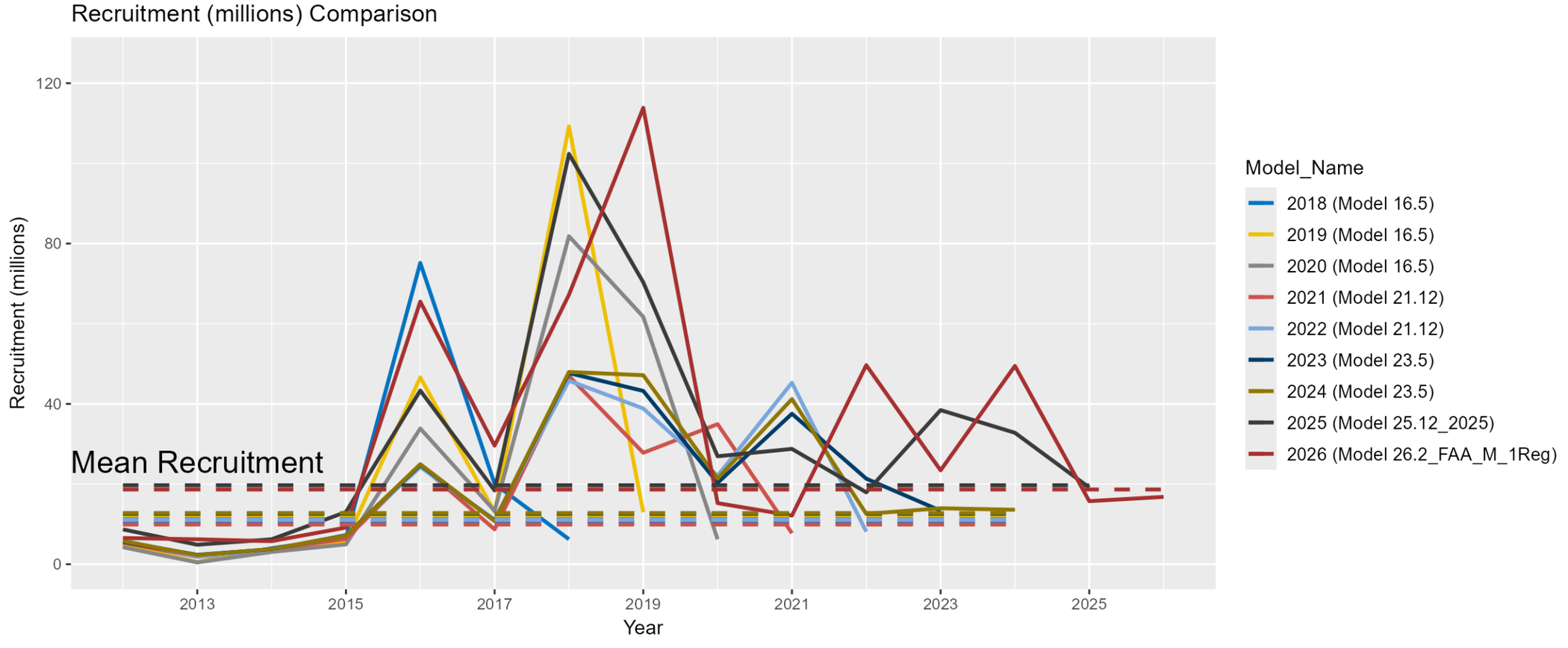

Recruitment (millions) Comparison
Recruitment (millions
100
50
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_M_1Reg)


Recruitment (millions) Comparison
Recruitment (millions)
100
50
0
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2024 (Model 25.12_2024)
2025 (Model 25.12_2025)
2025 (Model 26.1_1Reg_2025_v1.2)
2025 (Model 26.2_FAA_2025_v1.2)
2026 (Model 26.1_1Reg_2026_v1.2)
2026 (Model 26.2_FAA_M_1Reg)


SSB (kt) Comparison
SSB (kt)
300
200
100
0
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_M_1Reg)


SSB (kt) Comparison
B40
SSB (kt)
250
200
150
100
50
0
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_M_1Reg)


Recruitment (millions) Comparison
Recruitment (millions
0
30
60
90
1960
1965
1970
1975
1980
1985
1990
1995
2000
2005
2010
2015
2020
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_M_1Reg)


Recruitment (millions) Comparison
Recruitment (millions)
0
40
80
120
Mean Recruitment
2013
2015
2017
2019
2021
2023
2025
Year
Model_Name
2018 (Model 16.5)
2019 (Model 16.5)
2020 (Model 16.5)
2021 (Model 21.12)
2022 (Model 21.12)
2023 (Model 23.5)
2024 (Model 23.5)
2025 (Model 25.12_2025)
2026 (Model 26.2_FAA_M_1Reg)

Retrospective Year
1 2 3 4 5 6 7

Recruitment

Mohns Rho: -0.2902

SSB

Mohns Rho: 0.0361

AK

Relative Difference from Terminal Year

0.0

-0.5

1960 1980 2000 2020

1960 1980 2000 2020

Year

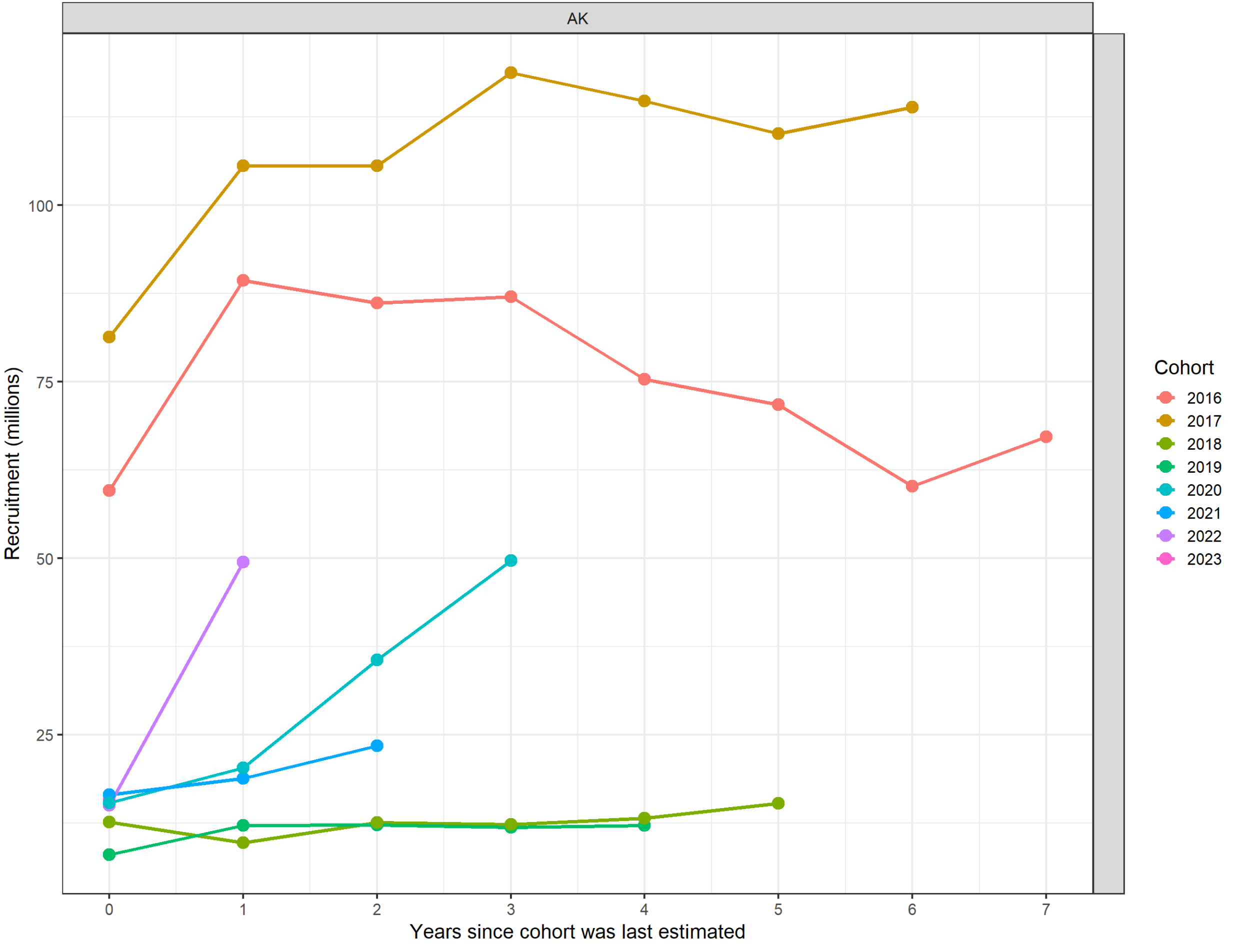
Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year


AK
Recruitment (millions)
100
75
50
25
0
1
2
3
4
5
6
7
Years since cohort was last estimated
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

Retrospective Year
1 2 3 4 5 6 7

Recruitment

SSB

AK

Mohns Rho: -0.2839

Mohns Rho: 0.0726

0.5

0.0

-0.5

Relative Difference from Terminal Year

1960 1980 2000 2020

1960 1980 2000 2020

Year

Peel
1 2 3 4 5 6 7

AK
Recruitment

AK
SSB

Value

Year

AK

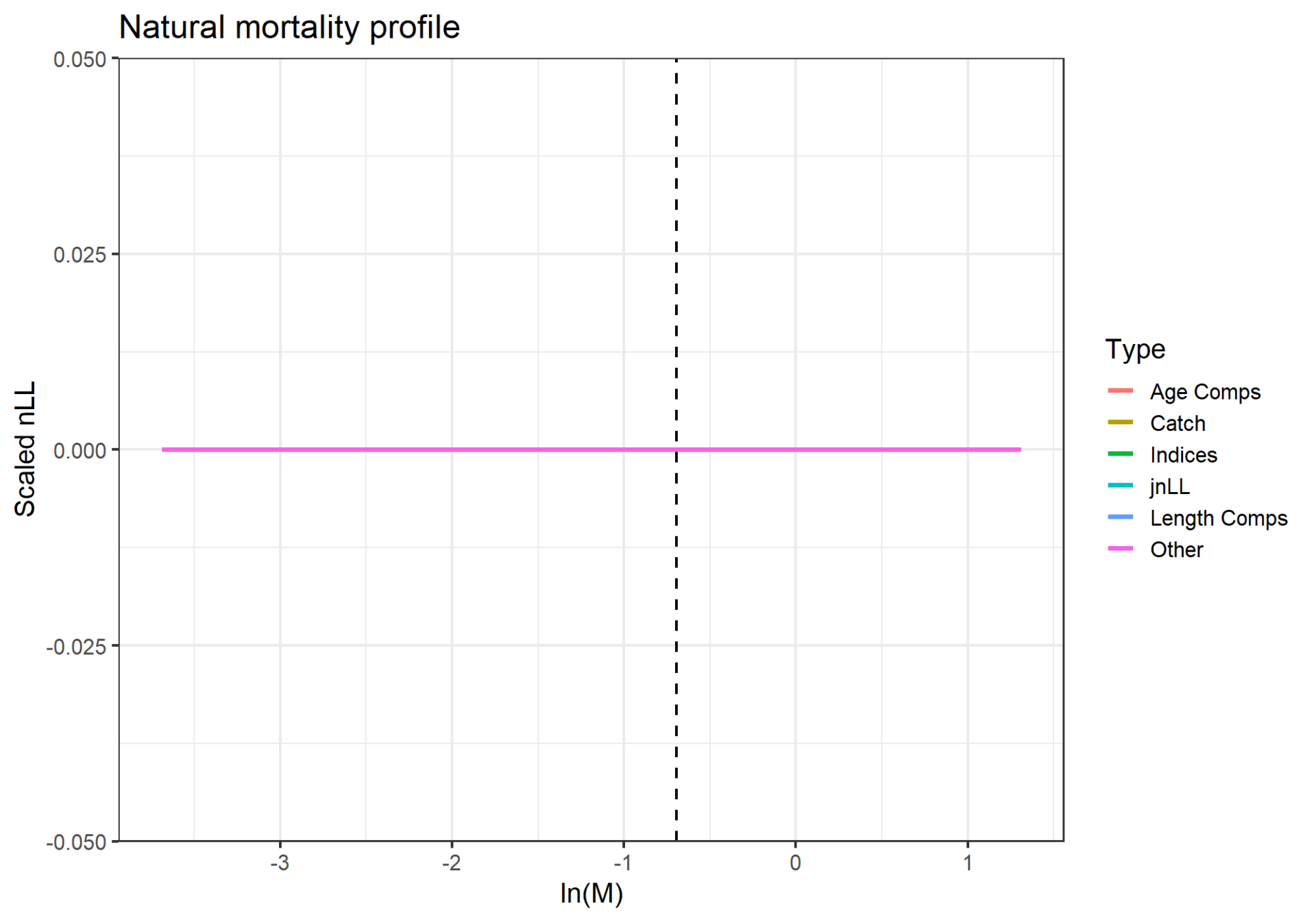
120
90
60
30

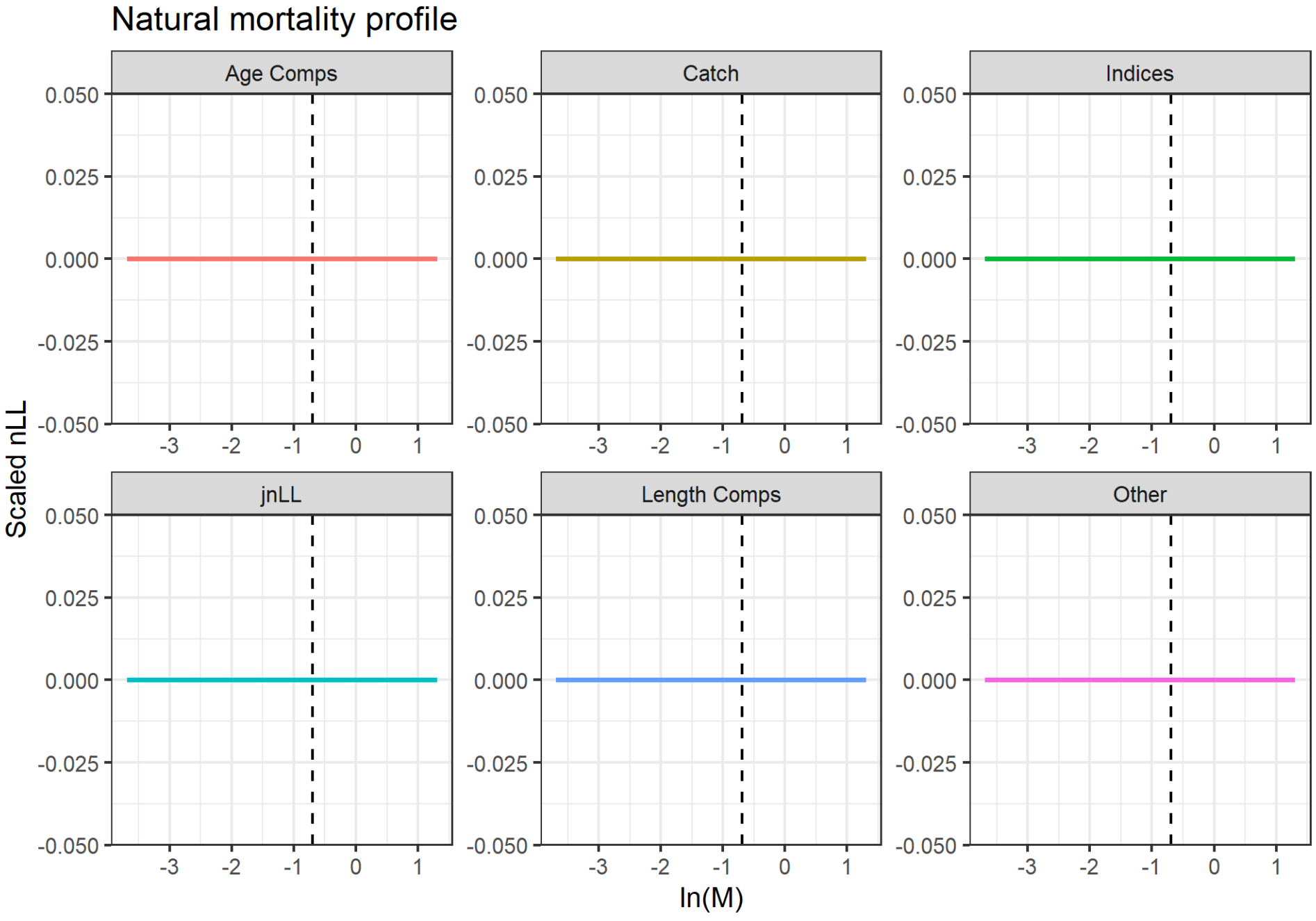
Recruitment (millions)

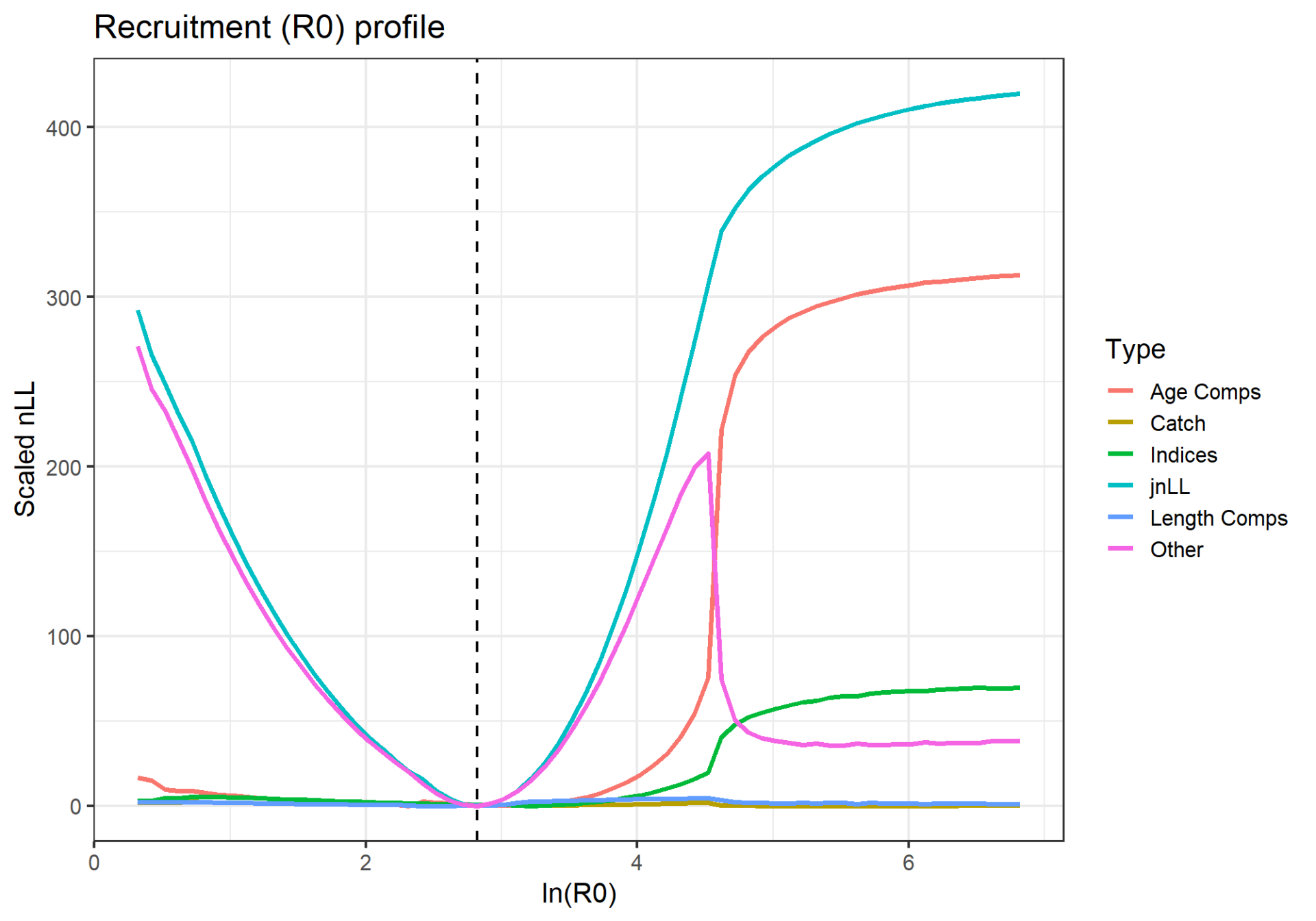
0
1
2
3
4
5
6
7

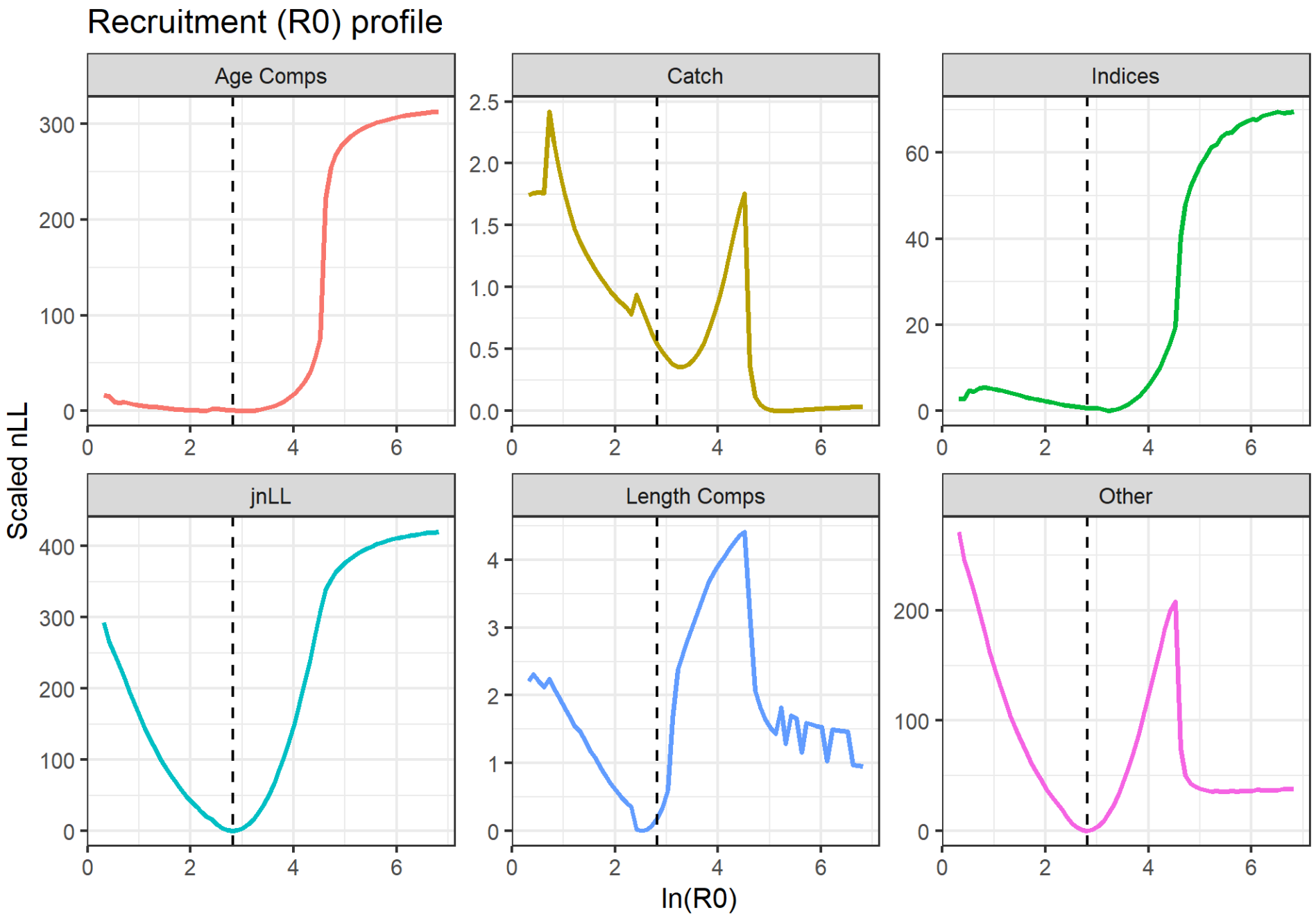
Years since cohort was last estimated

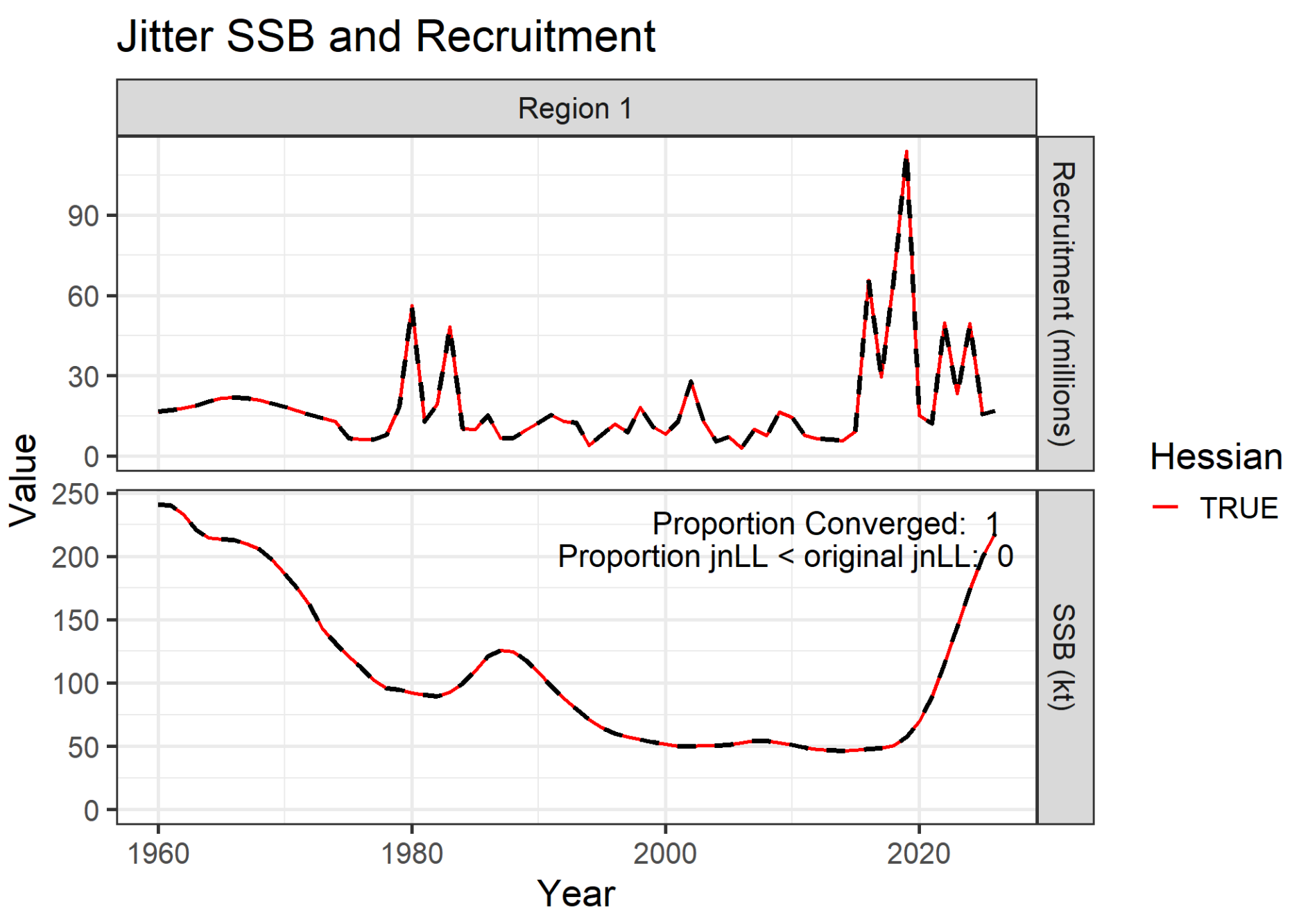
Cohort
2016
2017
2018
2019
2020
2021
2022
2023

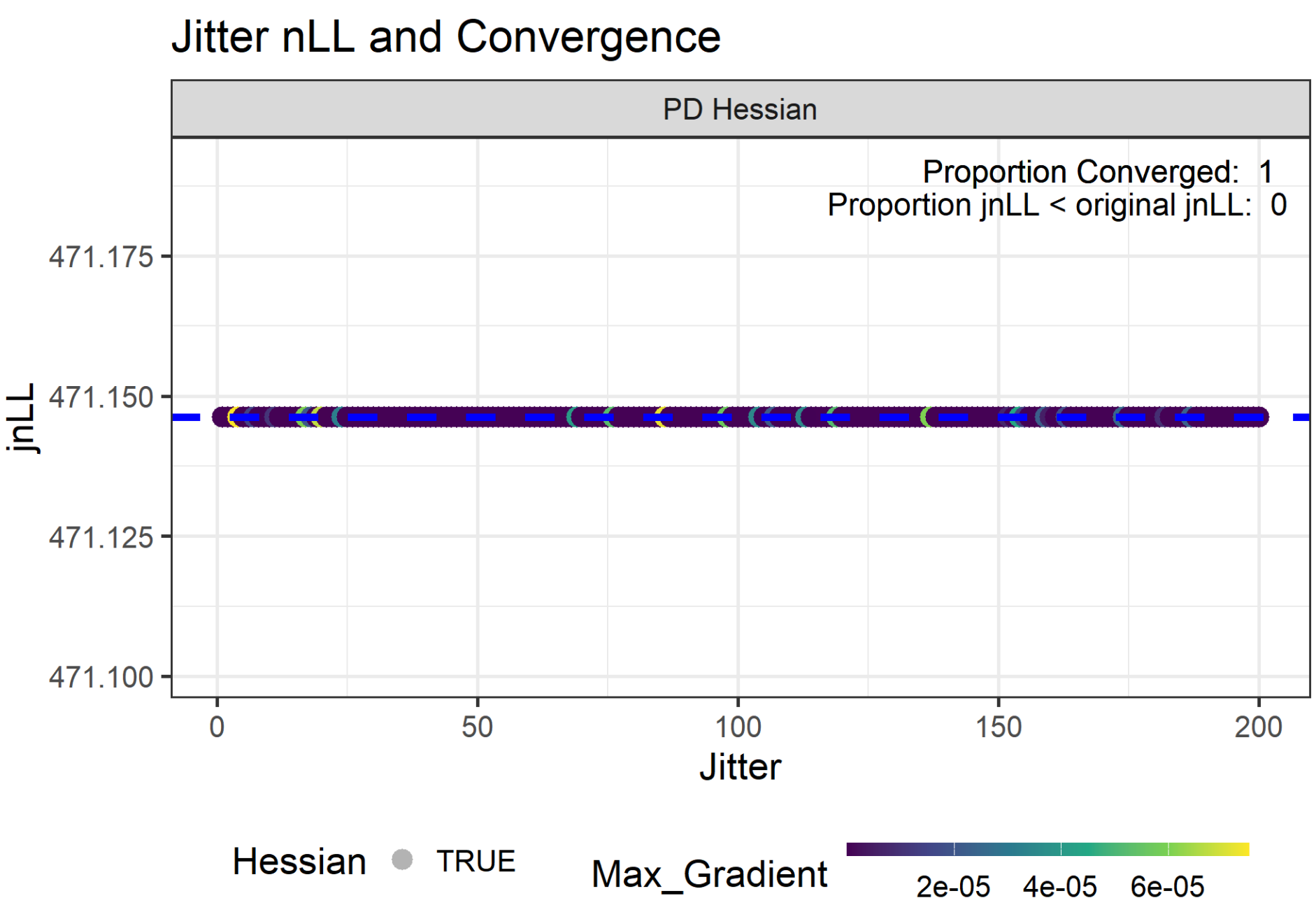

Natural mortality profile
0.050
0.025
0.000
-0.025
-0.050
Scaled nLL
-3
-2
-1
0
1
ln(M)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

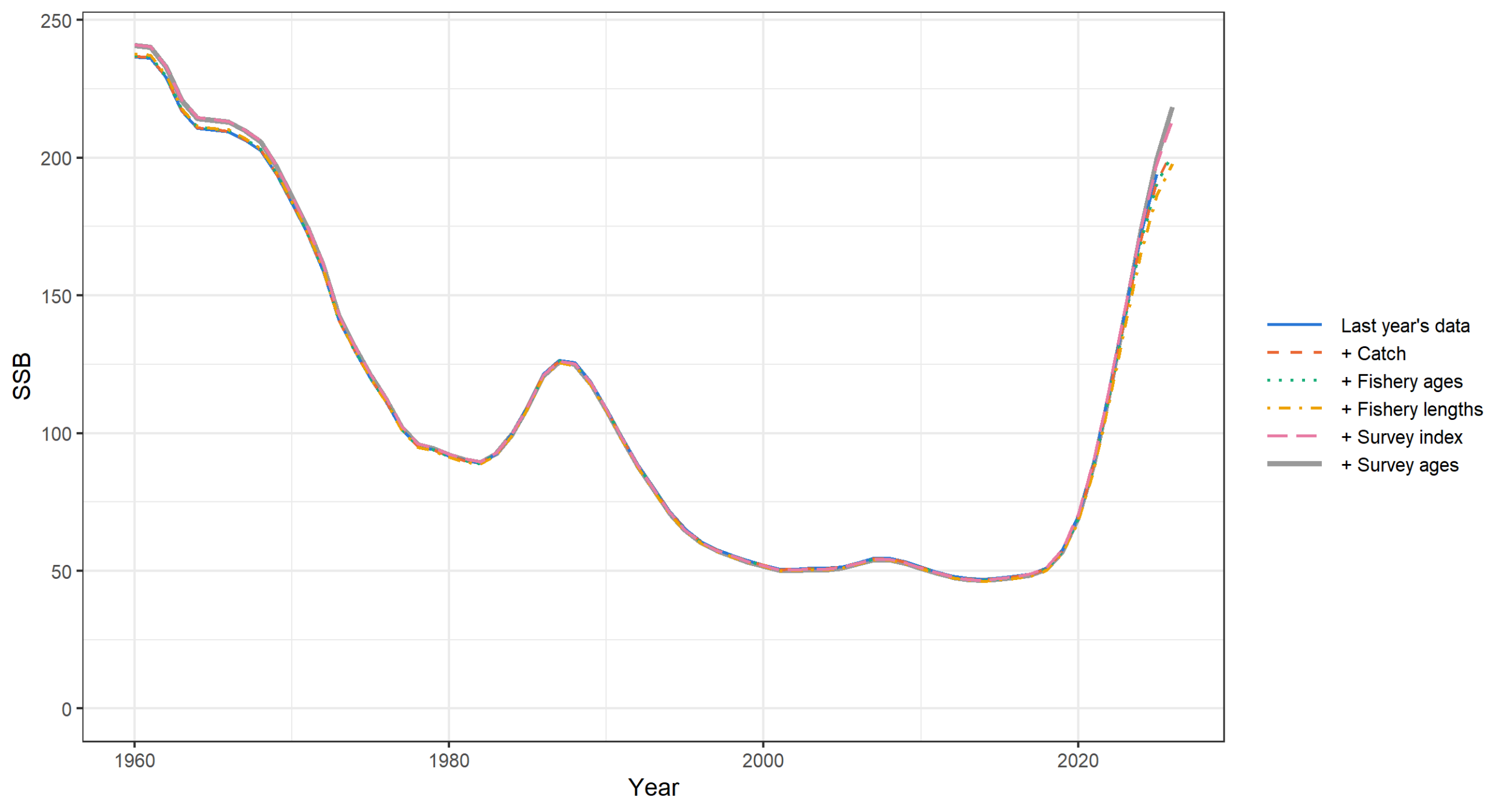

Natural mortality profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(M)
0.050
0.025
0.000
-0.025
-0.050
-3
-2
-1
0
1

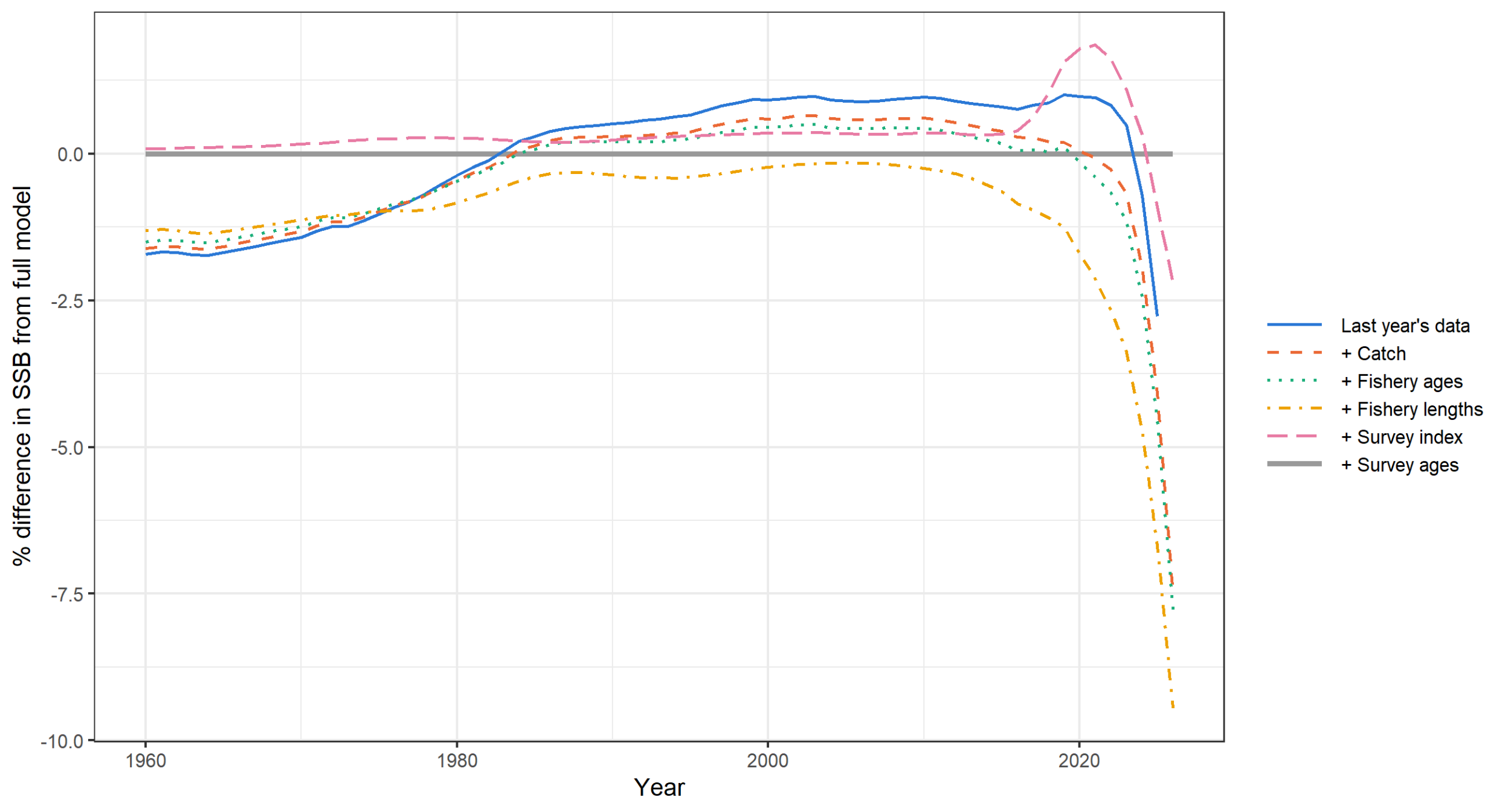

Recruitment (R0) profile
Scaled nLL
400
300
200
100
0
0
2
4
6
ln(R0)
Type
Age Comps
Catch
Indices
jnLL
Length Comps
Other

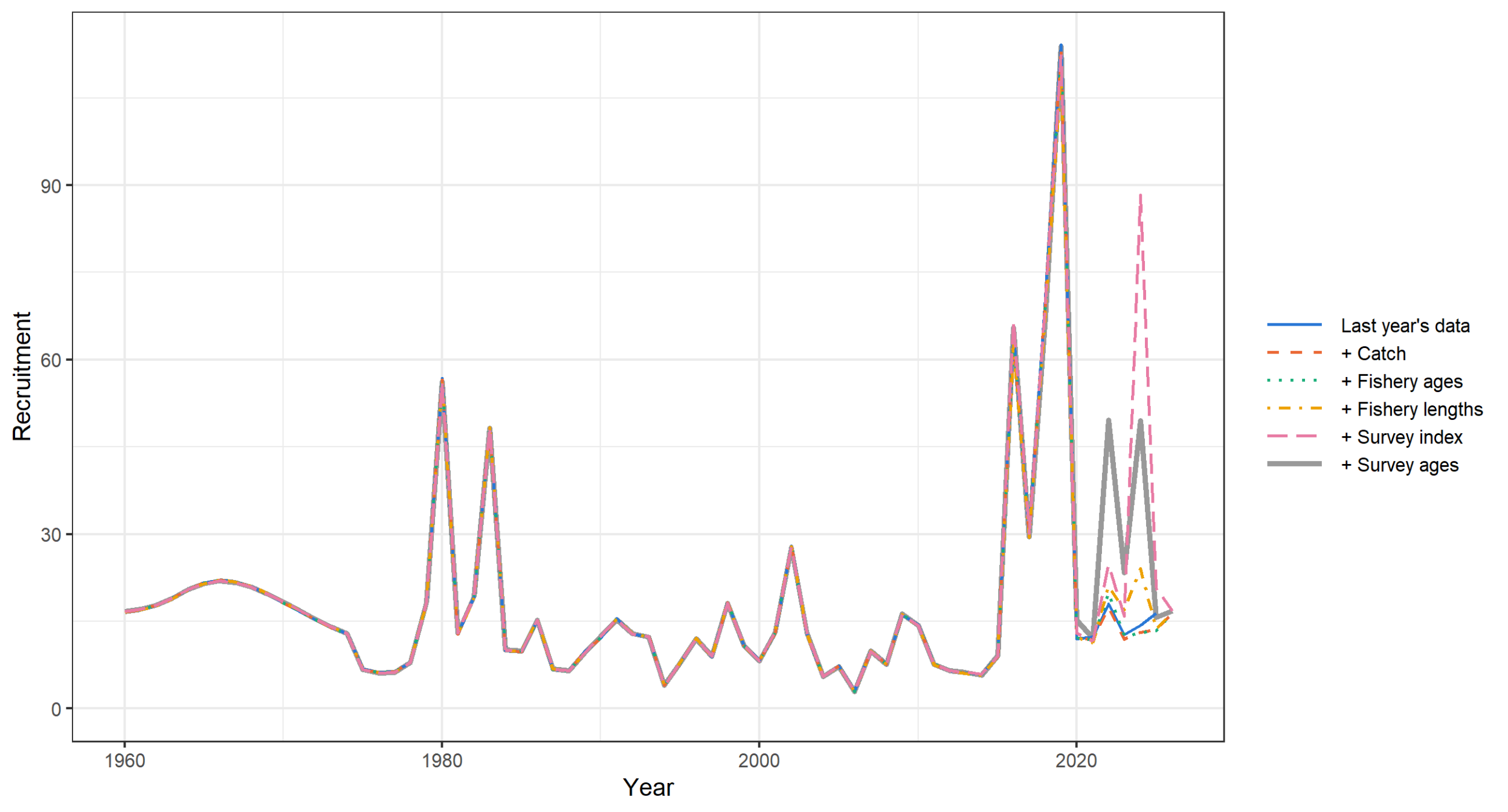

Recruitment (R0) profile
Age Comps
Catch
Indices
jnLL
Length Comps
Other
Scaled nLL
ln(R0)

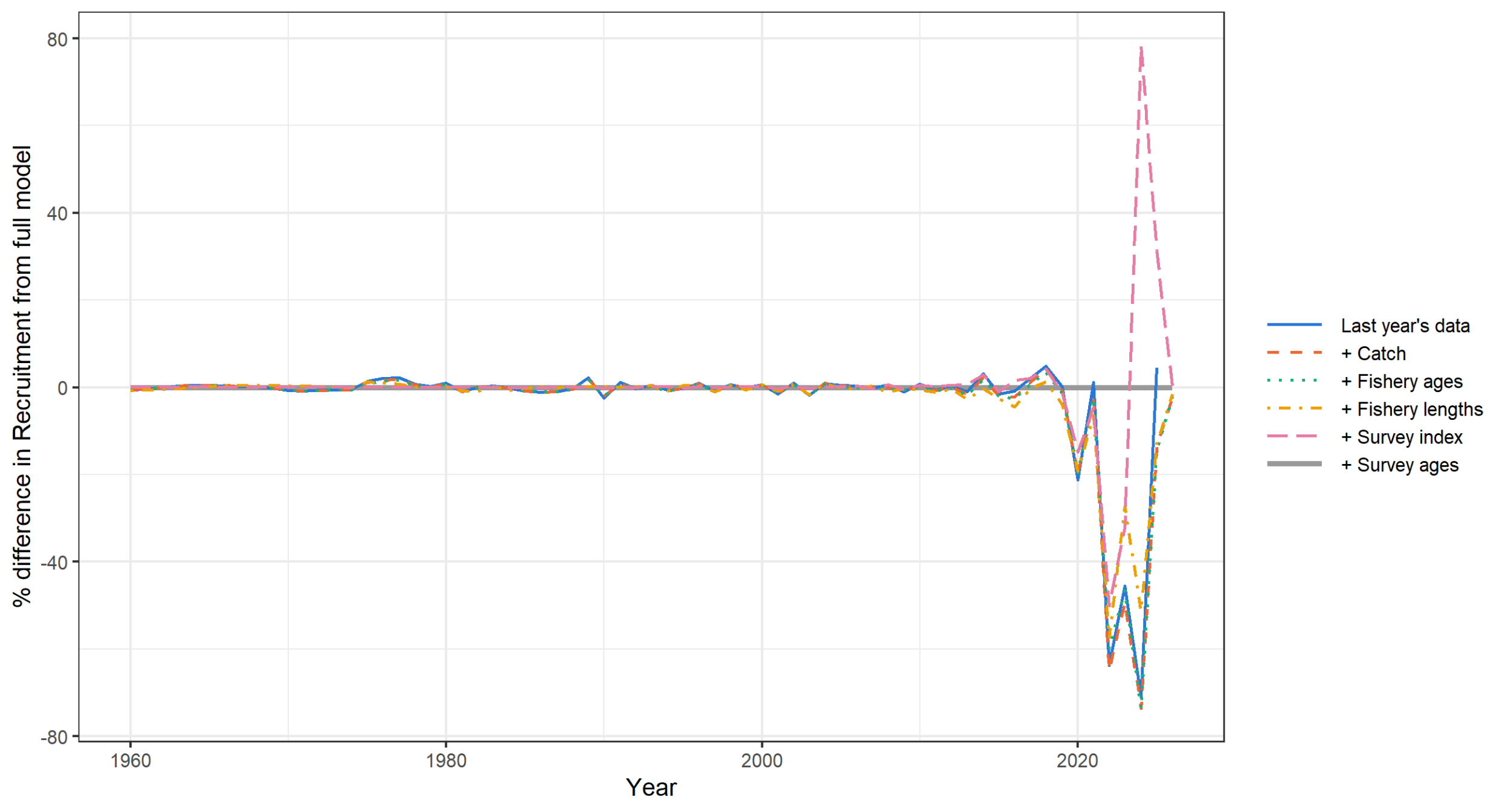

Jitter SSB and Recruitment
Region 1
Recruitment (millions)
SSB (kt)
Value
Year
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
Hessian
TRUE
0
30
60
90
0
50
100
150
200
250
1960
1980
2000
2020

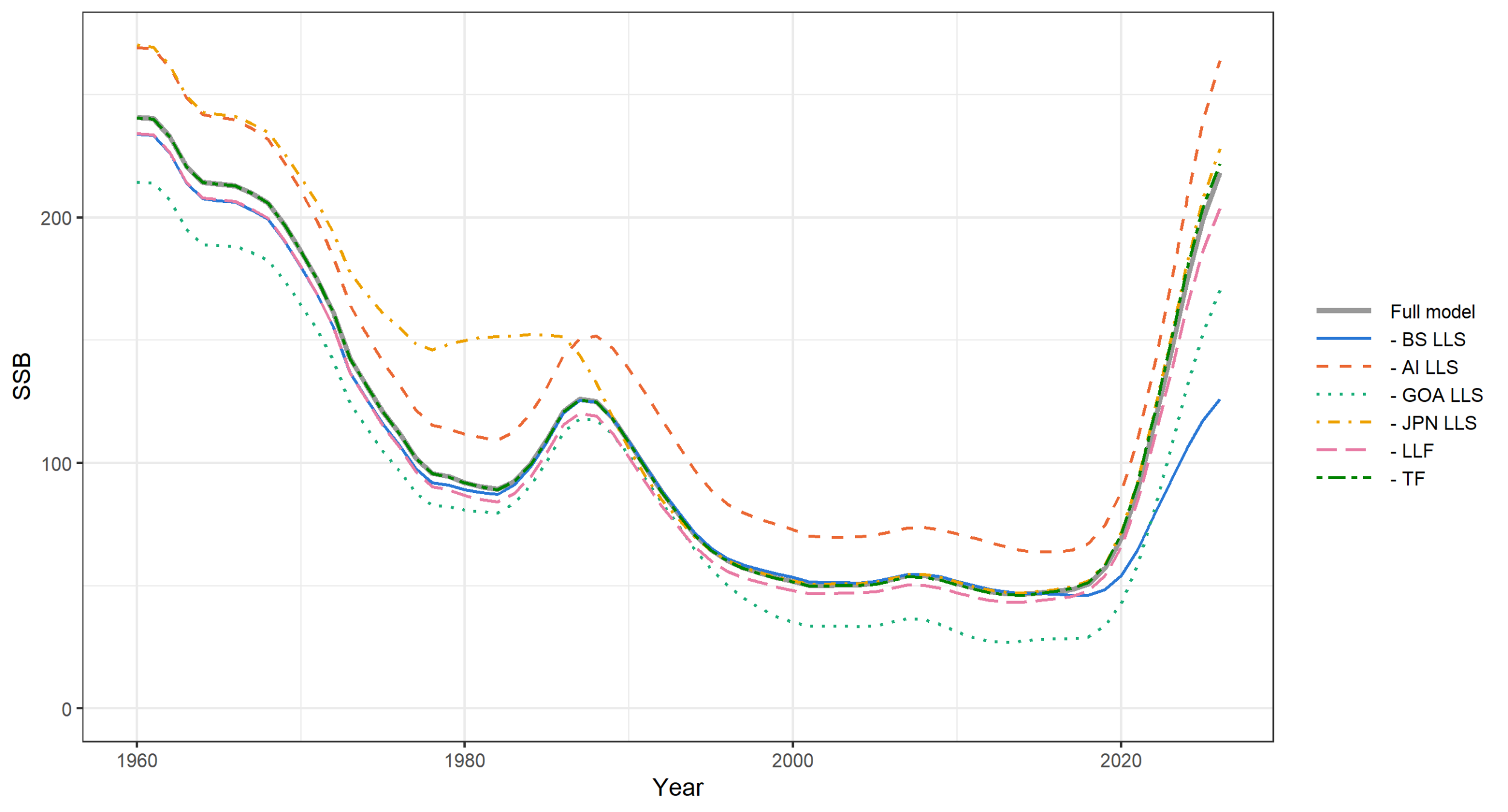

Jitter nLL and Convergence
PD Hessian
Proportion Converged: 1
Proportion jnLL < original jnLL: 0
471.175
471.150
471.125
471.100
jnLL
0
50
100
150
200
Jitter
Hessian
TRUE
Max_Gradient
2e-05
4e-05
6e-05

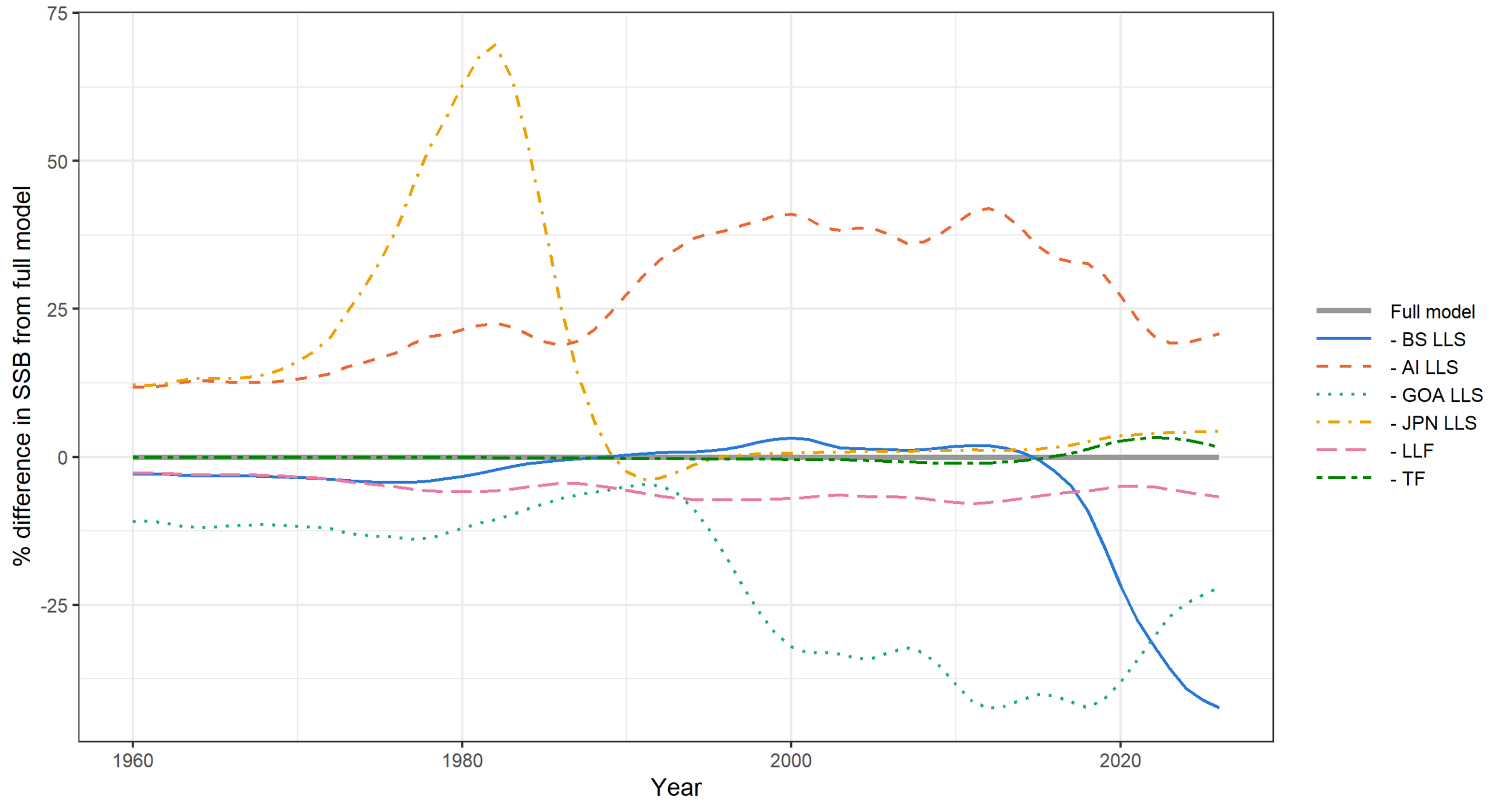

250
200
150
100
50
0
SSB
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

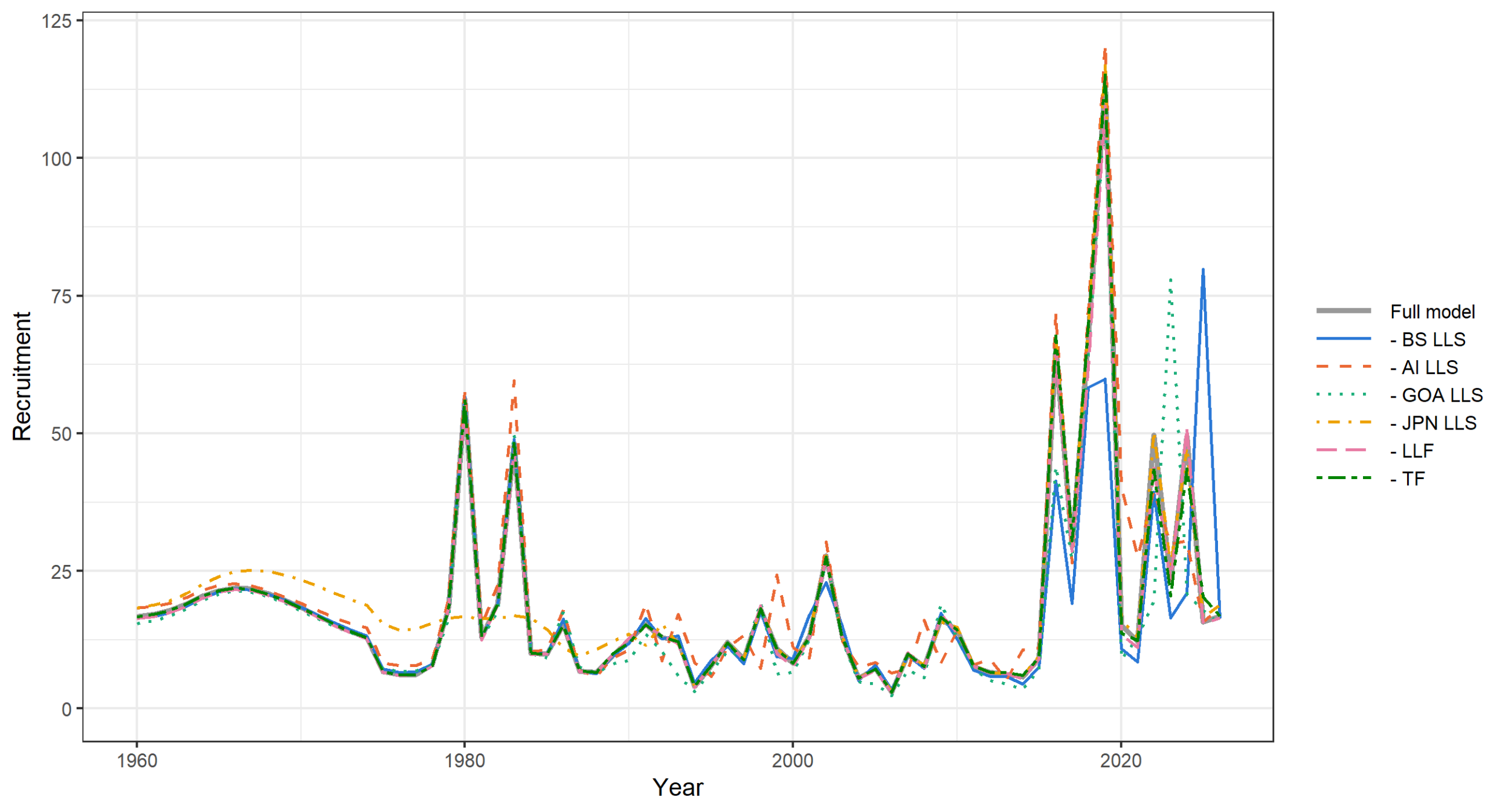

% difference in SSB from full model
0.0
-2.5
-5.0
-7.5
-10.0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages

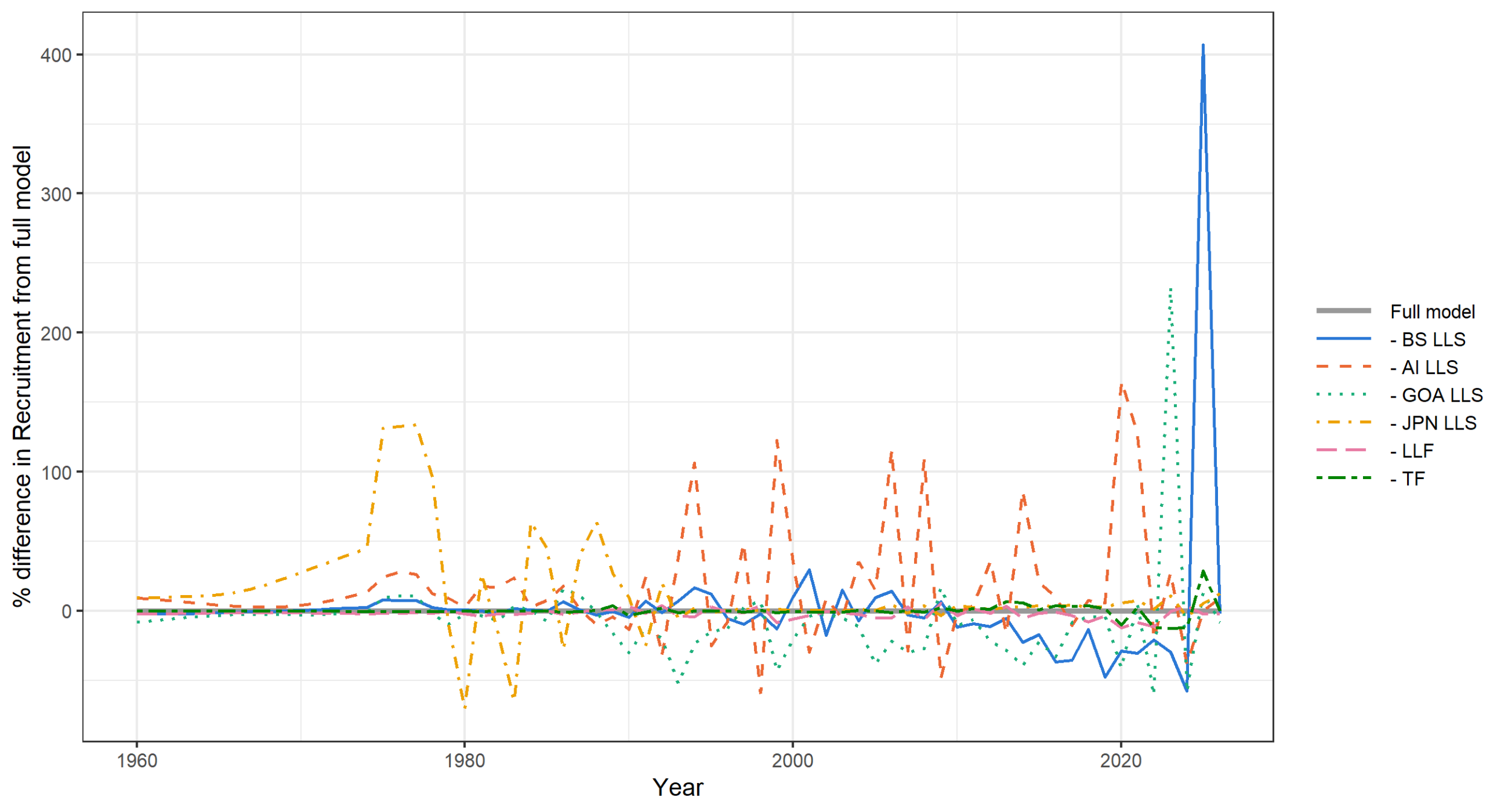

Recruitment
90
60
30
0
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


80
40
0
-40
-80
% difference in Recruitment from full model
1960
1980
2000
2020
Year
Last year's data
+ Catch
+ Fishery ages
+ Fishery lengths
+ Survey index
+ Survey ages


SSB
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


75
50
25
0
-25
% difference in SSB from full model
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

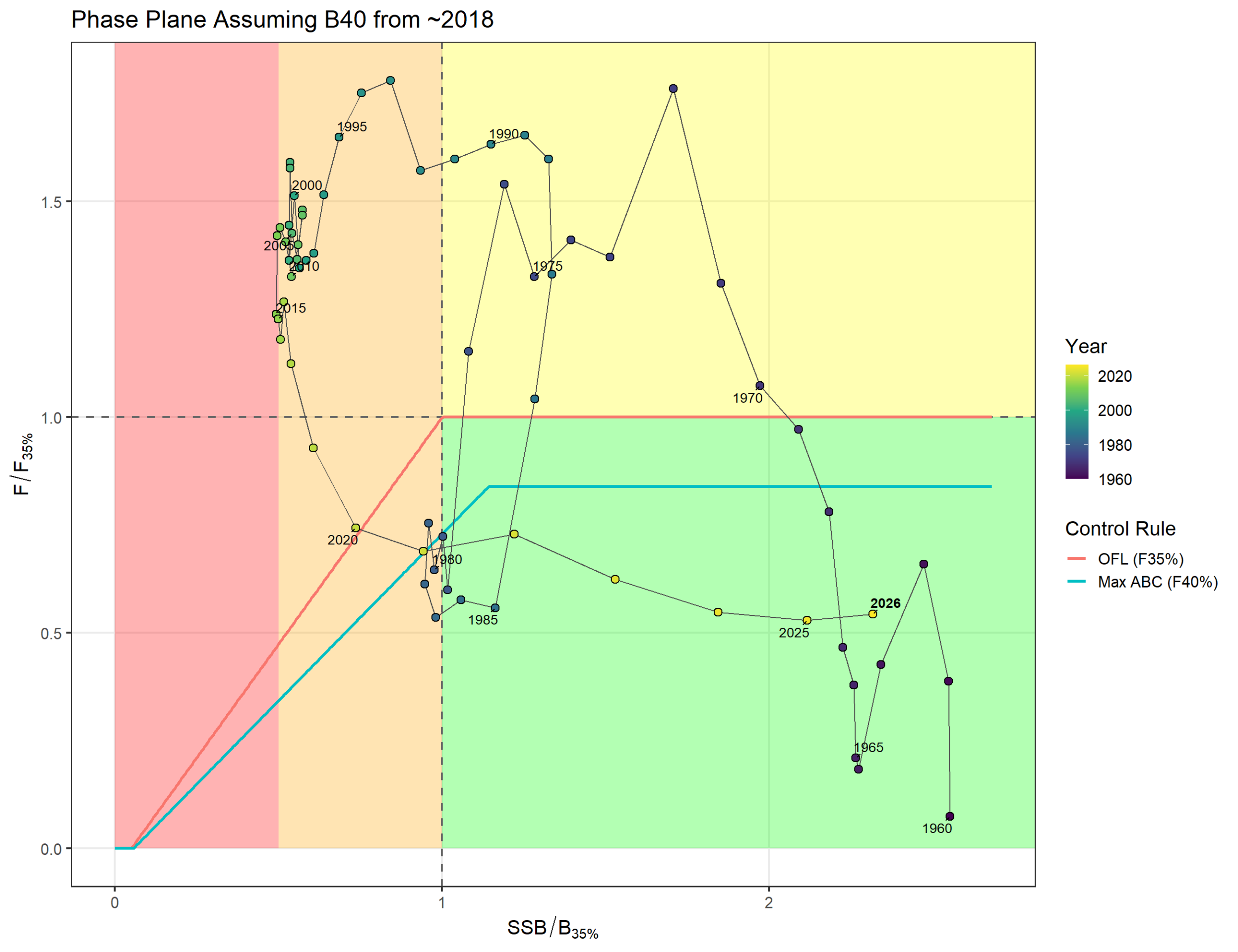

125
100
75
50
25
0
Recruitment
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF


% difference in Recruitment from full model
400
300
200
100
0
1960
1980
2000
2020
Year
Full model
- BS LLS
- AI LLS
- GOA LLS
- JPN LLS
- LLF
- TF

Source	SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	1.00	NA	NA
AI LL Survey Index	0.23	1.00	NA	NA
GOA LL Survey Index	0.08	1.00	NA	NA
Trawl Survey Index	NA	NA	NA	NA
JPN LL Survey Index	0.07	1.00	NA	NA
Fixed Gear Fishery CPUE	NA	NA	NA	NA
BS LL Survey Ages	NA	0.61	34.07	20.95
AI LL Survey Ages	NA	2.01	35.57	71.39
GOA LL Survey Ages	NA	0.37	67.17	24.53
JPN LL Survey Ages	NA	0.19	83.33	15.72
JPN LL Survey Lengths	NA	0.32	54.62	17.49
Fixed Gear Fishery Ages	NA	0.04	142.85	5.90
Trawl Fishery Lengths	NA	0.09	47.62	4.19

Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
26.2_FAA_M_1Reg	1	218.3976	389.2587	0.05159311	39.26729	131.6174	0.07965	1.65934	0.56106	0.64771

Phase Plane Assuming B40 from ~2018



S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

0
100
200
300
400

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

SSB (kt)

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100

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Biomass (kt)

0
250
500
750

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

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2010
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S6: Overfished (FOFL)

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S7: Approaching Overfishing (FABC -> FOFL)

Recruitment (millions)

0
30
60
90

1960
1980
2000
2020
2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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S7: Approaching Overfishing (FABC -> FOFL)

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0

30

60

90

2010

2020

2030

2040

Year

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S1a: Author's F (FABC Ratio) Year 1, FABC After

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S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

0.15

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

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S7: Approaching Overfishing (FABC -> FOFL)

Fishing Mortality

0.00

0.05

0.10

2010

2020

2030

2040

Year

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S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

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S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0

20

40

1960

1980

2000

2020

2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0
10
20
30
40

1960
1980
2000
2020
2040

Year

Fleet
LLF
TF

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

0 10 20 30 40 50

2010 2020 2030 2040

Year

S1: FABC (F40)

S1a: Author's F (FABC Ratio) Year 1, FABC After

S2: Author's F (FABC Ratio) Years 1+2

S3: 5-year Average F

S4: Half max FABC (1/2 * F40)

S5: No Fishing

S6: Overfished (FOFL)

S6a: Author's F (FABC Ratio) Year 1, FOFL After

S7: Approaching Overfishing (FABC -> FOFL)

Yield (kt)

Year

Fleet

LLF

TF

Quantity	2025	2026	2027	2028
M Female (estimated)	0.114	0.114	0.091	0.091
M Male (estimated)	0.114	0.114	0.091	0.091
Average Recruitment (millions)	25.9	25.9	19.9	19.9
Tier	3a	3a	3a	3a
Projected Total Biomass (Age-2+, t)	704,713	695,681	603114	599536
Projected Female Spawning Biomass (t)	219,714	241,217	231170	241867
B100% (t)	302,672	302,672	329044	329044
B40% (t)	121,069	121,069	131617	131617
B35% (t)	105,935	105,935	115165	115165
FOFL	0.102	0.102	0.095	0.095
max FABC	0.087	0.087	0.08	0.08
Total Depredation (t)	171	169	131	131
OFL (t)	58,731	57,993	46448	46549
OFLw (t)	58,532	57,797	46317	46418
max ABC (t)	50,283	49,651	39265	39354
ABC (t)	50,283	49,651	39265	39354
ABCw (t)	50,111	49,482	39134	39223
Status	2023	2024	2025	2026
Overfishing	No	NA	No	NA
Overfished	NA	No	NA	No
Approaching Overfished	NA	No	NA	No

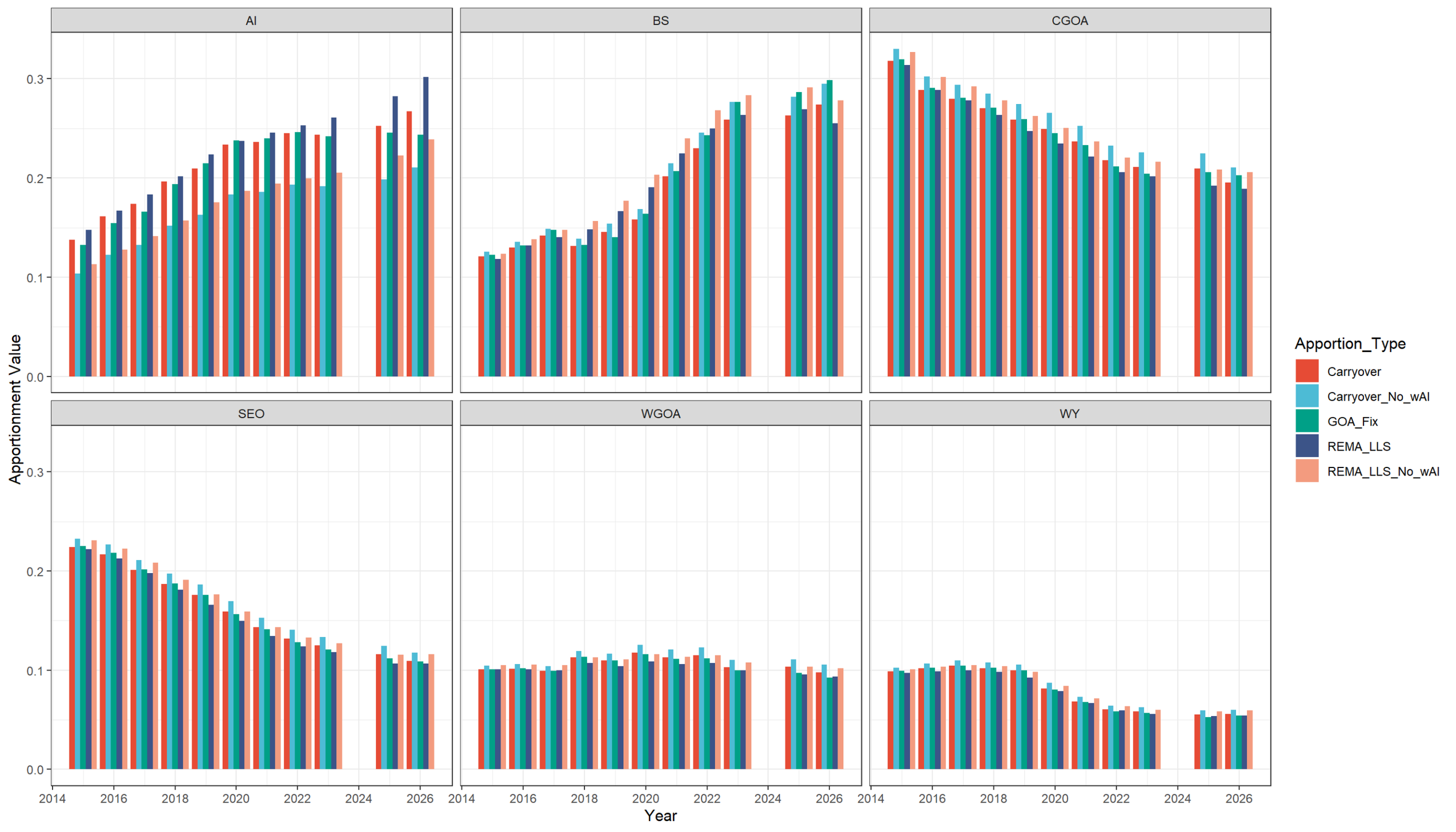
Year	Apportion_Type	BS	AI	WGOA	CGOA	WY	SEO
2026	Carryover_No_wAI	0.30	0.20	0.11	0.21	0.06	0.12
2026	REMA_LLS_No_wAI	0.28	0.23	0.10	0.21	0.06	0.12
2026	Carryover	0.27	0.26	0.10	0.20	0.06	0.11
2026	REMA_LLS	0.26	0.30	0.09	0.19	0.05	0.11
2026	GOA_Fix	0.30	0.24	0.09	0.20	0.05	0.12

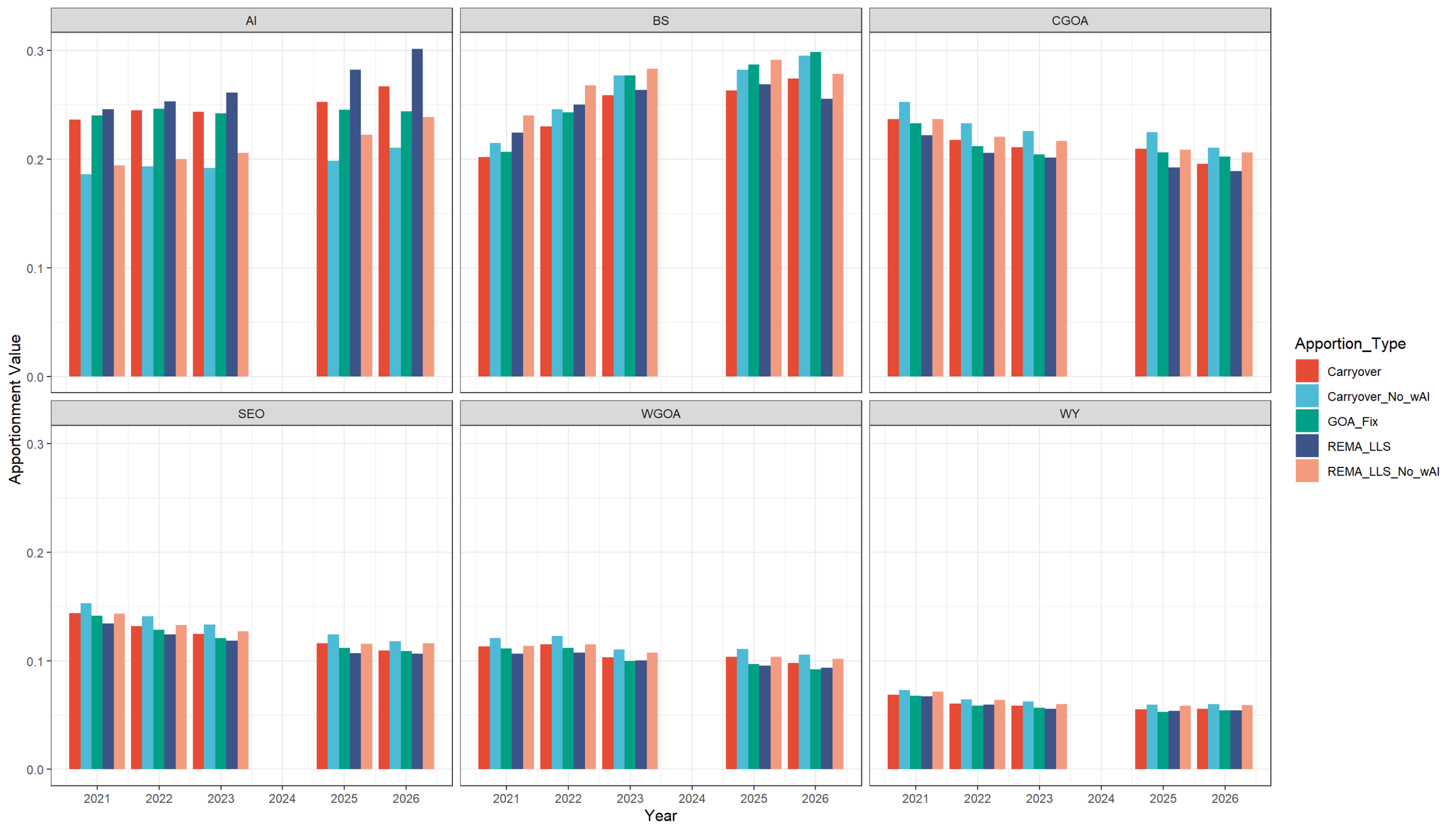
Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	13037	11421	22550	4687	9622	2652	5589	47008
2027	Carryover_No_wAI	11740	7827	19567	4305	8218	2348	4696	39134
2027	REMA_LLS_No_wAI	10958	9001	19176	3913	8218	2348	4696	39134
2027	Carryover	10566	10175	18393	3913	7827	2348	4305	39134
2027	REMA_LLS	10175	11740	17219	3522	7435	1957	4305	39134
2027	GOA_Fix	11740	9392	18002	3522	7827	1957	4696	39134
2028	Carryover_No_wAI	11767	7844	19612	4315	8237	2353	4707	39223
2028	REMA_LLS_No_wAI	10982	9021	19219	3922	8237	2353	4708	39223
2028	Carryover	10590	10198	18435	3922	7845	2353	4315	39223
2028	REMA_LLS	10198	11767	17258	3530	7452	1961	4315	39223
2028	GOA_Fix	11767	9413	18043	3530	7845	1961	4707	39223

Year	Apportion_Type	BS	AI	GOA	WGOA	CGOA	WY	SEO	ABCw
2026	GOA_Fix	0	0	0	0	0	0	0	0
2027	Carryover_No_wAI	-10	-31	-13	-8	-15	-11	-16	-17
2027	REMA_LLS_No_wAI	-16	-21	-15	-17	-15	-11	-16	-17
2027	Carryover	-19	-11	-18	-17	-19	-11	-23	-17
2027	REMA_LLS	-22	3	-24	-25	-23	-26	-23	-17
2027	GOA_Fix	-10	-18	-20	-25	-19	-26	-16	-17
2028	Carryover_No_wAI	-10	-31	-13	-8	-14	-11	-16	-17
2028	REMA_LLS_No_wAI	-16	-21	-15	-16	-14	-11	-16	-17
2028	Carryover	-19	-11	-18	-16	-18	-11	-23	-17
2028	REMA_LLS	-22	3	-23	-25	-23	-26	-23	-17
2028	GOA_Fix	-10	-18	-20	-25	-18	-26	-16	-17

Year	Region	OFLw	ABCw	TAC	Catch	Year2	Region	OFLw	ABCw	Year3	Region	OFLw	ABCw
2026	BS	NA	13037	8996	3339	2027	BS	NA	11740	2028	BS	NA	11767
2026	AI	NA	11421	7440	632	2027	AI	NA	7827	2028	AI	NA	7844
2026	GOA	NA	22550	22550	18191	2027	GOA	NA	19567	2028	GOA	NA	19612
2026	WGOA	NA	4687	4687	3458	2027	WGOA	NA	4305	2028	WGOA	NA	4315
2026	CGOA	NA	9622	9622	7597	2027	CGOA	NA	8218	2028	CGOA	NA	8237
2026	WY	NA	2652	2652	2271	2027	WY	NA	2348	2028	WY	NA	2353
2026	SEO	NA	5589	5589	4865	2027	SEO	NA	4696	2028	SEO	NA	4707
2026	Total	57797	47008	38986	22161	2027	Total	46317	39134	2028	Total	46418	39223

Region	Year	Biomass (4+)	OFLw	ABCw	TAC	Catch
GOA	2025	305000	NA	24038	22836	18387
GOA	2026	299000	NA	23737	22550	18191
GOA	2027	260693	NA	19567	NA	NA
GOA	2028	257175	NA	19612	NA	NA
BS	2025	175000	NA	13898	8496	3909
BS	2026	172000	NA	13723	8996	3339
BS	2027	142749	NA	11740	NA	NA
BS	2028	140823	NA	11767	NA	NA
AI	2025	153000	NA	12175	7940	948
AI	2026	151000	NA	12022	7440	632
AI	2027	157155	NA	7827	NA	NA
AI	2028	155035	NA	7844	NA	NA


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2014
2016
2018
2020
2022
2024
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI


AI
BS
CGOA
SEO
WGOA
WY
Apportionment Value
0.0
0.1
0.2
0.3
2021
2022
2023
2024
2025
2026
Year
Apportion_Type
Carryover
Carryover_No_wAI
GOA_Fix
REMA_LLS
REMA_LLS_No_wAI